

2. CPU

The MCU is based on the Arm® Cortex-M85 Processor and Arm® Cortex®-M33 Processor.

2.1 Overview

2.1.1 CPU

(1) CPU0

- Arm® Cortex-M85
 - Revision: r1p1-00rel0
 - ARMv8.1-M architecture profile
 - Floating Point Unit (FPU) compliant with the ANSI/IEEE Std 754-2008 Scalar half, single, and double-precision floating-point operation. Interrupt notification when FPU exception occurs.
 - M-profile Vector Extension (MVE) Integer, half-precision, and single-precision floating-point MVE (MVE-F)
 - Armv8-M Security Extension
- Security Attribution Unit (SAU): 8 regions
- Implementation Defined Attribution Unit (IDAU)
- Memory Protection Unit (MPU)
 - Protected Memory System Architecture (PMSAv8)
 - Secure MPU (MPU_S): 8 regions
 - Non-secure MPU (MPU_NS): 8 regions
- System Timer (SysTick)

Reference clock:

 - CPUCLK0
 - 1 MHz generated from on-chip oscillator (MOCO/8)

Two system timers are implemented:

 - Secure instance (SysTick_S)
 - Non-secure instance (SysTick_NS)
- Low power modes
 - Sleep mode
 - Deep Sleep mode
- CPU individual reset
- Cache
 - Instruction cache: 16 KB with ECC
 - Data cache: 16 KB with ECC
- Tightly Coupled Memory (TCM)
 - ITCM: 128 KB (16 blocks × 8 KB) with ECC
 - DTCM: 128 KB (16 blocks × 8 KB) with ECC
- Nested Vectored Interrupt Controller (NVIC): 96 IRQs
- Programmable MBIST Controller: Excluded
- Global exclusive monitor is not supported

See reference [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B), [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924) for the detail.

(2) CPU1

- Arm® Cortex®-M33 Processor
 - Revision: r0p4-00rel2
 - ARMv8-M architecture profile
 - Floating Point Unit (FPU) compliant with the ANSI/IEEE Std 754-2008 Single-precision floating-point operation.
Interrupt notification when FPU exception occurs.
 - Armv8-M Security Extension (Can be disabled by OTP. CPU1 can only be selected as primary CPU if its Security Extension is enabled.)
 - Armv8-DSP Extension.
- Security Attribution Unit (SAU): 8 regions
- Implementation Defined Attribution Unit (IDAU)
- Memory Protection Unit (MPU)
 - Protected Memory System Architecture (PMSAv8)
 - Secure MPU (MPU_S): 8 regions
 - Non-secure MPU (MPU_NS): 8 regions
- System Timer (SysTick)
Reference clock:
 - CPUCLK1
 - 1 MHz generated from on-chip oscillator (MOCO/8)

Two system timers are implemented:

 - Secure instance (SysTick_S)
 - Non-secure instance (SysTick_NS)
- Low power modes
 - Sleep mode
 - Deep Sleep mode
- Cache
 - Code-bus Cache (C-Cache): 16 KB with ECC
 - System-bus Cache (S-Cache): 16 KB with ECC
- Tightly Coupled Memory (TCM)
 - CTCM: 64 KB with ECC
 - STCM: 64 KB with ECC
- Nested Vectored Interrupt Controller (NVIC): 96 IRQs
- Global exclusive monitor is not supported

See the following references for the detail:

- [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B)
- [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230).

2.1.2 Debug

(1) CPU0 Debug Components

- Arm® CoreSight™ ETM-M85
 - Revision: r1p1-00rel0
 - ARM ETM Architecture version 4.5
- Instrumentation Trace Macrocell (ITM)
- Data Watchpoint and Trace unit (DWT)
 - 8 Data Watchpoint and Trace (DWT) comparators, up to 2 data value comparison
- Breakpoint Unit (BPU)
 - 8 instruction comparators
- Cross Trigger Interface (CTI0)
- Arm® CoreSight™ Trace Port Interface Unit for Cortex-M85 (TPIU)
 - TPIU Formatter output: Excluded
 - Serial Wire Output (SWO)

(2) CPU1 Debug Components

- Arm® CoreSight™ ETM-M33
 - Revision: r0p2-00rel0
 - ARM ETM Architecture version 4.2
- Instrumentation Trace Macrocell (ITM)
- Data Watchpoint and Trace unit (DWT)
 - 4 Data Watchpoint and Trace (DWT) comparators
- Breakpoint Unit (BPU)
 - 8 instruction comparators
- Cross Trigger Interface (CTI1)
- Arm® CoreSight™ Trace Port Interface Unit for Cortex-M33 (TPIU)
 - TPIU Formatter output: Excluded
 - Serial Wire Output (SWO)

(3) Common Debug Components

- Debug Access Port (DAP)
 - CoreSight SoC-400 Access Port :
 - #0: AHB-AP0
 - #1: APB-AP
 - #2: AHB-AP1
 - JTAG Debug Port and Serial Wire Debug Port (SWJ-DP)
Serial Wire protocol version 2 (Target instance is fixed value: 0000)
- Timestamp Generator (TSG)
 - CoreSight SoC-400 Timestamp Generator
 - Global Timestamp for ETM and ITM
 - Driven by ICLK
- Embedded Cross Trigger

- CoreSight SoC-400 Cross Trigger Interface (CTI)
 - CoreSight SoC-400 Cross Trigger Matrix (CTM)
- Embedded Trace FIFO (ETF)
 - CoreSight™ Trace Memory Controller with ETF configuration
 - FIFO size: 8 KB
- Debug Register Module (DBGREG)
- On-Chip Debug Register Module (OCDREG)
- Authentication Method using JTAG/SWD: Challenge response
- Trace Output Selector: Select between CPU0 and CPU1 SWO trace output.
- CoreSight SoC-400 Trace Port Interface Unit (TPIU) up to 4 bits TPIU Formatter output.

See the following references for the detail:

- [1] [ARM Limited., ARMv8-M Architecture Reference Manual \(ARM DDI 0553B\)](#)
- [2] [ARM Limited., Arm\(R\) Cortex-M85 Processor Technical Reference Manual \(ARM 101924\)](#)
- [4] [ARM Limited., Arm\(R\) CoreSight SoC-400 Technical Reference Manual \(ARM 0302-01 2nd release for r3p2\)](#)
- [9] [ARM Limited., Arm\(R\) Cortex-M33 Processor Technical Reference Manual \(ARM 100230\).](#)

2.2 Block Diagram

[Figure 2.1](#) shows the block diagram of CPUs and CoreSight Debug subsystem (DBGSS).

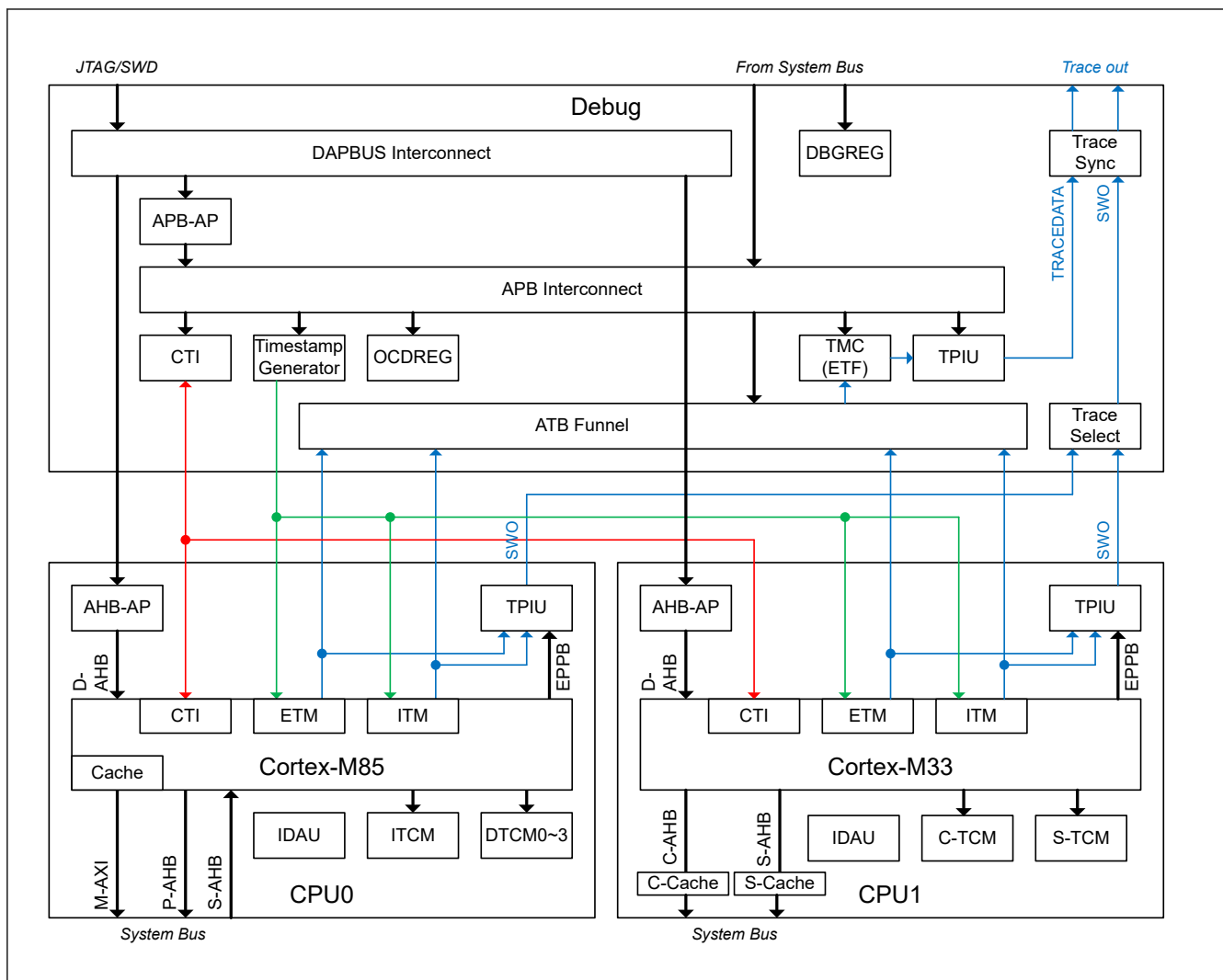


Figure 2.1 CPU and Debug block diagram

2.3 Implementation Options

Table 2.1 shows the implementation options related to CPU0 (Cortex-M85) and Table 2.2 shows the implementation options related to CPU1 (Cortex-M33).

Table 2.1 Implementation options related to CPU0 (1 of 2)

Option	Implementation
SAU	8 regions
IDAU	Included
MPU	MPU_S: 8 regions MPU_NS: 8 regions
MVE	Integer and half and single-precision floating-point MVE
FPU	Scalar half, single, and double-precision floating-point
Interrupts	96
Priority bits	4 bits (16 levels)
Sleep mode power-saving	Sleep mode, Deep Sleep mode
TCM	- ITCM : 128 KB (16 blocks × 8 KB) - DTCM : 128 KB (16 blocks × 8 KB) - ECC enable (determined by OFS1.INITECCEN) - ITCMCR.EN = 1 and DTCMCR.EN = 1, cannot be rewritten

Table 2.1 Implementation options related to CPU0 (2 of 2)

Option	Implementation
CACHE	- I-Cache: 16 KB - D-Cache: 16 KB - ECC enable (determined by OFS1.INITECCEN) - Automatic invalidation is enabled During debug, the cache is controlled by CACHEDBGCR.L1RSTDIS.
P-AHB	512 MB, always enabled (PAHBCR.EN is always 1)
SysTick	Included two system timers: SysTick_S and SysTick_NS External reference clock with frequency = 1 MHz (MOCO/8) SYST_CALIB = 0x0000270F NOREF, bit[31] = 0: Reference clock is implemented SKEW, bit[30] = 0: TENMS calibration value is exact Bits[29:24] = 0x00: Reserved TENMS, bits[23:0] = 0x00270F: $(0x270F + 1) \times 1 [\mu s] = 10 [ms]$
System reset request output	AIRCR.SYSRESETREQ bit causes the system reset
DWT	8 comparators
BPU	8 comparators
CTI	Included, CTI0 (for CPU0)
ITM	Included
ETM	Included
TPIU	Included - TPIU Formatter output: Excluded - Serial Wire Output (SWO)
Programmable MBIST Controller (PMC-100)	Excluded

Table 2.2 Implementation options related to CPU1 (1 of 2)

Option	Implementation
SAU	8 regions
IDAU	Included
MPU	MPU_S: 8 regions MPU_NS: 8 regions
SECEXT (Security Extension)	Supported. Can be enabled/disabled by setting OTP.
FPU	Single-precision floating-point
Interrupts	96
Priority bits	4 bits (16 levels)
Sleep mode power-saving	Sleep mode, Deep Sleep mode
SysTick	Included two system timers: SysTick_S and SysTick_NS External reference clock with frequency = 1 MHz (MOCO/8) SYST_CALIB = 0x0000270F NOREF, bit[31] = 0: Reference clock is implemented SKEW, bit[30] = 0: TENMS calibration value is exact Bit[29:24] = 0x00: Reserved TENMS, bits[23:0] = 0x00270F: $(0x270F + 1) \times 1 [\mu s] = 10 [ms]$
System reset request output	AIRCR.SYSRESETREQ bit causes the system reset request output
DWT	4 comparators
BPU	8 comparators
CTI	Included (CTI1 for CPU1)
ITM	Included
ETM	Included

Table 2.2 Implementation options related to CPU1 (2 of 2)

Option	Implementation
TPIU	Included - TPIU formatter output: Excluded - Serial Wire Output (SWO)

Table 2.3 DBGSS implementation options

Option	Implementation
TSG	Included Driven by clock with frequency = ICLK
TMC (ETF configuration)	8 KB
DAP	SWJ-DP AHB-AP0 (for CPU0): AP0 APB-AP (for CoreSight SoC-400): AP1 AHB-AP1 (for CPU1): AP2
TPIU	Included Up to 32-bits TPIU formatter output

2.4 Trace Interface

A Trace Port Interface Unit (TPIU) and Serial Wire Output (SWO) provide trace output. [Table 2.4](#) shows the MCU pins for the function. These pins are multiplexed with other functions.

Table 2.4 Trace function pins

Name	I/O	Function	When not in use
TCLK	Out	Trace clock	Open
TDATA0	Out	Trace data output	Open
TDATA1	Out	Trace data output	Open
TDATA2	Out	Trace data output	Open
TDATA3	Out	Trace data output	Open
SWO	Out	Serial Wire Output Mutiplexed with JTAG TDO pin	Open

2.5 JTAG/SWD Interface

[Table 2.5](#) shows JTAG/SWD chip pins.

Table 2.5 JTAG/SWD pins

Name	I/O	Function	When not in use
TCK/SWCLK	In	JTAG Clock pin Serial Wire Data Clock pin	Pull-up
TMS/SWDIO	In-Out	JTAG TMS pin Serial Wire Data In-Out pin	Pull-up
TDI	In	JTAG TDI pin	Pull-up
TDO	Out	JTAG TDO pin Mutiplexed with Serial Wire Output	Open

2.6 Boot

The MCU supports Single-chip mode, Boot modes (JTAG, SCI/USB). For details of each mode, see [section 4, Operating Modes](#).

2.6.1 Single-chip Mode

In this mode, CPUs execute user application code. However, when the First Stage Boot Loader (FSBL) execution is enabled, the primary CPU executes the FSBL code provided by Renesas before executing the user application code.

2.6.1.1 Boot CPU Selection

In the dual-core system, the primary CPU can be selected by the boot firmware command and booted first after a system reset release. By default, CPU0 is the primary CPU.

During the system reset release, the secondary CPU is in power gating state until it is activated by the primary CPU by writing to CPUACTCSR register. Then depends on CPUWAITCR.CPUWAIT, the secondary CPU may be in the wait state (CPUWAIT) or starts program execution. While in the wait state, code and data can be transferred to TCMs of the secondary CPU (when CPU0 is the secondary CPU, its activation status must be confirmed before code/data transferring or release of CPUWAIT). The primary CPU is not affected by CPUWAITCR.CPUWAIT.

2.6.2 Initial Vector Table Base Address

In the MCU, the secure vector table base address of each CPU is determined according to the operating mode and whether it is the primary or secondary CPU as shown in [Table 2.6](#), and the non-secure vector table base address is fixed at 0x0000_0000 for both CPUs except the case when CPU1 (CM33) with Security Extension is disabled, non-secure vector table base address is determined by CPUINITVTOR value.

Table 2.6 Initial secure vector table base address in Single-chip mode

Operating mode	Primary CPU INITSVTOR	Secondary CPU INITSVTOR
Single-chip mode	0x0200_0000	CPUINITVTOR

2.7 Sleep Modes for Low Power States

Each CPU supports two Sleep modes: Sleep mode and Deep Sleep mode.

2.7.1 Sleep Mode

2.7.1.1 Transition to Sleep Mode

When a WFI/WFE instruction or Sleep-on-exit function is executed while SCR.SLEEPDEEP bit is 0, the corresponding CPU enters Sleep mode. In this mode, the CPU stops operating but the contents of its internal registers are retained. Other peripheral functions do not stop. Available resets or interrupts in Sleep mode cause the CPU to cancel Sleep mode. All interrupt sources are available. If using an interrupt to cancel Sleep mode, you must set the associated IELSRn register before executing a WFI instruction. For details, see [section 14, Interrupt Controller Unit \(ICU\)](#).

[Table 2.7](#) shows the operation of IWDT when primary CPU enters Sleep mode.

Table 2.7 IWDT operation when primary CPU enters Sleep mode

IWDT operation	OFS0.IWDTSTRT	OFS0.IWDTSTPCTL* ¹	IWDTCTPR.SLCSTP* ²
Stop counting	0 (auto start mode)	1	—
Continue counting	0 (auto start mode)	0	—
Stop counting	1 (register start mode)	—	1
Continue counting	1 (register start mode)	—	0

Note 1. OFS0.IWDTSTPCTL controls IWDT stops/continues counting while IWDT is in auto start mode and the primary CPU is in Sleep mode.

Note 2. IWDTCTPR.SLCSTP controls IWDT stops/continues counting while IWDT is in register start mode and the primary CPU is in Sleep mode.

Counting by IWDT stops when the primary CPU enters Sleep mode while the IWDT is in auto start mode and the OFS0.IWDTSTPCTL bit is 1 (IWDT stops when the primary CPU is in Sleep mode).

Counting by IWDT continues when the primary CPU enters Sleep mode while the IWDT is in auto start mode and the OFS0.IWDTSTPCTL bit is 0 (IWDT does not stop when the primary CPU is in Sleep mode).

Counting by the IWDTCSTP stops when the primary CPU enters Sleep mode while the IWDTCSTP is in register start mode and the SLCSTP bit in IWDTCSTPR is 1 (IWDTCSTP stops when the primary CPU is in Sleep mode).

Counting by the IWDTCSTP continues when the primary CPU enters Sleep mode while the IWDTCSTP is in register start mode and the SLCSTP bit in IWDTCSTPR is 0 (IWDTCSTP does not stop when the primary CPU is in Sleep mode).

The operation of WDT0/1 when CPU0/1 enters Sleep mode is shown in Table 2.8 and Table 2.9.

Table 2.8 WDT0 operation when CPU0 enters Sleep mode

WDT0 operation	OFS0.WDTSTRT0	OFS0.WDTSTPCTL0*1	WDT0.WDTCSTPR.SLCSTP*2
Stop counting	0 (auto start mode)	1	—
Continue counting	0 (auto start mode)	0	—
Stop counting	1 (register start mode)	—	1
Continue counting	1 (register start mode)	—	0

Note 1. OFS0.WDTSTPCTL0 controls WDT0 stops/continues counting while WDT0 is in auto start mode and the CPU0 is in Sleep mode.

Note 2. WDT0.WDTCSTPR.SLCSTP controls WDT0 stops/continues counting while WDT0 is in register start mode and the CPU0 is in Sleep mode.

Counting by WDT0 stops when CPU0 enters Sleep mode while the WDT0 is in auto start mode and the OFS0.WDTSTPCTL0 bit is 1 (WDT0 stops when CPU0 is in Sleep mode). Similarly, counting by WDT0 stops when CPU0 enters Sleep mode while the WDT0 is in register start mode and the SLCSTP bit in WDTCSTPR is 1 (WDT0 stops when CPU0 is in Sleep mode).

Counting by WDT0 continues when the CPU0 enters Sleep mode while the WDT0 is in auto start mode and the OFS0.WDTSTPCTL0 bit is 0 (WDT0 does not stop when CPU0 is in Sleep mode). Similarly, counting by WDT0 continues when CPU0 enters Sleep mode while the WDT0 is in register start mode and the SLCSTP bit in WDTCSTPR is 0 (WDT0 does not stop when CPU0 is in Sleep mode).

Table 2.9 WDT1 operation when CPU1 enters Sleep mode

WDT1 operation	OFS3.WDTSTRT1	OFS3.WDTSTPCTL1*1	WDT1.WDTCSTPR.SLCSTP*2
Stop counting	0 (auto start mode)	1	—
Continue counting	0 (auto start mode)	0	—
Stop counting	1 (register start mode)	—	1
Continue counting	1 (register start mode)	—	0

Note 1. OFS3.WDTSTPCTL1 controls WDT1 stops/continues counting while WDT1 is in auto start mode and the CPU1 is in Sleep mode.

Note 2. WDT1.WDTCSTPR.SLCSTP controls WDT1 stops/continues counting while WDT1 is in register start mode and the CPU1 is in Sleep mode.

Counting by WDT1 stops when CPU1 enters Sleep mode while the WDT1 is in auto start mode and the OFS3.WDTSTPCTL1 bit is 1 (WDT1 stops when CPU1 is in Sleep mode). Similarly, counting by WDT1 stops when CPU1 enters Sleep mode while the WDT1 is in register start mode and the SLCSTP bit in WDTCSTPR is 1 (WDT1 stops when CPU1 is in Sleep mode).

Counting by WDT1 continues when the CPU1 enters Sleep mode while the WDT1 is in auto start mode and the OFS3.WDTSTPCTL1 bit is 0 (WDT1 does not stop when CPU1 is in Sleep mode). Similarly, counting by WDT1 continues when CPU1 enters Sleep mode while the WDT1 is in register start mode and the SLCSTP bit in WDTCSTPR is 0 (WDT1 does not stop when CPU1 is in Sleep mode).

For details on WFI/WFE instruction and sleep-on-exit function to enter Sleep mode, see [1] [ARM Limited., ARMv8-M Architecture Reference Manual \(ARM DDI 0553B\)](#).

2.7.1.2 Cancelling Sleep Mode

Sleep mode of each CPU is canceled by:

- An interrupt/event
- A reset:
 - RES pin reset
 - Power-on reset

- Voltage monitor 0 reset
- Overcurrent protection reset
- Core voltage monitor reset
- Independent watchdog timer reset
- CPU_n (n = 0, 1) watchdog timer reset
- Voltage Monitor m reset (m = 1, 2 and 4, 5)
- Software reset
- CPU_n (n = 0, 1) Lockup reset
- Common memory error reset
- Local memory n (n = 0, 1) error reset
- Bus error reset

The operations are as follows:

Canceling by an interrupt/event

- When an available interrupt request is generated, Sleep mode is canceled, and the CPU_n (n = 0, 1) starts the interrupt handling.
- When the debugger writes 1 to MCUCTRL.EDBGRQ_n (n = 0, 1 for CPU0 and CPU1), Sleep mode is canceled, and the CPU_n (n = 0, 1) enters Debug state*1.
- When the CPU_n (n = 0, 1) receives an event issued by an SEV instruction executed by another CPU and the transition to Sleep mode is caused by WFE instruction, Sleep mode is canceled, and the CPU_n (n = 0, 1) executes the next instruction after WFE.

Canceling by RES pin reset

When the RES pin is driven low, the MCU enters the reset state. Be sure to keep the RES pin low for a period specified in [section 70, Electrical Characteristics](#). When the RES pin is driven high after the specified period, the primary CPU starts the reset exception handling.

Canceling by IWDT reset

Sleep mode is canceled by a system reset generated by an IWDT underflow and the primary CPU starts the reset exception handling. However, IWDT stops in Sleep mode and a system reset for canceling Sleep mode is not generated in the following conditions:

- OFS0.IWDTSTRT = 0 (auto start mode) and OFS0.IWDTSTPCTL = 1
- OFS0.IWDTSTRT = 1 (register start mode) and IWDTCSTPR.SLCSTP = 1

Canceling by WDT0 reset

Sleep mode is canceled by a system reset generated by a WDT0 underflow and the primary CPU starts the reset exception handling. However, WDT0 stops in Sleep mode even when counting in Normal mode and a system reset for canceling Sleep mode is not generated in the following conditions:

- OFS0.WDTSTRT0 = 0 (auto start mode) and OFS0.WDTSTPCTL0 = 1
- OFS0.WDTSTRT0 = 1 (register start mode) and WDT0.WDTCSTPR.SLCSTP = 1

Canceling by WDT1 reset

Sleep mode is canceled by a system reset generated by a WDT1 underflow and the primary CPU starts the reset exception handling. However, WDT1 stops in Sleep mode even when counting in Normal mode and a system reset for canceling Sleep mode is not generated in the following conditions:

- OFS3.WDTSTRT1 = 0 (auto start mode) and OFS3.WDTSTPCTL1 = 1
- OFS3.WDTSTRT1 = 1 (register start mode) and WDT1.WDTCSTPR.SLCSTP = 1

Canceling by other resets available in Sleep mode

Sleep mode is canceled by other resets and the Primary CPU starts the reset exception handling.

Note: For details on proper setting of the interrupts, see [section 14, Interrupt Controller Unit \(ICU\)](#).

Note 1. CPU cannot enter Debug state if it is in Secure state while debugger is in Non-Secure state.

2.7.2 Deep Sleep Mode

2.7.2.1 Transition to Deep Sleep Mode

When a WFI instruction or sleep-on-exit function is executed while SCR.SLEEPDEEP is set to 1, the corresponding CPU enters Deep Sleep mode. In this mode, the CPU stops operating but the contents of its internal registers are retained. Other peripheral functions do not stop. However, accessing the TCM of CPU0 (when CPU0 is in Deep Sleep mode) is not available and SysTick also stops in this mode. Available resets or all interrupts in Deep Sleep mode cause the MCU to cancel Deep Sleep mode.

In Deep Sleep mode, the CPU is in power gating state (for CPU0, it depends on CPDLPSTATE setting).

IWDT and WDT0/1 operations in Deep Sleep mode is the same in Sleep mode.

2.7.2.2 Cancelling Deep Sleep Mode

Deep Sleep mode is canceled by:

- Interrupts shown in [Table 14.5](#)
- A reset:
 - RES pin reset
 - Power-on reset
 - Voltage monitor 0 reset
 - Overcurrent protection reset
 - Core voltage monitor reset
 - Independent watchdog timer reset
 - CPU_n (n = 0, 1) watchdog timer reset
 - Voltage monitor m reset (m = 1, 2 and 4, 5)
 - Software reset
 - CPU_n (n = 0, 1) lockup reset
 - Common memory error reset
 - Local memory n (n = 0, 1) error reset
 - Bus error reset

The operations are as follows:

Canceling by an interrupt

When an interrupt request is generated, Deep Sleep mode is canceled and the CPU_n (n = 0, 1) starts the interrupt handling.

Canceling by RES pin reset

When the RES pin is driven low, the MCU enters the reset state. Be sure to keep the RES pin low for a period specified in [section 70, Electrical Characteristics](#). When the RES pin is driven high after the specified period, the CPUs start the reset exception handling.

Canceling by IWDT reset

Deep Sleep mode is canceled by a system reset generated by an IWDT underflow and the primary CPU starts the reset exception handling. However, IWDT stops in Deep Sleep mode and a system reset for canceling Deep Sleep mode is not generated in the following conditions:

- OFS0.IWDTSTRT = 0 (auto start mode) and OFS0.IWDTSTPCTL = 1
- OFS0.IWDTSTRT = 1 (register start mode) and IWDCSTPR.SLCSTP = 1

Canceling by WDT0 reset

Deep Sleep mode is canceled by a system reset generated by a WDT0 underflow and the primary CPU starts the reset exception handling. However, WDT0 stops in Deep Sleep mode even when counting in Normal mode and a system reset for canceling Deep Sleep mode is not generated in the following conditions:

- OFS0.WDTSTRT0 = 0 (auto start mode) and OFS0.WDTSTPCTL0 = 1
- OFS0.WDTSTRT0 = 1 (register start mode) and WDT0.WDCSTPR.SLCSTP = 1

Canceling by WDT1 reset

Deep Sleep mode is canceled by a system reset generated by a WDT1 underflow and the primary CPU starts the reset exception handling. However, WDT1 stops in Deep Sleep mode even when counting in Normal mode and a system reset for canceling Deep Sleep mode is not generated in the following conditions:

- OFS3.WDTSTRT1 = 0 (auto start mode) and OFS3.WDTSTPCTL1 = 1
- OFS3.WDTSTRT1 = 1 (register start mode) and WDT1.WDCSTPR.SLCSTP = 1

Canceling by other resets available in Deep Sleep mode

Deep Sleep mode is canceled by other resets and the primary CPU starts the reset exception handling.

Note: For details on proper setting of the interrupts, see [section 14, Interrupt Controller Unit \(ICU\)](#).

2.8 Debug Function

The MCU provides ability to debug software by either external debug or self-hosted debug.

The external debug function is enabled by either JTAG/SWD or software programming. See [section 2.8.1. External Debugger Connectivity, Authentication by JTAG/SWD](#) and [section 2.8.2. Debug Protection Mechanism by Software](#) for details.

Self-hosted debug function can be enabled by software programming. See [section 2.8.2. Debug Protection Mechanism by Software](#) for details.

When debug function is enabled, operation of system can be affected. See [section 2.8.3. Effect of Debug Function](#) for details.

2.8.1 External Debugger Connectivity, Authentication by JTAG/SWD

In this MCU, three Protection Levels (PL) can be set:

- PL0 — Debug is not permitted
- PL1 — Non-secure debug is permitted
- PL2 — Secure debug is permitted.

When debug function is enabled by performing authentication using JTAG/SWD, three debug levels corresponding to three Authentication Level (AL) are available:

- AL0 — No debug functions are available
- AL1 — Only non-secure debug functions (both invasive and non-invasive debug) are enabled, and debugger can access only defined non-secure debug accessible regions
- AL2 — Non-secure and secure debug functions (both invasive and non-invasive debug) are enabled and accessible from the debugger.

Note: See reference [1] [ARM Limited., ARMv8-M Architecture Reference Manual \(ARM DDI 0553B\)](#) for details about invasive, non-invasive, secure, and non-secure debug.

Authentication Level is determined as follows:

- When Device Life Cycle state (DLM state) is not OEM (LCK_BOOT, RMA_RET, RMA_REQ, RMA_ACK), AL is determined by DLM state. See [Table 2.10](#).
- When Device Life Cycle state (DLM state) is equal to OEM, AL can be determined by PL set by user and authentication performed using JTAG/SWD. Authentication method of the MCU is the challenge response method with the authentication key installed by user.

The initial value of Authentication Level (AL) is determined by the value of the Protection Level (PL). See [Table 2.10](#).

For the authentication sequence, see [section 2.14.2.2. Connecting Sequence and JTAG/SWD Authentication](#).

In addition, using authentication key AL1 and AL2 can be prohibited by setting Protection LCKS and LCKNS, respectively. Setting LCKS and LCKNS can be performed by a boot firmware command.

[Table 2.10](#) shows condition for authentication level that can be used for serial programmer.

Table 2.10 Condition for authentication level

Condition				Available authentication level
DLM state	PL	LCKS	LCKNS	
OEM	PL2	Don't care	Don't care	AL2 (default)
OEM	PL1	111	Don't care	AL1 (default) AL2 (authentication using AL2_KEY)
OEM	PL1	000 (using AL2_KEY prohibited)	Don't care	AL1 (default)
OEM	PL0	111	111	AL0 (default) AL1 (authentication using AL1_KEY) AL2 (authentication using AL2_KEY)
OEM	PL0	111	000 (using AL1_KEY prohibited)	AL0 (default) AL2 (authentication using AL2_KEY)
OEM	PL0	000 (using AL2_KEY prohibited)	111	AL0 (Default) AL1 (authentication using AL1_KEY)
OEM	PL0	000 (using AL2_KEY prohibited)	000 (using AL1_KEY prohibited)	AL0 (default)
LCK_BOOT	PL0	Don't care	Don't care	AL0 (default)
RMA_RET	PL0	Don't care	Don't care	AL0 (default)
RMA_REQ	PL0	Don't care	Don't care	AL0 (default)
RMA_ACK	PL2	Don't care	Don't care	AL2 (default)

Note: When debugging using SWJ-DP, set CDBGPWUPREQ = 1.

Note: Debug connection from external emulator to CoreSight components using Debug APB is enabled as default setting, and as don't care for authentication mechanism result. However, this connection can be disabled by software programming. See DBGAUTH0.DEVICEEN for details.

Debug capability is not available while MCU is operating with the Boot mode or during the FSBL execution.

2.8.2 Debug Protection Mechanism by Software

In this mechanism, the challenge and response authentication procedure is not performed, instead software can be programmed to enable debug function of the MCU directly. In other words, debug function is controlled by software completely.

The debug function can be enabled by setting OFS1.SWDBG = 0 and one of the following bits:

- DBGAUTH0.NIDEN_n = 1: Non-invasive debug enable (n = 0,1 for CPU0 and CPU1)
- DBGAUTH0.DBGEN_n = 1: Invasive debug enable (n = 0,1 for CPU0 and CPU1)
- DAUTHCTRL.SPENIDENSEL = 1 and DAUTHCTRL.INTSPENIDEN = 1: Secure non-invasive debug enable. See [\[1\] ARM Limited., ARMv8-M Architecture Reference Manual \(ARM DDI 0553B\)](#) for details.
- DAUTHCTRL.SPIDENSEL = 1 and DAUTHCTRL.INTSPIDEN = 1: Secure invasive debug enable. See [\[1\] ARM Limited., ARMv8-M Architecture Reference Manual \(ARM DDI 0553B\)](#) for details.

Note: Authentication using software programming is not restricted by DLM state, PL, LCKS and LCKNS.

Note: Except DAUTHCTRL.INTSPNIDEN and DAUTHCTRL.INTSPIDEN overrides processor external secure invasive/non-invasive debug authentication interface, when debug function enabled by authentication using JTAG/SWD, it cannot be disabled by software.

2.8.3 Effect of Debug Function

The debug function affects inside and outside of CPU. This section describes the effects excluding debug authentication.

2.8.3.1 Low Power Mode

All CoreSight debug components can store the register settings even when the CPU enters Deep Sleep mode, or when the MCU enters Software Standby or Deep Software Standby mode if SYOCDCCR.DBGEN is 1 (see [section 11, Low Power Mode](#)) and any of the following condition is true:

- CDBGPWRUPREQ = 1 and AL is AL2 or AL1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN0 or DBGAUTH0.NIDEN0 is 1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN1 or DBGAUTH0.NIDEN1 is 1

However, AHB-APn (n = 0, 1 for AHB-AP0, AHB-AP1 of each CPU0, 1) cannot respond to On-Chip Debug (OCD) access in these low power modes. The OCD must wait for cancellation of the low power mode to access the CoreSight debug components.

The MCU does not respond to On-Chip Debug (OCD), when OCD starts connecting to the MCU while it is in Software Standby mode or Deep Software Standby mode.

2.8.3.2 Reset/Interrupt

During debug when CPU enters debug state, the IWDT and corresponding WDTn (n = 0, 1) is stopped^{*1}. During debug when CPU does not enter debug state and debug function is not AL0 or self-hosted debug is enabled, then some resets are restrained depend on DBGSTOPPCR setting as WDT0, WDT1, IWDT^{*2}. When PVDn (n! = 0), interrupt and reset depends on DBGSTOPPCR whether CPU is in halting mode or not. Some access error is not generated by external debugger (DAP) access, see [Table 2.11](#).

Table 2.11 Controlling of resets and interrupts generation in OCD mode

Reset/Interrupt name	Control in On-Chip Debug (OCD) mode	
	OCD break mode	OCD run mode
RES pin reset	Same as user mode	
Power-on reset	Same as user mode	
Independent watchdog timer reset/interrupt ^{*2}	Does not occur ^{*1}	Depends on DBGSTOPPCR setting ^{*2}
Watchdog timer 0/1 reset/interrupt	Does not occur ^{*1}	Depends on DBGSTOPPCR setting ^{*2}
CPU Lockup reset/interrupt	Does not occur	Same as user mode
Voltage monitor 0 reset	Same as user mode	
Voltage monitor 1 reset/interrupt	Depends on DBGSTOPPCR setting ^{*3}	
Voltage monitor 2 reset/interrupt	Depends on DBGSTOPPCR setting ^{*3}	
CPU0 Cache/TCM ECC error reset	Depends on DBGSTOPPCR setting	
CPU1 Cache/TCM ECC error reset/interrupt	Cache: Depends on CAPOAD.OAD TCM: Depends on TCMCRC.OAD and TCMCRS.OAD	
MRAM ECC error reset/interrupt	Depends on DBGSTOPPCR setting	
Cache/TCM ECC error BusFaults	Same as user mode ^{*4}	
SRAM ECC error reset/interrupt	Same as user mode ^{*4}	
Deep Software Standby reset	Same as user mode	
Software reset	Same as user mode	
Bus error reset/interrupt	Same as user mode ^{*4}	

Note 1. In OCD Break mode

IWDT always stops if any of the following condition is true:

- CDBGWRUPREQ is 1, and AL is AL2 or AL1
- OSF1.SWDBG is 0, DBGAUTH0.DBGEN0 is 1, and CPU0 is the primary CPU
- OSF1.SWDBG is 0, DBGAUTH0.DBGEN1 is 1, and CPU1 is the primary CPU.

WDT0 always stops if any of the following condition is true:

- CDBGWRUPREQ is 1, and AL is AL2 or AL1
- OSF1.SWDBG is 0 and DBGAUTH0.DBGEN0.

WDT1 always stops if any of the following condition is true:

- CDBGWRUPREQ is 1, and AL is AL2 or AL1
- OSF1.SWDBG is 0 and DBGAUTH0.DBGEN1.

Note 2. In OCD run mode, IWDT and WDTn (n = 0, 1) operation depends on DBGSTOPPCR setting if any of the following condition is true:
IWDT:

- CDBGWRUPREQ is 1, and AL is AL2 or AL1
- OSF1.SWDBG is 0, DBGAUTH0.DBGEN0 is 1, and CPU0 is the primary CPU
- OSF1.SWDBG is 0, DBGAUTH0.DBGEN1 is 1, and CPU1 is the primary CPU.

WDT0:

- CDBGWRUPREQ is 1, and AL is AL2 or AL1
- OSF1.SWDBG is 0 and DBGAUTH0.DBGEN0.

WDT1:

- CDBGWRUPREQ is 1, and AL is AL2 or AL1
- OSF1.SWDBG is 0 and DBGAUTH0.DBGEN1.

Note 3. Reset or interrupt masking depends on DBGSTOPPCR setting.

Note 4. The system reset request or interrupt request is not generated when an error is detected for DAP access (applies to the Cache and TCM of CPU1).

2.8.3.3 First Stage Bootloader (FSBL)

During FSBL execution, debug is prohibited regardless of Authentication Level (AL).

2.8.4 Hot Plug-in Debug

Hot plug-in debugging is supported if challenge-response authentication is not required. Also, it is not supported in Software Standby and Deep Software Standby modes.

2.8.5 Unprivileged Debug

Unprivileged debug is supported only in CPU0 (Armv8.1-M).

2.9 Register Descriptions

2.9.1 CPU Control Registers

In addition to the internal registers of the CPU, the MCU also provides additional specific control and status registers that are considered as CPU control register.

Table 2.12 shows a list of CPU control registers.

Table 2.12 List of CPU control registers (1 of 2)

Name	Abbreviation	Address (Secure)	Address (Non-secure)	Access size	R/W
CPU Security Attribution Register	CPUSAR	0x4000_8170	0x5000_8170	32-bit	Read/Write
CPU0 Lockup Control Register	CPU0LCKUPCR	0x4000_F030	0x5000_F030	8-bit	Read/Write
CPU1 Lockup Control Register	CPU1LCKUPCR	0x4000_F034	0x5000_F034	8-bit	Read/Write
CPU0 Control Register Protection Register	CPU0CRPT	0x4000_F840	0x5000_F840	16-bit	Read/Write

Table 2.12 List of CPU control registers (2 of 2)

Name	Abbreviation	Address (Secure)	Address (Non-secure)	Access size	R/W
CPU1 Control Register Protection Register	CPU1CRPT	0x4000_F844	0x5000_F844	16-bit	Read/Write
CPU0 Function Lock Control Register	CPU0LOCKCR	0x4000_F400	—	8-bit	Read/Write
CPU1 Function Lock Control Register	CPU1LOCKCR	0x4000_F404	—	8-bit	Read/Write
CPU0 Non-secure Function Lock Control Register	CPU0LOCKCRNS	—	0x5000_F500	8-bit	Read/Write
CPU1 Non-secure Function Lock Control Register	CPU1LOCKCRNS	—	0x5000_F504	8-bit	Read/Write
CPU0 Initial Vector Base Address Register	CPU0INITVTOR	0x4000_F040	—	32-bit	Read/Write
CPU1 Initial Vector Base Address Register	CPU1INITVTOR	0x4000_F044	0x5000_F044	32-bit	Read/Write
CPU0 CPUWAIT Control Register	CPU0WAITCR	0x4000_F050	—	8-bit	Read/Write
CPU1 CPUWAIT Control Register	CPU1WAITCR	0x4000_F054	0x5000_F054	8-bit	Read/Write
CPU0 Activation Control and Status Register	CPU0ACTCSR	0x4000_F060	—	16-bit	Read/Write
CPU1 Activation Control and Status Register	CPU1ACTCSR	0x4000_F064	0x5000_F064	16-bit	Read/Write
CPU0 Local Memory Error Control Register	CPU0LMECR	0x4000_F070	0x5000_F070	8-bit	Read/Write
CPU Identification Register	CPUIDR	0x4000_F078	0x5000_F078	8-bit	Read
CPU0 Status Monitor Register	CPU0STATM	0x4000_F080	0x5000_F080	8-bit	Read
CPU1 Status Monitor Register	CPU1STATM	0x4000_F084	0x5000_F084	8-bit	Read
CPU SECEXT Monitor Register	SECEXTMON	0x4000_F090	—	8-bit	Read
Non-secure CPU Control Register	NSCPUCR	0x4000_F094	—	32-bit	Read/Write

2.9.1.1 CPUSAR : CPU Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x170

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CPUS A1	CPUS A0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPUSA0	CPU Security Attribution 0 (CPU0) 0: Secure 1: Non-secure	R/W
1	CPUSA1	CPU Security Attribution 1 (CPU1) 0: Secure 1: Non-secure	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1
P-TYPE-1

CPUSAn bit

Security attributes of registers for CPU control. The target registers are as follow.

- CPU.CPU_nLCKUPCR

- CPU.CPU_nCRPT

CPUSA0 bit

The CPUSA0 bit controls the security attributes of the following registers.

- CPU0RSTCR
- CPU0LMECR

CPUSA1 bit

The CPUSA1 bit controls the security attributes of the following registers.

- CPU1INITVTOR
- CPU1WAITCR
- CPU1ACTCSR

2.9.1.2 CPU_nLCKUPCR : CPU_n Lockup Control Register (n = 0, 1)

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x030 + 0x004 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	OAD

Value after reset: 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	OAD	Operation after detection of CPU _n lockup 0: Non-maskable interrupt 1: System reset	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3 (CPUSAR.CPUSAn)
P-TYPE-2
n = 0, 1

When CPU_n enters the lockup state, it can be selected to generate either a system reset or an interrupt by writing to CPU_nLCKUPCR.OAD.

The CPU_nLCKUPCR security attribution is controlled by CPUSAR.

Writing to CPU_nLCKUPCR.OAD bit is protected by CPU Control Register Protection Register (CPU_nCRPT).

2.9.1.3 CPU_nCRPT : CPU_n Control Register Protection Register (n = 0, 1)

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x840 + 0x004 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								—	—	—	—	—	—	—	PROTECT

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	PROTECT	Protection of register 0: Writing to CPU _n LCKUPCR register is available. 1: Writing to CPU _n LCKUPCR register is not available. Read access is available.	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
15:8	KEY[7:0]	The KEY[7:0] bits enable or disable writing to the PROTECT bit. When writing to the PROTECT bit, write 0xA5 in KEY[7:0] bits at the same time. When values other than 0xA5 are written to KEY[7:0] bits, the PROTECT bit is not updated. The KEY[7:0] bits are always read as 0x00.	W

Note: S-TYPE-3 (CPUSAR.CPUSAn)
P-TYPE-2
n = 0, 1

The CPU_nCRPT.PROTECT bit enables writing to CPU_nLCKUPCR, CPU_nWAITCR, and CPU_nLMECR registers (CPU1LMECR is not available).

The security attribution of CPU_nCRPT is controlled by CPUSAR.

2.9.1.4 CPU0LOCKCR : CPU0 Function Lock Control Register

Base address: CPU_CTRL = 0x4000_F000

Offset address: 0x400

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	LCKD CAIC	LCKD TGU	LCKIT GU	LCKS AU	LCKS MPU	LCKS VTAIR
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LCKSVTAIR	Disables writes to the following secure registers from software or from a debug agent that is connected to the processor. VTOR_S, AIRCR.PRIS, AIRCR.BFHFNMINS	R/W ^{*1}
1	LCKSMPU	Disables writes to registers that are associated with the Secure MPU region from software or from a debug agent connected to the processor. MPU_CTRL, MPU_RNR, MPU_RBAR, MPU_RLAR, MPU_RBAR_An, MPU_RLAR_An	R/W ^{*1}
2	LCKSAU	This signal disables writes to registers that are associated with the SAU region from software or from a debug agent connected to the processor. SAU_CTRL, SAU_RNR, SAU_RBAR, SAU_RLAR	R/W ^{*1}
3	LCKITGU	Disables writes to registers that are associated with the ITCM interface security gating from software or from a debug agent connected to the processor. ITGUCTRL, ITGU_LUTn	R/W ^{*1}
4	LCKDTGU	Disables writes to registers that are associated with the DTCM interface security gating from software or from a debug agent connected to the processor. DTGUCTRL, DTGU_LUTn	R/W ^{*1}
5	LCKDCAIC	Disables access to the instruction cache direct cache access registers DCAICLR and DCAICRR	R/W ^{*1}
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6
P-TYPE-2

Note 1. Writing 0 is ignored.

The CPU0LOCKCR register controls some miscellaneous signals of CPU0.

Reset is the only clear condition.

2.9.1.5 CPU1LOCKCR : CPU1 Function Lock Control Register

Base address: CPU_CTRL = 0x4000_F000

Offset address: 0x404

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	LCKS AU	LCKS MPU	LCKS VTAIR
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LCKSVTAIR	Disables writes to the following secure registers from software or from a debug agent that is connected to the processor. VTOR_S, AIRCR.PRIS, AIRCR.BFHFNMINS	R/W ^{*1}
1	LCKSMPU	Disables writes to registers that are associated with the Secure MPU region from software or from a debug agent connected to the processor. MPU_CTRL, MPU_RNR, MPU_RBAR, MPU_RLAR, MPU_RBAR_An, MPU_RLAR_An	R/W ^{*1}
2	LCKSAU	This signal disables writes to registers that are associated with the SAU region from software or from a debug agent connected to the processor. SAU_CTRL, SAU_RNR, SAU_RBAR, SAU_RLAR	R/W ^{*1}
7:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6

P-TYPE-2

Note 1. Writing 0 is ignored.

The CPU1LOCKCR register controls some miscellaneous signals of CPU1.

Reset is the only clear condition.

2.9.1.6 CPU_nLOCKCRNS : CPU_n Non-secure Function Lock Control Register (n = 0, 1)

Base address: CPU CTRL NS = 0x5000 F000

Offset address: $0x500 + 0x004 \times n$

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	LCKN SMPU	LCKN SVTO R
------------	---	---	---	---	---	---	--------------	-------------------

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	LCKNSVTOR	Disables writes to the VTOR_NS register.	R/W*1
1	LCKNSMPU	Disables writes to registers that are associated with the Non-secure MPU region from software or from a debug agent connected to the processor. MPU_CTRL_NS, MPU_RNR_NS, MPU_RBAR_NS, MPU_RLAR_NS, MPU_RBAR_A_NS[n], MPU_RLAR_A_NS[n]	R/W*1
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-7

P-TYPE-2

Note 1. Writing 0 is ignored.

The CPU_nLOCKCRNS register controls some miscellaneous signals of CPU_n.

Reset is the only clear condition.

2.9.1.7 CPUnINITVTOR : CPU Initial Vector Base Address Register (n = 0, 1)

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: $0x040 + 0x004 \times n$

Bit position: 31 0

Bit field:	CPUINITVTOR[31:0]
------------	-------------------

Value after reset: 0 0 0 0 0 0 1 0

Bit	Symbol	Function	R/W
31:0	CPUINITVTOR[31:0]	CPU _n Initial Vector Base Address Writing to CPUINITVTOR[6:0] is discarded and read as 0.	R/W

Note: S-TYPE-6 (CPU0INITVTOR) / S-TYPE-3 (CPU1INITVTOR)
P-TYPE-2 (CPU_nINITVTOR)

The CPU_nINITVTOR register controls Vector Base Address bit field in the Secure/Non-secure Vector Table Offset.

2.9.1.8 CPU_nWAITCR : CPU_n CPUWAIT Control Register (n = 0, 1)

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x050 + 0x004 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CPUW AIT
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPUWAIT	Writing 1 to stall the CPU _n when it is out of reset. This bit is valid only when CPU _n is the secondary CPU. 0: CPU _n starts instruction execution out of reset 1: CPU _n is forced into a quiescent state out of reset	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6 (CPU0WAITCR) / S-TYPE-3 (CPU1WAITCR)
P-TYPE-2 (CPU_nWAITCR)

The CPU_nWAITCR register controls the execution state of the secondary CPU when it is out of reset.

Writing to CPU_nWAITCR.CPUWAIT bit is protected by CPU Control Register Protection Register (CPU_nCRPT).

2.9.1.9 CPU_nACTCSR : CPU_n Activation Control and Status Register (n = 0, 1)

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x060 + 0x004 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KEY[7:0]								ACT	—	—	—	—	—	—	ACTR EQ
Value after reset:	0	0	0	0	0	0	0	0	X	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ACTREQ	CPU _n activation request This bit is read as 0. This bit is valid only when CPU _n is the secondary CPU. 0: No effect 1: Request CPU _n activation if ACT bit is 0	R/W
6:1	—	These bits are read as 0. The write value should be 0.	R/W
7	ACT	CPU _n activation status 0: CPU _n is in inactive state (power gating/reset state) 1: CPU _n is in active state (out of reset)	R
15:8	KEY[7:0]	Key Code These bits control the write access to this register. To modify this register, write 0xA5 to the upper 8 bits and the target value to the lower 8 bits as a 16-bit unit. These bits are read as 0.	R/W

Note: The initial value of ACT bit is:
1: If CPU_n is the primary CPU.
0: If CPU_n is the secondary CPU.

Note: S-TYPE-6 (CPU0ACTCSR) / S-TYPE-3 (CPU1ACTCSR)
P-TYPE-2 (CPUnACTCSR)

The CPUUnACTCSR register controls the activation of the secondary CPU after a system reset. It also indicates the activation status of CPUn.

2.9.1.10 CPU0LMECR : CPU0 Local Memory Error Control Register

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x070

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SYRS TEN
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SYRSTEN	System Reset Request Enable 0: Disables system reset request when multiple bits error occurs in data cache or TCM 1: Enables system reset request when multiple bits error occurs in data cache or TCM	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3
P-TYPE-2

The CPU0LMECR register controls a system reset request when multiple bits error occurs in data cache or in TCM of CPU0.

Writing to CPU0LMECR.SYRSTEN bit is protected by CPU Control Register Protection Register (CPU0CRPT).

TCM must be initialized before setting CPU0LMECR.SYRSTEN bit because ECC error may occur if CPU0 and CPU1 issue a speculative access to the uninitialized TCM.

2.9.1.11 CPUIDR : CPU Identification Register

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x078

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	CPUID
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CPUID	CPU Identification 0: CPU0 1: CPU1	R
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-5

The CPUIDR register shows the ID of CPUn. This register is expected to be read by CPUn. The read value is undefined if read by a bus master other than CPUn.

2.9.1.12 CPU0STATM : CPU0 Status Monitor Register

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x080

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	SAHB STP	—	—	SLEE PDEE P	SLEE PING
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SLEEPING	CPU0 Sleeping State 0: CPU0 is running 1: CPU0 is at Sleep mode	R
1	SLEEPDEEP	CPU0 SLEEPDEEP state Indicates that the processor is at a Deep Sleep mode.	R
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	SAHBSTP	CPU0 S-AHB Status Flag 0: CPU0 S-AHB is active 1: CPU0 S-AHB is stopped	R
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-5

The CPU0STATM register shows status of CPU0 related to Sleep, SleepDeep, and S-AHB Stop.

2.9.1.13 CPU1STATM : CPU1 Status Monitor Register

Base address: CPU_CTRL = 0x4000_F000
CPU_CTRL_NS = 0x5000_F000

Offset address: 0x084

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SLEE PDEE P	SLEE PING
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SLEEPING	CPU1 Sleeping State 0: CPU1 is running 1: CPU1 is at Sleep mode	R
1	SLEEPDEEP	CPU1 SLEEPDEEP state Indicates that the processor is at a Deep Sleep mode.	R
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-5

The CPU1STATM register shows status of CPU1 related to Sleep and SleepDeep.

2.9.1.14 SECEXTMON : CPU SECEXT Monitor Register

Base address: CPU_CTRL = 0x4000_F000

Offset address: 0x090

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	SECEXT1	SECEXT0
Value after reset:	0	0	0	0	0	0	0/1	1

Bit	Symbol	Function	R/W
0	SECEXT0	CPU0 Security Extension CPU0 (Cortex-M85) always includes Security Extension. 0: CPU0 does not include Security Extension 1: CPU0 includes Security Extension	R
1	SECEXT1	CPU1 Security Extension 0: CPU1 does not include Security Extension 1: CPU1 includes Security Extension	R
7:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6
P-TYPE-2

The SECEXTMON register shows status of CPU0/1 related to Security Extension.

2.9.1.15 NSCPUCR : Non-secure CPU Control Register

Base address: CPU_CTRL = 0x4000_F000

Offset address: 0x094

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RSTR EQEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	RSTREQEN	System Reset Request Enable 0: System reset request caused by non-secure CPU without SECEXT is not allowed 1: System reset request caused by non-secure CPU without SECEXT is allowed	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6
P-TYPE-2

The NSCPUCR register manages the system reset request of the CPU which does not have Security Extension.

2.9.2 Address Spaces for External Debugger

The MCU debug system includes three CoreSight Access Ports (AP):

- AHB-AP_n, which is connected to the CPU_n bus matrix, has the same accessibility to the system address space as CPU_n (n = 0, 1)
- APB-AP, which has OCD address space, is connected to the CoreSight components and OCDREG registers.

Figure 2.2 shows a block diagram of the AP connection and address spaces.

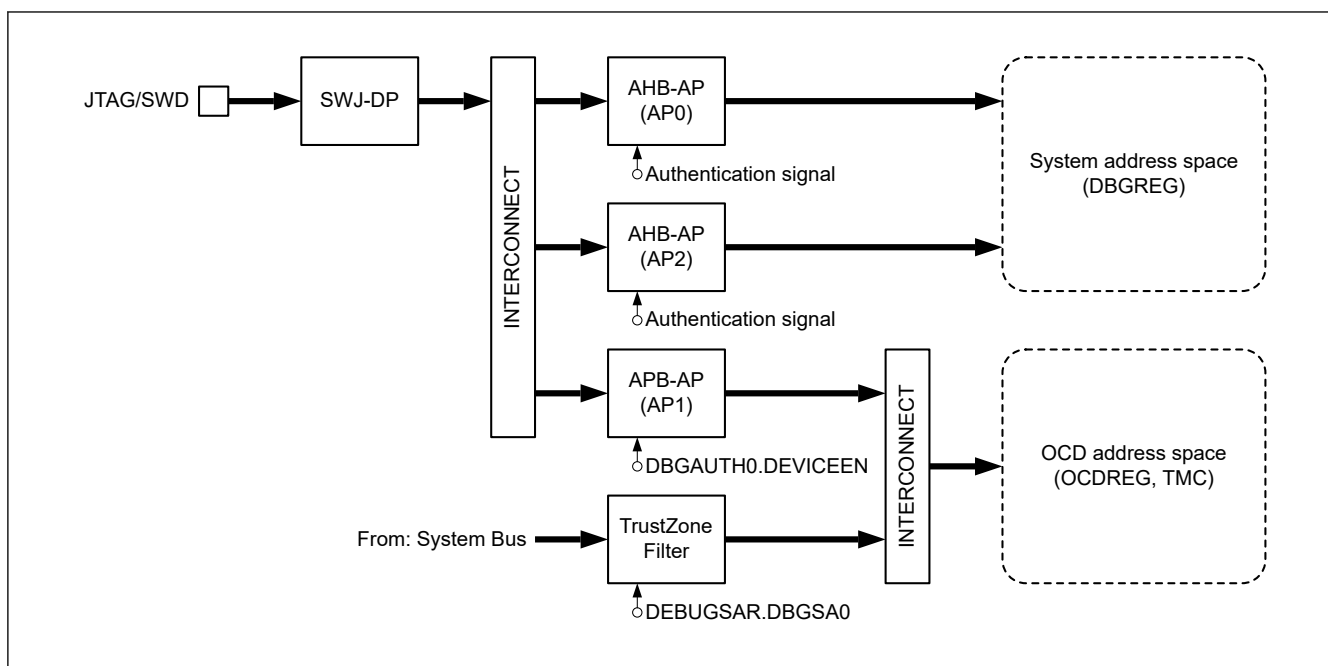


Figure 2.2 Block diagram of the AP connection and address spaces

For debugging purpose, there are two address spaces, DBGREG and OCDREG.

DBGREG is located in the system address space and can be accessed from either OCD emulator, CPUs or other bus masters in the MCU.

OCDREG is located in the OCD address space and can be accessed from either OCD emulator, CPUs or other bus masters in the MCU. See [Table 2.15](#) for details.

2.9.3 CPU Peripheral

Each CPU core has its Private Peripheral Bus (PPB) which can be accessed only from that CPU and OCD emulator. CoreSight components are accessible from either CPUs or OCD emulator. [Table 2.13](#) and [Table 2.14](#) show dedicated Peripheral address of CPU_n (n = 0, 1). [Table 2.15](#) describes CoreSight component address.

Table 2.13 Peripheral address of CPU0 (1 of 2)

Component name	Start address	End address	Note
ITM	0xE000_0000	0xE000_0FFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
DWT	0xE000_1000	0xE000_1FFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
BPU	0xE000_2000	0xE000_2FFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
PMU	0xE000_3000	0xE000_3FFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
SCS	0xE000_E000	0xE000_EFFF	See reference [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B)
SCS Non-Secure Alias	0xE002_E000	0xE002_EFFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
TPIU	0xE004_0000	0xE004_0FFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
ETM	0xE004_1000	0xE004_1FFF	See section 2.11. CoreSight ATB Funnel and reference [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B)
CTIO	0xE004_2000	0xE004_2FFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)

Table 2.13 Peripheral address of CPU0 (2 of 2)

Component name	Start address	End address	Note
EPPB ROM Table	0xE00F_E000	0xE00F_EFFF	—
Processor ROM Table	0xE00F_F000	0xE00F_FFFF	See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)

Table 2.14 Peripheral address of CPU1

Component name	Start address	End address	Note
ITM	0xE000_0000	0xE000_0FFF	See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230)
DWT	0xE000_1000	0xE000_1FFF	See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230)
BPU	0xE000_2000	0xE000_2FFF	See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230)
SCS	0xE000_E000	0xE000_EFFF	See reference [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B)
SCS Non-Secure Alias	0xE002_E000	0xE002_EFFF	See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230)
TPIU	0xE004_0000	0xE004_0FFF	See reference [11] ARM Limited., Arm(R) CoreSight ETM-M33 Technical Reference Manual (ARM 100232)
ETM	0xE004_1000	0xE004_1FFF	See section 2.11. CoreSight ATB Funnel and reference [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B)
CTI1	0xE004_2000	0xE004_2FFF	See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230)
EPPB ROM Table	0xE00F_E000	0xE00F_EFFF	—
Processor ROM Table	0xE00F_F000	0xE00F_FFFF	See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230)

Table 2.15 CoreSight component address

Component name	Start address*1	End address	Note
System ROM Table	0x4001_0000	0x4001_0FFF	See reference [3]
OCDREG	0x4001_1000	0x4001_1FFF	See reference [4] ARM Limited., Arm(R) CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2)
CTI	0x4001_2000	0x4001_2FFF	See reference [4] ARM Limited., Arm(R) CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2)
Funnel	0x4001_3000	0x4001_3FFF	See section 2.11. CoreSight ATB Funnel and reference [4] ARM Limited., Arm(R) CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2)
TMC (ETF configuration)	0x4001_4000	0x4001_4FFF	See reference [12] ARM Limited., Arm(R) CoreSight Trace Memory Controller Technical Reference Manual (ARM DDI0461B)
Time Stamp Generator	0x4001_5000	0x4001_5FFF	See section 3.1. Overview and reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924)
TPIU	0x4001_6000	0x4001_6FFF	See reference [4] ARM Limited., Arm(R) CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2)
Reserved area	0x4001_7000	0x4001_AFFF	—

Note: Do not access TPIU if trace clock is not supplied.

Note 1. The base address 0x4001_0000 is dedicated for CPU secure access.

The base address for CPU non-secure access is 0x5001_0000.

The base address for OCD emulator access is 0x8001_0000.

Do not access reserved area (secure: 0x4001_7000-0x4001_AFFF, non-secure: 0x5001_7000-0x5001_AFFF).

If reserved area is accessed, the access is not guaranteed.

2.9.4 ROM Table

There are five ROM Tables in the MCU:

- CPU0 processor ROM table
- CPU0 EPPB ROM table
- CPU1 processor ROM table
- CPU1 EPPB ROM table
- System ROM Table.

The CPU_n EPPB ROM Table points to the TPIU component and processor ROM table inside each CPU. The EPPB ROM table does not point to or be pointed by the System ROM Table.

The Processor ROM Table contains the entries which hold a list of debug components implemented in CPU_n (n = 0, 1). The System ROM Table contains the entries which hold a list of debug components implemented as CoreSight debug components of the MCU.

2.9.4.1 ROM Entries

ROM entries holds a list of CoreSight debug components in the system. OCD emulator can use the ROM entries to determine which components are implemented in the system.

The entries of debug components are described as follow:

- [Table 2.16](#): System ROM entries
- [Table 2.17](#): CPU0 EPPB ROM entries
- [Table 2.18](#): CPU0 Processor ROM entries (see reference [\[2\] ARM Limited., Arm\(R\) Cortex-M85 Processor Technical Reference Manual \(ARM 101924\)](#))
- [Table 2.19](#): CPU1 EPPB ROM entries
- [Table 2.20](#): CPU1 Processor ROM entries (see reference [\[9\] ARM Limited., Arm\(R\) Cortex-M33 Processor Technical Reference Manual \(ARM 100230\)](#)).

Table 2.16 System ROM entries

#	Address*1	Access size	R/W	Value	Target module
0	0x4001_0000	32-bit	Read-only	0x00001003	OCDREG
1	0x4001_0004	32-bit	Read-only	0x00002003	CTI
2	0x4001_0008	32-bit	Read-only	0x00003003	Funnel
3	0x4001_000C	32-bit	Read-only	0x00004003	ETF
4	0x4001_0010	32-bit	Read-only	0x00005003	TSG
5	0x4001_0014	32-bit	Read-only	0x00006003	TPIU
6	0x4001_00148	32-bit	Read-only	0x00000000	End of entries

Note 1. The base address 0x4001_0000 is dedicated for secure access of CPUs. The base address for non-secure access of CPUs is 0x5001_0000. The base address for OCD emulator access is 0x8001_0000.

Table 2.17 CPU0 EPPB ROM entries

#	Address	Access size	R/W	Value	Target module
0	0xE00F_E000	32-bit	Read-only	0x00001003	CPU0 Processor ROM Table
1	0xE00F_E004	32-bit	Read-only	0xFFF42003	CPU0 TPIU
2	0xE00F_E008	32-bit	Read-only	0x00000000	End of entries

Note: The base address of CPU0 EPPB ROM Table is 0xE00F_E000.

Note: TPIU clock enable should be supplied before reading ROM table.

Table 2.18 CPU0 Processor ROM entries

#	Address	Access size	R/W	Value	Target module
0	0xE00F_F000	32-bit	Read-only	0xFFFF0F003	SCS
1	0xE00F_F004	32-bit	Read-only	0xFFFF02003	DWT
2	0xE00F_F008	32-bit	Read-only	0xFFFF03003	BPU
3	0xE00F_F00C	32-bit	Read-only	0xFFFF01003	ITM
5	0xE00F_F014	32-bit	Read-only	0xFFFF42003	ETM
6	0xE00F_F018	32-bit	Read-only	0xFFFF04003	PMU
7	0xE00F_F01C	32-bit	Read-only	0xFFFF43003	CTI0
8	0xE00F_F020	32-bit	Read-only	0xFFFF47002	PMC
9	0xE00F_F024	32-bit	Read-only	0x00000000	End of entries

Note: The base address of CPU0 Processor ROM Table is 0xE00F_F000.

Table 2.19 CPU1 EPPB ROM entries

#	Address	Access size	R/W	Value	Target module
0	0xE00F_E000	32-bit	Read-only	0x00001003	CPU1 Processor ROM Table
1	0xE00F_E004	32-bit	Read-only	0xFFFF42003	CPU1 TPIU
2	0xE00F_E008	32-bit	Read-only	0x00000000	End of entries

Note: The base address of CPU1 EPPB ROM Table is 0xE00F_E000.

Note: TPIU clock enable should be supplied before reading ROM table.

Table 2.20 CPU1 Processor ROM entries

#	Address	Access size	R/W	Value	Target module
0	0xE00F_F000	32-bit	Read-only	0xFFFF0F003	SCS
1	0xE00F_F004	32-bit	Read-only	0xFFFF02003	DWT
2	0xE00F_F008	32-bit	Read-only	0xFFFF03003	BPU
3	0xE00F_F00C	32-bit	Read-only	0xFFFF01003	ITM
5	0xE00F_F014	32-bit	Read-only	0xFFFF42003	ETM
6	0xE00F_F018	32-bit	Read-only	0xFFFF43003	CTI1
7	0xE00F_F01C	32-bit	Read-only	0xFFFF44002	MTB
8	0xE00F_F020	32-bit	Read-only	0x00000000	End of entries

Note: The base address of CPU1 Processor ROM Table is 0xE00F_F000.

2.9.4.2 CoreSight Component Registers

ROM Table has CoreSight Component registers defined in Arm® CoreSight™ Architecture.

Table 2.21 shows the registers of System ROM Table.

Table 2.22 shows CPU0 EPPB ROM Table.

Table 2.23 shows CPU0 Processor ROM Table. See reference [2] ARM Limited., Arm(R) Cortex-M85 Processor Technical Reference Manual (ARM 101924).

Table 2.24 shows CPU1 EPPB ROM Table.

Table 2.25 shows CPU1 Processor ROM Table. See reference [9] ARM Limited., Arm(R) Cortex-M33 Processor Technical Reference Manual (ARM 100230).

Table 2.21 Registers of System ROM Table (1 of 2)

Name	Address	Access size	R/W	Initial value
PID4	0x8001_0FD0	32-bit	Read-only	0x00000004

Table 2.21 Registers of System ROM Table (2 of 2)

Name	Address	Access size	R/W	Initial value
PID5	0x8001_0FD4	32-bit	Read-only	0x00000000
PID6	0x8001_0FD8	32-bit	Read-only	0x00000000
PID7	0x8001_0FDC	32-bit	Read-only	0x00000000
PID0	0x8001_0FE0	32-bit	Read-only	0x00000050
PID1	0x8001_0FE4	32-bit	Read-only	0x00000030
PID2	0x8001_0FE8	32-bit	Read-only	0x0000000A
PID3	0x8001_0FEC	32-bit	Read-only	0x00000000
CID0	0x8001_0FF0	32-bit	Read-only	0x0000000D
CID1	0x8001_0FF4	32-bit	Read-only	0x00000010
CID2	0x8001_0FF8	32-bit	Read-only	0x00000005
CID3	0x8001_0FFC	32-bit	Read-only	0x000000B1

Table 2.22 CPU0 EPPB ROM Table

Name	Address	Access size	R/W	Initial value
PID4	0xE00F_EFD0	32-bit	Read-only	0x00000004
PID5	0xE00F_EFD4	32-bit	Read-only	0x00000000
PID6	0xE00F_EFD8	32-bit	Read-only	0x00000000
PID7	0xE00F_EFDC	32-bit	Read-only	0x00000000
PID0	0xE00F_EFE0	32-bit	Read-only	0x00000050
PID1	0xE00F_EFE4	32-bit	Read-only	0x00000030
PID2	0xE00F_EFE8	32-bit	Read-only	0x0000000A
PID3	0xE00F_EFEC	32-bit	Read-only	0x00000000
CID0	0xE00F_EFF0	32-bit	Read-only	0x0000000D
CID1	0xE00F_EFF4	32-bit	Read-only	0x00000010
CID2	0xE00F_EFF8	32-bit	Read-only	0x00000005
CID3	0xE00F_EFFC	32-bit	Read-only	0x000000B1

Table 2.23 CPU0 processor ROM Table

Name	Address	Access size	R/W	Initial value
PID4	0xE00F_FFD0	32-bit	Read-only	0x00000004
PID5	0xE00F_FFD4	32-bit	Read-only	0x00000000
PID6	0xE00F_FFD8	32-bit	Read-only	0x00000000
PID7	0xE00F_FFDC	32-bit	Read-only	0x00000000
PID0	0xE00F_FFE0	32-bit	Read-only	0x000000D4
PID1	0xE00F_FFE4	32-bit	Read-only	0x000000B4
PID2	0xE00F_FFE8	32-bit	Read-only	0x0000000B
PID3	0xE00F_FFEC	32-bit	Read-only	0x00000000
CID0	0xE00F_FFF0	32-bit	Read-only	0x0000000D
CID1	0xE00F_FFF4	32-bit	Read-only	0x00000010
CID2	0xE00F_FFF8	32-bit	Read-only	0x00000005
CID3	0xE00F_FFFC	32-bit	Read-only	0x000000B1

Table 2.24 CPU1 EPPB ROM Table

Name	Address	Access size	R/W	Initial value
PID4	0xE00F_EFD0	32-bit	Read-only	0x00000004
PID5	0xE00F_EFD4	32-bit	Read-only	0x00000000
PID6	0xE00F_EFD8	32-bit	Read-only	0x00000000
PID7	0xE00F_EFDC	32-bit	Read-only	0x00000000
PID0	0xE00F_EFE0	32-bit	Read-only	0x00000050
PID1	0xE00F_EFE4	32-bit	Read-only	0x00000030
PID2	0xE00F_EFE8	32-bit	Read-only	0x0000000A
PID3	0xE00F_EFEC	32-bit	Read-only	0x00000000
CID0	0xE00F_EFF0	32-bit	Read-only	0x0000000D
CID1	0xE00F_EFF4	32-bit	Read-only	0x00000010
CID2	0xE00F_EFF8	32-bit	Read-only	0x00000005
CID3	0xE00F_EFFC	32-bit	Read-only	0x000000B1

Table 2.25 CPU1 processor ROM Table

Name	Address	Access size	R/W	Initial value
PID4	0xE00F_FFD0	32-bit	Read-only	0x00000004
PID5	0xE00F_FFD4	32-bit	Read-only	0x00000000
PID6	0xE00F_FFD8	32-bit	Read-only	0x00000000
PID7	0xE00F_FFDC	32-bit	Read-only	0x00000000
PID0	0xE00F_FFE0	32-bit	Read-only	0x000000C9
PID1	0xE00F_FFE4	32-bit	Read-only	0x000000B4
PID2	0xE00F_FFE8	32-bit	Read-only	0x0000000B
PID3	0xE00F_FFEC	32-bit	Read-only	0x00000000
CID0	0xE00F_FFF0	32-bit	Read-only	0x0000000D
CID1	0xE00F_FFF4	32-bit	Read-only	0x00000010
CID2	0xE00F_FFF8	32-bit	Read-only	0x00000005
CID3	0xE00F_FFFC	32-bit	Read-only	0x000000B1

2.9.5 DBGREG

Base_Address: 0x4001_B000 (Secure), 0x5001_B000 (Non-secure)

The DBGREG module controls the debug functionalities.

[Table 2.26](#) shows a list of DBGREG registers.

Table 2.26 DBGREG registers (1 of 2)

Name		DAP port	Address	Access size	R/W
Debug Status Register	DBGSTR	Port 0/2	0x4001_B000 0x5001_B000	32-bit	Read/Write
Debug Stop Control Register	DBGSTOPCR	Port 0/2	0x4001_B010 0x5001_B010	32-bit	Read/Write
Debug Authentication Control Register 0	DBGAUTH0	Port 0/2	0x4001_B020 0x5001_B020	32-bit	Read/Write
Trace Port Control Register	TRPORTCR	Port 0/2	0x4001_B030 0x5001_B030	32-bit	Read/Write

Table 2.26 DBGREG registers (2 of 2)

Name		DAP port	Address	Access size	R/W
Trace Port Size Control Register	TRPORTSZ	Port 0/2	0x4001_B038 0x5001_B038	32-bit	Read/Write
Cache Debug Control Register	CACHEDBGCR	Port 0/2	0x4001_B040 0x5001_B040	32-bit	Read/Write
Debug Non-volatile Memory Control Register	DBGNVMCR	Port 0/2	0x4001_B050 0x5001_B050	32-bit	Read/Write
Authentication Level Control Register	ALCTRL	Port 0/2	0x4001_B100 0x5001_B100	32-bit	Read/Write
First Stage Boot Loader Status Register	FSBLSTAT	Port 0/2	0x4001_B200 0x5001_B200	32-bit	Read/Write

2.9.5.1 DEBUGSAR : Debug Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x180

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DBGS A0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DBGSA0	Debug Security Attribution 0 0: Secure 1: Non-secure	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-1
P-TYPE-1

DBGSA0 bit (Debug Security Attribution 0)

Security attributes of registers for debug control. The target registers are as follow:

- DBGREG.DBGAUTH0
- The CoreSight registers connected to APB-AP and mapped to the system address space.

Note: This is not affected to the access from external by APB-AP.

2.9.5.2 DBGSTR : Debug Status Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	CDBG PWRU PACK	CDBG PWRU PREQ	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	x	x	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
27:0	—	These bits are read as 0.	R
28	CDBGPWRUPREQ	Debug power-up request 0: OCD is not requesting debug power-up 1: OCD is requesting debug power-up	R
29	CDBGPWRUPACK	Debug power-up acknowledge 0: Debug power-up request is not acknowledged 1: Debug power-up request is acknowledged	R
31:30	—	These bits are read as 0.	R

Note: S-TYPE-5
P-TYPE-2

The DBGSTR register is a status register to notify OCD status to programs. This register monitors the debug connecting state.

2.9.5.3 DBGSTOPCR : Debug Stop Control Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	DBGS TOP_ CTER R1	DBGS TOP_ CTER R0	—	DBGS TOP_ NVME RR	—	—	—	—	—	—	—	—	DBGS TOP_ PVD	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	DBGS TOP_ WDT1	DBGS TOP_ WDT0	DBGS TOP_ I WDT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
0	DBGSTOP_IWDT	Mask bit for IWDT reset/interrupt in the OCD run mode In the OCD break mode, the reset/interrupt is masked and IWDT counter is stopped, regardless of this bit value. 0: Enable IWDT reset/interrupt 1: Mask IWDT reset/interrupt and stop IWDT counter	R/W

Bit	Symbol	Function	R/W
1	DBGSTOP_WDT0	Mask bit for WDT0 reset/interrupt in the OCD run mode In the OCD break mode, the reset/interrupt is masked and WDT0 counter is stopped, regardless of this bit value. 0: Enable WDT0 reset/interrupt 1: Mask WDT0 reset/interrupt and stop WDT0 counter	R/W
2	DBGSTOP_WDT1	Mask bit for WDT1 reset/interrupt in the OCD run mode In the OCD break mode, the reset/interrupt is masked and WDT1 counter is stopped, regardless of this bit value. 0: Enable WDT1 reset/interrupt 1: Mask WDT1 reset/interrupt and stop WDT1 counter	R/W
16:3	—	These bits are read as 0. The write value should be 0.	R/W
17	DBGSTOP_PVD	Mask bit for PVDn (n! = 0) reset/interrupt 0: Enable PVDn (n! = 0) reset/interrupt 1: Mask PVDn (n! = 0) reset/interrupt	R/W
25:18	—	These bits are read as 0. The write value should be 0.	R/W
26	DBGSTOP_NVMER R	Mask bit for MRAM ECC error reset/interrupt 0: Enable MRAM ECC error reset/interrupt 1: Mask MRAM ECC error reset/interrupt	R/W
27	—	This bit is read as 0. The write value should be 0.	R/W
28	DBGSTOP_CTERR0	Mask bit for Cache/TCM ECC error reset of CPU0 0: Enable Cache/TCM ECC error reset 1: Mask Cache/TCM ECC error reset	R/W
29	DBGSTOP_CTERR1	This bit is reserved. It can be R/W but has no effect.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-2

The DBGSTOPCR register specifies functional stop during debugging.

This register is valid only when any of the following conditions are true:

- CDBGPWRUPREQ is 1, and AL is AL2 or AL1
- OFS1.SWDBG is 0, and DBGAUTH0.DBGEN0 is 1
- OFS1.SWDBG is 0, and DBGAUTH0.DBGEN1 is 1.

When none of the above conditions are true, the MCU ignores this register setting and regards all bits as 0.

2.9.5.4 DBGAUTH0 : Debug Authentication Control Register0

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	SWDBG	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DEVIC EEN
	OFS1. SWDBG															
Value after reset:	G	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	NIDEN 1	NIDEN 0	—	—	DBGE N1	DBGE N0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DBGEN0*1	CPU0 invasive debug enable 0: Disabled 1: Enabled	R/W
1	DBGEN1*1	CPU1 invasive debug enable 0: Disabled 1: Enabled	R/W
3:2	—	These bits are read as 0. The write value should be 0.	R/W
4	NIDEN0	CPU0 non-invasive debug enable 0: Disabled 1: Enabled	R/W
5	NIDEN1	CPU1 non-invasive debug enable 0: Disabled 1: Enabled	R/W
15:6	—	These bits are read as 0. The write value should be 0.	R/W
16	DEVICEEN*2	APB-AP (AP1) authentication 0: Disabled 1: Enabled	R/W
30:17	—	These bits are read as 0. The write value should be 0.	R/W
31	SWDBG	Software control of debug function 0: Disabled 1: Enabled	R

Note: S-TYPE-3(DEBUGSAR.DBGSA0)
P-TYPE-2

Note: MCU behaves according to the setting of DBGSTOPCR, SYOCDRCR, TRCKCR, TRACECTR, and MCUCTRL when DBGENn = 1 or NIDENn = 1 is TRUE or when CDBGPWRUPREQ = 1 and AL ≠ AL0 is TRUE (n = 0, 1).

Note 1. When AL2 or AL1, the MCU ignores this bit and regards as if it is set to 1.

Note 2. When AL2, the MCU ignores this bit and regards as if it is set to 1.

When software control of debug function (OFS1.SWDBG = 0) is enabled, non-secure invasive and non-invasive debug function of CPU_n can be controlled by DBGENn and NIDENn bits of this register (n = 0, 1).

Access from APB-AP to the CoreSight components through Debug APB can be enabled by DEVICEEN bit of this register.

When OFS1.SWDBG is 1, the MCU ignores this register and regards it as if it is set to its initial value.

2.9.5.5 CACHEDBGCR : Cache Debug Control Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	L1RST DIS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	L1RSTDIS	Disable L1 cache automatic invalidation of CPU0 0: Enable automatic invalidation of the L1 cache 1: Disable automatic invalidation of the L1 cache	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-6
P-TYPE-2

Note: The setting of the register is valid only if any of the following conditions apply:

- CDBGPWUPREQ = 1 and AL is AL2 or AL1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN0 or DBGAUTH0.NIDEN0 is 1.

The CACHEDBGCR register provides capability of control Disable L1 cache invalidation out of reset during debugging.

2.9.5.6 TRPORTCR : Trace Port Control Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x030

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	SWOSEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PORTSEL[1:0]	—	—	—	—	—	DRV[1:0]	—	—	OE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
0	OE	Data Out Enable bit indicates whether trace clock, trace data and SWO outputs are enabled or not. OE bit should be enabled before trace clock is enabled. 0: Output disabled 1: Output enabled	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
3:2	DRV[1:0]	Port drive capability control indicates trace port buffer speed. 0 0: Low 0 1: Middle 1 0: High-Speed high-drive 1 1: High	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
9:8	PORTSEL[1:0]	Port select. Only A port setting is supported. 0 0: A port 0 1: B port	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
16	SWOSEL	Select SWO between CPU0 and CPU1 0: SWO of CPU0 is selected (default) 1: SWO of CPU1 is selected	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5

P-TYPE-2

Note: The setting of the register is valid only when any of the following conditions applies:

- CDBGPWUPREQ = 1 and AL is AL2 or AL1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN0 or DBGAUTH0.NIDEN0 is 1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN1 or DBGAUTH0.NIDEN1 is 1.

Trace data control when OE bit = 1.

Target CPU	Trace Protocol	TPIU_SPPR (CPU0)	TPIU_SPPR (CPU1)	SWOSEL
CPU0	SWO	01b or 10b	Don't care	0 (CPU0)
	Parallel	00b	Don't care	0 (CPU0)

Target CPU	Trace Protocol	TPIU_SPPR (CPU0)	TPIU_SPPR (CPU1)	SWOSEL
CPU1	SWO	Don't care	01b or 10b	1 (CPU1)
	Parallel	Don't care	00b	1 (CPU1)
CPU0 and CPU1	SWO	01b or 10b	01b or 10b	Select CPU output SWO
	Parallel	00b to CPU0 or CPU1		Select CPU when TPIU_SPPR = 00b

For more information about TPIU_SPPR register (CPU0: bit PProtocol, CPU1: bit TXMODE), see [1] [ARM Limited., ARMv8-M Architecture Reference Manual \(ARM DDI 0553B\)](#).

2.9.5.7 ALCTRL : Authentication Level Control Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	FAILCNT[1:0]		—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	AL[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	x*1	x*1	x*1	x*1	x*1	x*1	x*1	x*1

Note 1. Determined by PL value.

Bit	Symbol	Function	R/W
7:0	AL[7:0]	AL monitor Power-on reset initializes the AL to AL0. When the current device lifecycle is not OEM, AL is updated to the level corresponding to the device lifecycle. When the current device lifecycle is OEM: - Without debugger connection: AL is updated to the same level as the PL - With debugger connection: AL is updated to the same level as the PL only if PL is higher than AL The Renesas firmware updates the AL in Boot mode when the debug authentication is successful. The Renesas firmware updates the AL to a lower level in Boot mode regardless of authentication status. A debugger disconnection updates the AL to the same level as the PL. When FAILCNT[1:0] = 11, AL[7:0] stops updating (it is controlled by software). 0x00: AL0 0x05: AL1 0xA5: AL2	R/W
29:8	—	These bits are read as 0. The write value should be 0.	R/W
31:30	FAILCNT[1:0]	Number of times responding to incorrect response data 0 0: No responding to incorrect response data 0 1: First time of responding to incorrect response data 1 0: Second time of responding to incorrect response data 1 1: Third time of responding to incorrect response data	R/W

Note: S-TYPE-5

P-TYPE-5

- Note:
- When CDBGWUPREQ changes from 1 to 0, the AL value is initialized to PL value
 - Exiting from Boot mode when DBGWUPREQ = 0 also initializes the AL value
 - Only accessible in Boot mode.

The ALCTRL register monitors the Authentication Level.

2.9.5.8 FSBLSTAT : First Stage Boot Loader Status Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x200

[illegible]

Bit	Symbol	Function	R/W
0	CS	FSBL completion status 0: FSBL is executing 1: FSBL execution is complete	R/W
1	RS	FSBL result status 0: FSBL failed 1: FSBL passed	R/W
7:2	—	These bits are read as 0. The write value should be 0.	R/W
10:8	FSBLCLK[2:0]	System clock frequency selection during FSBL execution The selection reflects the value of FSBLCTRL0.FSBLCLK[2:0]. FSBLCTRL0 specifications are described in section 7, Option-Setting Memory .	R
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-5

The FSBLSTAT register notifies FSBL completion to the debugger.

This register is initialized by a system reset and is accessible only while FSBL is executing.

2.9.5.9 TRPORTSZ : Trace Port Size Control Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x038

[illegible]

Bit	Symbol	Function	R/W
31:0	PORTSIZE[31:0]	Indicates how many pins of TRACEDATA are available. Only set 1 bit to this register, all others must be 0. Writing value with more than 1 bit set can cause Unpredictable behavior. The value set to this register must be the same as the value setting for CoreSight TPIU.CPSR register. Writing different value can cause Unpredictable behavior. Bit N is set: N + 1 bit(s) of TRACEDATA is supported (N = 0 to 31).	R/W

Note: S-TYPE-5
P-TYPE-2

Note: The setting of the register is valid only if any of the following conditions apply:

- CDBGPWUPREQ = 1 and AL is AL2 or AL1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN0 or DBGAUTH0.NIDEN0 is 1
- OFS1.SWDBG is 0 and DBGAUTH0.DBGEN1 or DBGAUTH0.NIDEN1 is 1.

The TRPORTSZ register controls how many pins of trace data are used.

2.9.5.10 DBGNVMCR : Debug Non-volatile Memory Control Register

Base address: CPU_DBG = 0x4001_B000
CPU_DBG_NS = 0x5001_B000

Offset address: 0x050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	NVMW E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	NVMWE	Non-volatile memory write enable 0: MRAM write is not allowed 1: MRAM write is allowed	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5

P-TYPE-2

Note: NVMWE is valid only when CPU is in halted state

2.9.5.11 CoreSight Component Registers

DBGREG has CoreSight Component registers defined in ARM® CoreSight™ Architecture. [Table 2.27](#) shows a list of DBGREG CoreSight Component registers.

Table 2.27 CoreSight Component registers of DBGREG

Name	Address	Access size	R/W	Initial value
PID4	0x4001_BFD0	32-bit	Read-only	0x00000004
PID5	0x4001_BFD4	32-bit	Read-only	0x00000000
PID6	0x4001_BFD8	32-bit	Read-only	0x00000000
PID7	0x4001_BFDC	32-bit	Read-only	0x00000000
PID0	0x4001_BFE0	32-bit	Read-only	0x00000005
PID1	0x4001_BFE4	32-bit	Read-only	0x00000030
PID2	0x4001_BFE8	32-bit	Read-only	0x0000000A
PID3	0x4001_BFEC	32-bit	Read-only	0x00000000
CID0	0x4001_BFF0	32-bit	Read-only	0x0000000D
CID1	0x4001_BFF4	32-bit	Read-only	0x000000F0
CID2	0x4001_BFF8	32-bit	Read-only	0x00000005
CID3	0x4001_BFFC	32-bit	Read-only	0x000000B1

2.9.6 OCDREG

OCDREG is a register module provided for OCD (On-Chip Debug) Emulator functionalities. OCDREG is implemented as a CoreSight compliant component. [Table 2.28](#) shows a list of OCDREG registers.

Note: Accesses to OCDREG from CPU and other bus masters is only permitted by privileged access, and one of the following conditions must be satisfied:

- Debugger is connected

- Self-hosted debug is enabled.

Accesses to OCDREG from external debug through APB-AP is permitted by either privileged access or unprivileged access.

Base address:

- CPU 0x4001_1000 (Secure), 0x5001_1000 (Non-Secure)
- OCD: 0x8001_1000

Table 2.28 **OCDREG registers list**

Name		DAP Port	OCD address	CPU address	Access size	R/W
MCU Status Register* ¹	MCUSTAT	Port 1	0x8000_0400	Cannot be accessed from CPU	32-bit	Read-only
MCU Error Status Register	MCUERRSTAT	Port 1	0x8001_1000	0x4001_1000 0x5001_1000	32-bit	Read-only
MCU Control Register	MCUCTRL	Port 1	0x8001_1004	0x4001_1004 0x5001_1004	32-bit	Read/Write
JTAG Boot Mode Entry Register	JBMDR	Port 1	0x8001_1100	0x4001_1100 0x5001_1100	32-bit	Read/Write
JTAG Boot Receive Data Register	JBRDR	Port 1	0x8001_1120	0x4001_1120 0x5001_1120	32-bit	Read/Write
JTAG Boot Transmit Data Register	JBTDR	Port 1	0x8001_1130	0x4001_1130 0x5001_1130	32-bit	Read/Write
JTAG Boot Status Register	JBSTR	Port 1	0x8001_1140	0x4001_1140 0x5001_1140	32-bit	Read/Write
JTAG Boot Interrupt Control Register	JBICR	Port 1	0x8001_1150	0x4001_1150 0x5001_1150	32-bit	Read/Write
First Stage Boot Loader Status Monitor Register	FSBLSTATM	Port 1	0x8001_1300	0x4001_1300 0x5001_1300	32-bit	Read-only

Note: Do not write to MCUCTRL using JTAG/SWD while the CPU is accessing MCUCTRL, JBMDR, JBRDR, JBTDR, and JBICR.
Do not write to JBMDR using JTAG/SWD except during RES pin reset.
Do not write to JBRDR using JTAG/SWD while the CPU is accessing MCUCTRL, JBMDR, JBRDR, JBTDR, and JBICR.
Do not write to JBTDR using JTAG/SWD.
Do not write to JBSTR using JTAG/SWD.
Do not write to JBICR using JTAG/SWD.

Note 1. See [section 2.9.6.1. MCUSTAT : MCU Status Register](#).

2.9.6.1 MCUSTAT : MCU Status Register

Address: 0x8000_0400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	AP[1:0]		—	—	—	—	—	—	—	DSTB Y3	DSTB Y2	DSTB Y1	—	—	STBY2	—
Value after reset:	0	x	0	0	0	0	0	0	0	x	x	x	0	0	x	x
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	AL[1:0]		—	—	—	SECE XT	—	—	—	CPU1 STOP CLK	CPU1 SLEEP	CPU0 STOP CLK	CPU0 SLEEP	—
Value after reset:	1	0	0	0	0	0	0	1	0	0	0	x	x	x	x	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0.	R
1	CPU0SLEEP	Indicates CPU0 is in Sleep mode. 0: CPU0 is not in Sleep mode 1: CPU0 is in Sleep mode	R

Bit	Symbol	Function	R/W
2	CPU0STOPCLK	0: CPU0 clock is not stopped. It indicates that the CPU0 is in Normal or Sleep mode. 1: CPU0 clock is stopped. It indicates that the CPU0 is in Deep Sleep, Or the MCU is in Software Standby mode or Deep Software Standby power mode.	R
3	CPU1SLEEP	Indicates CPU1 is in Sleep mode. 0: CPU1 is not in Sleep mode 1: CPU1 is in Sleep mode	R
4	CPU1STOPCLK	0: CPU1 clock is not stopped. It indicates CPU1 is in Normal or Sleep mode. 1: CPU1 clock is stopped. It indicates CPU1 is in Deep Sleep, or the MCU is in Software Standby mode or Deep Software Standby power mode.	R
7:5	—	These bits are read as 0.	R
8	SECEXT	This bit is read as 1. 0: Secure debug is not supported 1: Secure debug is supported	R
11:9	—	These bits are read as 0.	R
13:12	AL[1:0]	Authentication Level monitor 0x0: AL0 0x1: AL1 0x3: AL2	R
14	—	This bit is read as 0.	R
15	—	This bit is read as 1.	R
16	—	The read value is undefined.	R
17	STBY2	Indicates the MCU is in Software Standby mode. 0: MCU is not in Software Standby mode 1: MCU is in Software Standby mode	R
19:18	—	These bits are read as 0.	R
20	DSTBY1	Indicates the MCU is in Deep Software Standby 1 mode. 0: MCU is not in Deep Software Standby 1 mode 1: MCU is in Deep Software Standby 1 mode	R
21	DSTBY2	Indicates the MCU is in Deep Software Standby 2 mode. 0: MCU is not in Deep Software Standby 2 mode 1: MCU is in Deep Software Standby 2 mode	R
22	DSTBY3	Indicates the MCU is in Deep Software Standby 3 mode. 0: MCU is not in Deep Software Standby 3 mode 1: MCU is in Deep Software Standby 3 mode	R
29:23	—	These bits are read as 0.	R
31:30	AP[1:0]	Indicates which access ports are available. An external host must check AP[1:0] and must not access AP that is unavailable because unavailable AP cannot respond. 0 0: AP0 and AP1 are available 0 1: AP0, AP1 and AP2 are available	R

Note: MCUSTAT register address is independent of OCDREG space. Only OCD emulator can access this register.

The MCUSTAT register indicates MCU status and authentication status. All bits in the register are monitoring bits so cannot be reset. The register has no security protection.

2.9.6.2 MCUERRSTAT : MCU Error Status Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ZERO
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ZERO	Zeroization status flag 0: Zeroization caused by tamper detection has not been performed 1: Zeroization caused by tamper detection has been performed	R
31:1	—	These bits are read as 0.	R

Note: S-TYPE-5
P-TYPE-2

The MCUERRSTAT register indicates the error status of the MCU caused by tamper attack. All bits in the register are monitoring bits so cannot be reset. The register has no security protection.

2.9.6.3 MCUCTRL : MCU Control Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CPUW AIT1	CPUW AIT0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EDBG RQ1	EDBG RQ0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EDBGRQ0	External Debug Request for CPU0. Writing 1 to the bit causes CPU Halt or Debug Monitor exception request. 0: Not request debug event 1: Request debug event	R/W
1	EDBGRQ1	External Debug Request for CPU1. Writing 1 to the bit causes CPU Halt or Debug Monitor exception request. 0: Not request debug event 1: Request debug event	R/W
15:2	—	These bits are read as 0. The write value should be 0.	R/W
16	CPUWAIT0	CPU0 WAIT SETTING. Write 1 to assert CPUWAIT0, write 0 to deassert CPUWAIT0. 0: Clear CPUWAIT0 to low 1: Set CPUWAIT0 to high	R/W

Bit	Symbol	Function	R/W
17	CPUWAIT1	CPU1 WAIT SETTING. Write 1 to assert CPUWAIT1, write 0 to deassert CPUWAIT1. 0: Clear CPUWAIT1 to low 1: Set CPUWAIT1 to high	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: This register is valid only when any of the following conditions are true:

- CDBGPWRUPREQ is 1, and AL is AL2 or AL1
- OFS1.SWDBG is 0, and DBGAUTH0.DBGEN0 is 1
- OFS1.SWDBG is 0, and DBGAUTH0.DBGEN1 is 1.

When none of the above conditions are true, the MCU ignores this register setting and regards all bits as 0.

Note: S-TYPE-5
P-TYPE-2

The MCUCTRL register provides external debugger ability to wake up system from low power state, and ability to enter debug state by external event.

2.9.6.4 Authentication Control Registers

Authentication registers accessible from the debugger are placed in OCDREG.

2.9.6.4.1 JBMDR : JTAG Boot Mode Entry Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	KEY							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	KEY	Mode entry key Pin reset releases after 0xA5 is set, then system enters the JTAG boot mode. MDSR.JBOTS = 1 when system transitions to JTAG boot mode. Setting this value does not have any effect when debugger is not connected.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-2

The JBMDR register sets command from the debugger.

This register is initialized by the debugger disconnection.

2.9.6.4.2 JBRDR : JTAG Boot Receive Data Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x120

Bit position:	31																																	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
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Bit	Symbol	Function	R/W
31:0	RDAT[31:0]	Received data register R/W is possible for both the external host and CPU, but the following usage is recommended: W: External host R: CPU (Boot Firmware) When JBSTR.RDF = 1, write is not possible and an error occurs.	R/W

Note: S-TYPE-5
P-TYPE-2

The JBRDR register is used for receiving data from the debugger.

2.9.6.4.3 JBTDR : JTAG Boot Transmit Data Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x130

Bit position: 31

0

Bit field:

TDAT[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	TDAT[31:0]	Transmitted data register R/W is possible for both the external host and CPU, but the following usage is recommended: W: CPU (Boot Firmware) R: External host When JBSTR.TDE = 1, read is not possible and an error occurs.	R/W

Note: S-TYPE-5
P-TYPE-2

The JBTDR register is used for transmitting data from the debugger.

2.9.6.4.4 JBSTR : JTAG Boot Status Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x140

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	TDE	RDF
---	---	---	---	---	---	---	---	---	---	---	---	---	---	-----	-----

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	RDF	Receive buffer full [Set conditions] - Write access to JBRDR - Write 1 to JBSTR.RDF [Clear conditions] - Read access to JBRDR - Write 0 to JBSTR.RDF 0: No receiving data 1: There is receiving data	R/W

Bit	Symbol	Function	R/W
1	TDE	Transmit data empty [Set conditions] - Read access to JBTDR - Write 1 to JBSTR.TDE [Clear conditions] - Write access to JBTDR - Write 0 to JBSTR.TDE 0: There is data transmission 1: No data transmission	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-2

The JBSTR register is used for monitoring boot status.

2.9.6.4.5 JBICR : JTAG Boot Interrupt Control Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x150

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RDFIE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RDFIE	Receive buffer full interrupt enabled. 0: Interrupt request disabled by RDF = 1 1: Enable interrupt request by RDF = 1	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-5
P-TYPE-2

The JBICR register is used for controlling interrupt during JTAG boot.

2.9.6.4.6 FSBLSTATM : First Stage Boot Loader Status Monitor Register

Base address: CPU_OCD = 0x4001_1000
CPU_OCD_NS = 0x5001_1000

Offset address: 0x300

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RS	CS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CS	FSBL completion status 0: FSBL is not complete 1: FSBL is complete	R
1	RS	FSBL result status 0: FSBL failed 1: FSBL passed	R
31:2	—	These bits are read as 0.	R

Note: S-TYPE-5
P-TYPE-2

The FSBLSTATM register is mirror of the FSBLSTAT register.

2.9.6.5 CoreSight Component Registers

OCDREG has CoreSight Component registers defined in ARM® CoreSight™ Architecture. Table 2.29 shows a list of OCDREG CoreSight Component registers.

See reference [6] ARM Limited., Arm(R) CoreSight Architecture Specification (ARM IHI 0029D) for the detail of each register.

Table 2.29 CoreSight Component registers of OCDREG

Name	Address	Access size	R/W	Initial value
PID4	0x8001_1FD0	32-bit	Read-only	0x00000004
PID5	0x8001_1FD4	32-bit	Read-only	0x00000000
PID6	0x8001_1FD8	32-bit	Read-only	0x00000000
PID7	0x8001_1FDC	32-bit	Read-only	0x00000000
PID0	0x8001_1FE0	32-bit	Read-only	0x00000004
PID1	0x8001_1FE4	32-bit	Read-only	0x00000030
PID2	0x8001_1FE8	32-bit	Read-only	0x0000000A
PID3	0x8001_1FEC	32-bit	Read-only	0x00000000
CID0	0x8001_1FF0	32-bit	Read-only	0x0000000D
CID1	0x8001_1FF4	32-bit	Read-only	0x000000F0
CID2	0x8001_1FF8	32-bit	Read-only	0x00000005
CID3	0x8001_1FFC	32-bit	Read-only	0x000000B1

Note: S-TYPE-5
P-TYPE-2

2.10 CoreSight Cross Trigger Interface

Table 2.30 Cross Trigger Interface (1 of 2)

Number of CTI channel	CTITRIGIN		CTITRIGOUT	
CTI	0	ETF ACQCOMP	0	TPIU FLUSHIN
	1	ETF FULL	1	TPIU TRIGIN
	2	—	2	ETF FLUSHIN
	3	—	3	ETF TRIGIN
	4	—	4	—
	5	—	5	—
	6	—	6	—
	7	—	7	—

Table 2.30 Cross Trigger Interface (2 of 2)

Number of CTI channel	CTITRIGIN		CTITRIGOUT	
CTI0	0	Processor Halted	0	Processor debug request
	1	DWT Comparator Output 0	1	Processor Restart
	2	DWT Comparator Output 1	2	CTIIRQ[0] (Connected to ICU0)
	3	DWT Comparator Output 2	3	CTIIRQ[1] (Connected to ICU0)
	4	ETM Event Output 0	4	ETM Event Input 0
	5	ETM Event Output 1	5	ETM Event Input 1
	6	—	6	ETM Event Input 2
	7	—	7	ETM Event Input 3
CTI1	0	Processor Halted	0	Processor debug request
	1	DWT Comparator Output 0	1	Processor Restart
	2	DWT Comparator Output 1	2	CTIIRQ[0] (Connected to ICU1)
	3	DWT Comparator Output 2	3	CTIIRQ[1] (Connected to ICU1)
	4	ETM Event Output 0	4	ETM Event Input 0
	5	ETM Event Output 1	5	ETM Event Input 1
	6	—	6	ETM Event Input 2
	7	—	7	ETM Event Input 3

The MCU supports processor CTI0/CTI1 and CoreSight Common CTI which is implemented in Debug module. The ICU setting is necessary when CTIIRQ is used.

2.11 CoreSight ATB Funnel

There is one CoreSight ATB Funnel in the MCU. The funnel has four ATB slaves and one ATB master, and it is used to select the debug trace sources from ETMn and ITMn to ETF. [Figure 2.3](#) shows the CoreSight ATB connection in the MCU (n = 0, 1).

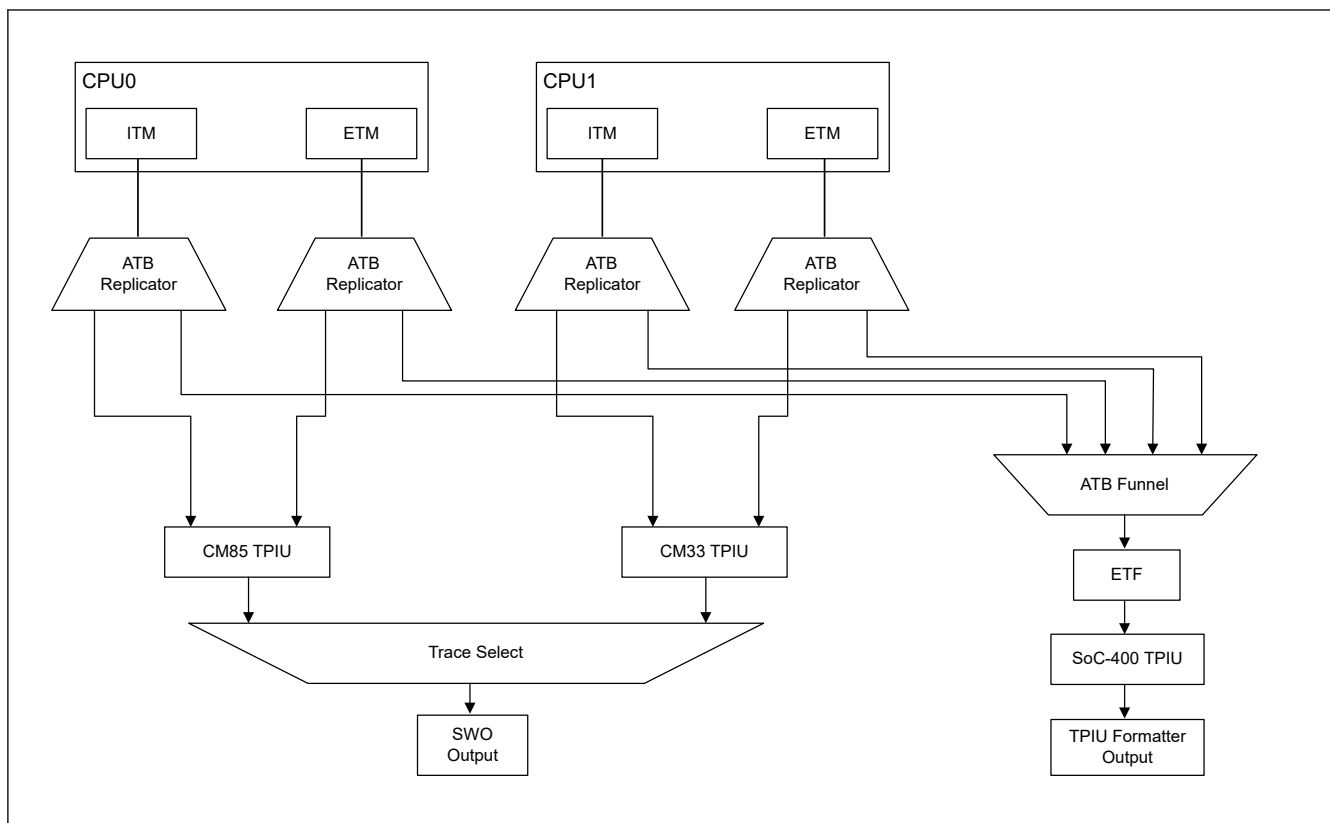


Figure 2.3 CoreSight ATB connection

Table 2.31 shows funnel ATB slave connection.

Table 2.31 ATB slave connection

ATB Slave No.	Connected Trace Source
#0	CPU0-ITM
#1	CPU0-ETM
#2	CPU1-ITM
#3	CPU1-ETM

See reference [4] ARM Limited., Arm(R) CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2) for the detail of ATB and Funnel.

2.12 SysTick System Timer

The MCU has two SysTick system timers for each CPU_n (n = 0, 1), secure SysTick and non-secure SysTick.

The SysTick source is CPUCCLK_n or 1 MHz clock generated from the on-chip oscillator. When using 1 MHz clock generated from the on-chip oscillator for the SysTick source, the CPUCCLK_n frequency must be 2 MHz or higher (n = 0, 1).

2.13 CoreSight Timestamp Generator

The MCU has a CoreSight Timestamp Generator to provide timestamp to ITM and ETM for each CPU_n (n = 0, 1).

The timestamp is generated by a 64-bit counter operating.

See reference [4] ARM Limited., Arm(R) CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2) for the detail.

2.14 OCD Emulator Connection

In this product, the MCU confirms access permission for MCU resources by checking current Authentication Level stored in OCDREG. There are two ways to control Authentication Level, through:

- JTAG/SWD. See [Figure 2.4](#).
- Software. See [Figure 2.5](#).

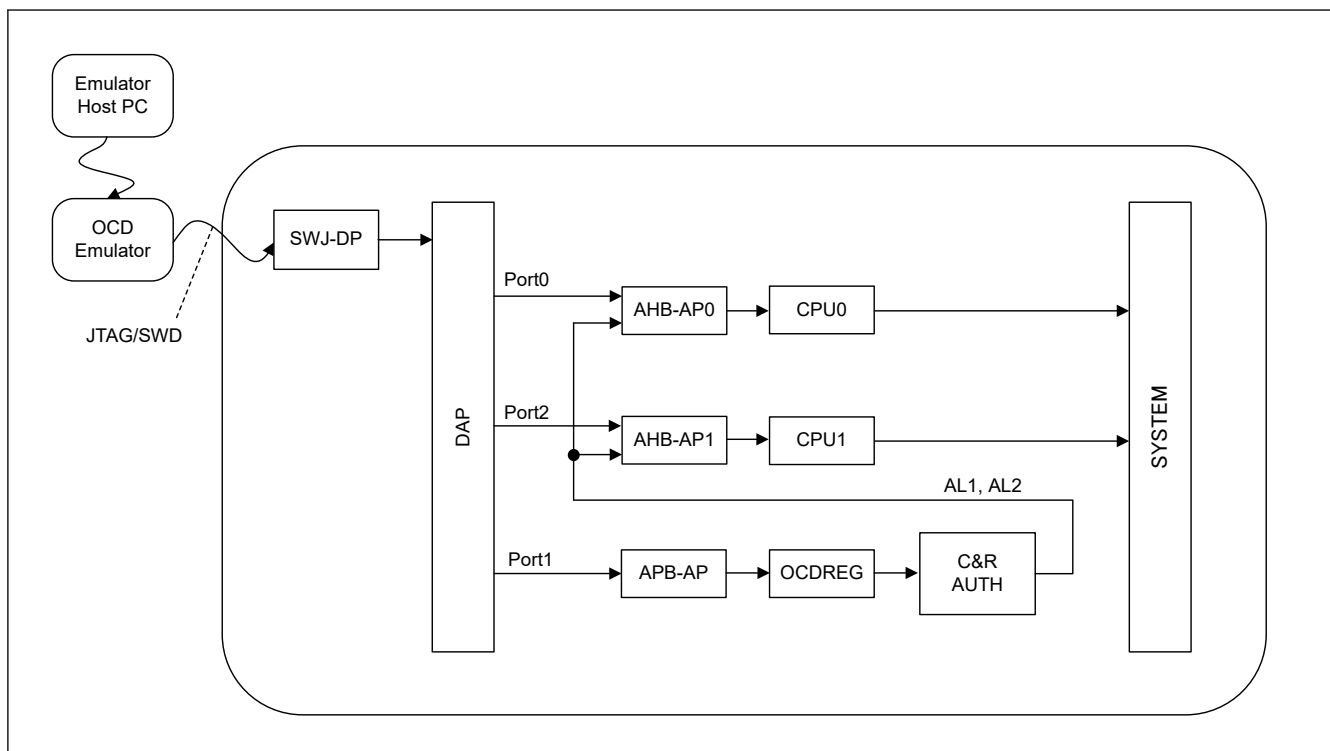


Figure 2.4 Using JTAG/SWD to control authentication level

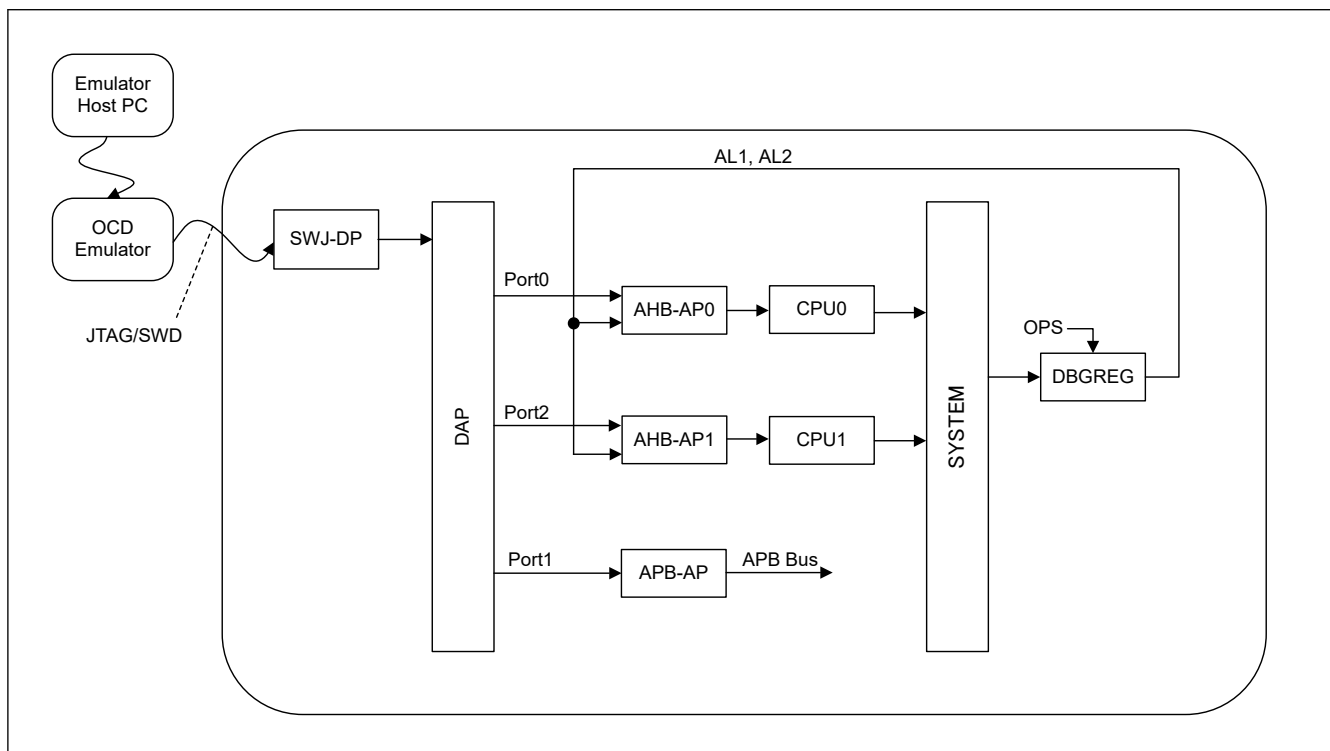


Figure 2.5 Using software to control authentication level

2.14.1 Restrictions in OCD Emulator Connecting

There are some restrictions regarding emulator access. This section describes the restrictions.

2.14.1.1 Starting Connection While in Low Power Mode

When starting a JTAG/SWD connection from an OCD emulator, the MCU must be in Normal, Sleep mode or Deep Sleep mode. If the MCU is in Software Standby or Deep Software Standby mode, the OCD emulator can cause the MCU to hang.

2.14.1.2 Low Power Mode Change During Debugging

When the MCU is in OCD mode, the low power mode can be changed. However, system bus access from AHB-APn (n = 0, 1) is prohibited in Deep Sleep, Software Standby, Deep Software Standby mode. Only SWJ-DP, APB-AP and OCDREG can be accessed from OCD emulator in these modes. [Table 2.32](#) shows the limitations.

Table 2.32 Limitations of SWJ-DP, APB-AP and OCDREG in low power mode

Current mode	Start OCD emulator connection	Change low power mode	Access AHB-APn (n = 0, 1) and system bus	Access APB-AP and OCDREG
Normal	Available	Available	Available	Available
Sleep	Available	Available	Available	Available
Deep Sleep	Available	Available	N/A	Available
Software Standby	N/A	Available	N/A	Available
Deep Software Standby	N/A	Available	N/A	Available

2.14.2 Connecting Sequence and JTAG/SWD Authentication

This section describes how to authenticate the MCU. [section 2.14.2.1. JTAG/SWD Authentication Mechanism](#) introduces which mechanism is used for authentication, and [section 2.14.2.2. Connecting Sequence and JTAG/SWD Authentication](#) provides steps for connecting sequence and authenticating the MCU using JTAG/SWD. See [section 2.9.5. DBGREG](#) and [section 2.9.6. OCDREG](#) for details about the registers control JTAG/SWD authentication.

2.14.2.1 JTAG/SWD Authentication Mechanism

The MCU supports challenge-response authentication using the boot firmware.

2.14.2.1.1 JTAG Boot Mode and Authentication Request

JTAG boot mode request is asserted if the debugger sets 0xA5 command to JBMDR while RES pin is asserted and MD pin is high when RES pin is deasserted. After entered JTAG boot mode, the debugger can issue authentication procedure by sending Authentication command to JBRDR register.

2.14.2.1.2 Auth Circuit

Auth circuit updates initial Authentication Level (AL) by Protection Level (PL).

When challenge-response authentication is complete, the authentication level is updated by the boot firmware

(1) Initial Authentication Level Update

Auth circuit updates initial authentication level by protection level.

[Table 2.33](#) shows auth circuit update of initial authentication level.

Other than OEM, the initial authentication level is not dependent on the protection level.

Table 2.33 Auth circuit update of initial authentication level (1 of 2)

Input		Output
Lifecycle	Protection level	Authentication level
OEM	PL0	AL0
	PL1	AL1
	PL2	AL2
RMA_REQ	*	AL0

Table 2.33 Auth circuit update of initial authentication level (2 of 2)

Input		Output
Lifecycle	Protection level	Authentication level
RMA_ACK	*	AL2
RMA_RET	*	AL0
LCK_BOOT	*	AL0

(2) Authentication at Challenge-response Authentication

The boot firmware authenticates and writes authentication result to authentication control registers.

2.14.2.2 Connecting Sequence and JTAG/SWD Authentication

Challenge-response authentication

1. Assert RES pin by connecting the OCD debugger to the MCU through the JTAG or SWD interface.
2. Set up SWJ-DP to access DAP bus.
In the setup, the OCD emulator must assert CDBGPWRUPREQ in the SWJDP Control Status Register (CTRL/STAT), and then wait until CDBGPWRUPACK in the same register is asserted.
3. Set the APB-AP to access OCDREG. This APB-AP is connected to DAP bus port 1.
4. Set boot mode request command (0xA5) to JBMDR and set MD pin to 1.
5. Negate RES pin.
6. Check MCUSTAT.AL:
 - If MCUSTAT.AL is higher than or equal to request AL, there is no need to authenticate. Go to step 11..
7. Set Authentication command to JBRDR.
8. Read challenge value from JBTDR.
9. Set the response data to JBRDR.
10. Read JBTDR and check MCUSTAT.AL.
 - If MCUSTAT.AL is not the same as the requested AL, authentication failed
 - If retry authentication: go to step 7..
11. Set up the AHB-APn to access the system address space. The AHB-APn is connected to DAP bus port 0, 2 (n = 0, 1).
12. Start accessing the CPU debug resources using the AHB-APn (n = 0, 1).

For CPU individual reset, FSBL is never executed.

For FSBL execution, IWDT and WDT0/1 function as the settings for OEM selected mode.

2.15 References

- [1] ARM Limited., ARMv8-M Architecture Reference Manual (ARM DDI 0553B)
- [2] ARM Limited., Arm[®] Cortex-M85 Processor Technical Reference Manual (ARM 101924)
- [3] ARM Limited., Arm[®] Cortex-M85 Processor User Guide (ARM 101927)
- [4] ARM Limited., Arm[®] CoreSight SoC-400 Technical Reference Manual (ARM 0302-01 2nd release for r3p2)
- [5] ARM Limited., Arm[®] CoreSight ETM-M85 Technical Reference Manual (ARM 101926)
- [6] ARM Limited., Arm[®] CoreSight Architecture Specification (ARM IHI 0029D)
- [7] ARM Limited., Arm[®] Embedded Trace Macrocell Architecture Specification (ARM IHI 0064H.a)
- [8] ARM Limited., Arm[®] CoreSight SoC-400 User Guide (ARM 100490)
- [9] ARM Limited., Arm[®] Cortex-M33 Processor Technical Reference Manual (ARM 100230)
- [10] ARM Limited., Arm[®] Cortex-M33 Processor User Guide (ARM 100234)

[11] ARM Limited., Arm® CoreSight ETM-M33 Technical Reference Manual (ARM 100232)

[12] ARM Limited., Arm® CoreSight Trace Memory Controller Technical Reference Manual (ARM DDI0461B)

2.16 CM33 CACHE (C-Cache and S-Cache)

2.16.1 Overview

There are two types of caches:

- C-Cache on Cortex-M33 Code AHB interface (C-AHB)
- S-Cache on Cortex-M33 System AHB interface (S-AHB).

2.16.1.1 Specifications of Cache

Table 2.34 Specifications of cache

Item	C-Cache	S-Cache
Capacity	16 KB	16 KB
Way	4-way set associative	4-way set associative
Line size	256 bits	256 bits
Number of entry	128 entries/ways (4-way set associative)	128 entries/ways (4-way set associative)
Write way	Write-through/write-back Non write-allocate/write-allocate	Write-through/write-back Non write-allocate/write-allocate
Replace way	LRU	LRU
Error detection	ECC Data memory: SECDED Tag memory: SEDDED	ECC Data memory: SECDED Tag memory: SEDDED
Cache support area	0x0000_0000 – 0x1FFF_FFFF	0x2000_0000 – 0xDFFF_FFFF

Note: Transient, non-transient and shareability of memory attributes (set by Arm MPU) do not affect cache behavior.

2.16.1.2 Block Diagram

Figure 2.6 shows the block structure of cache.

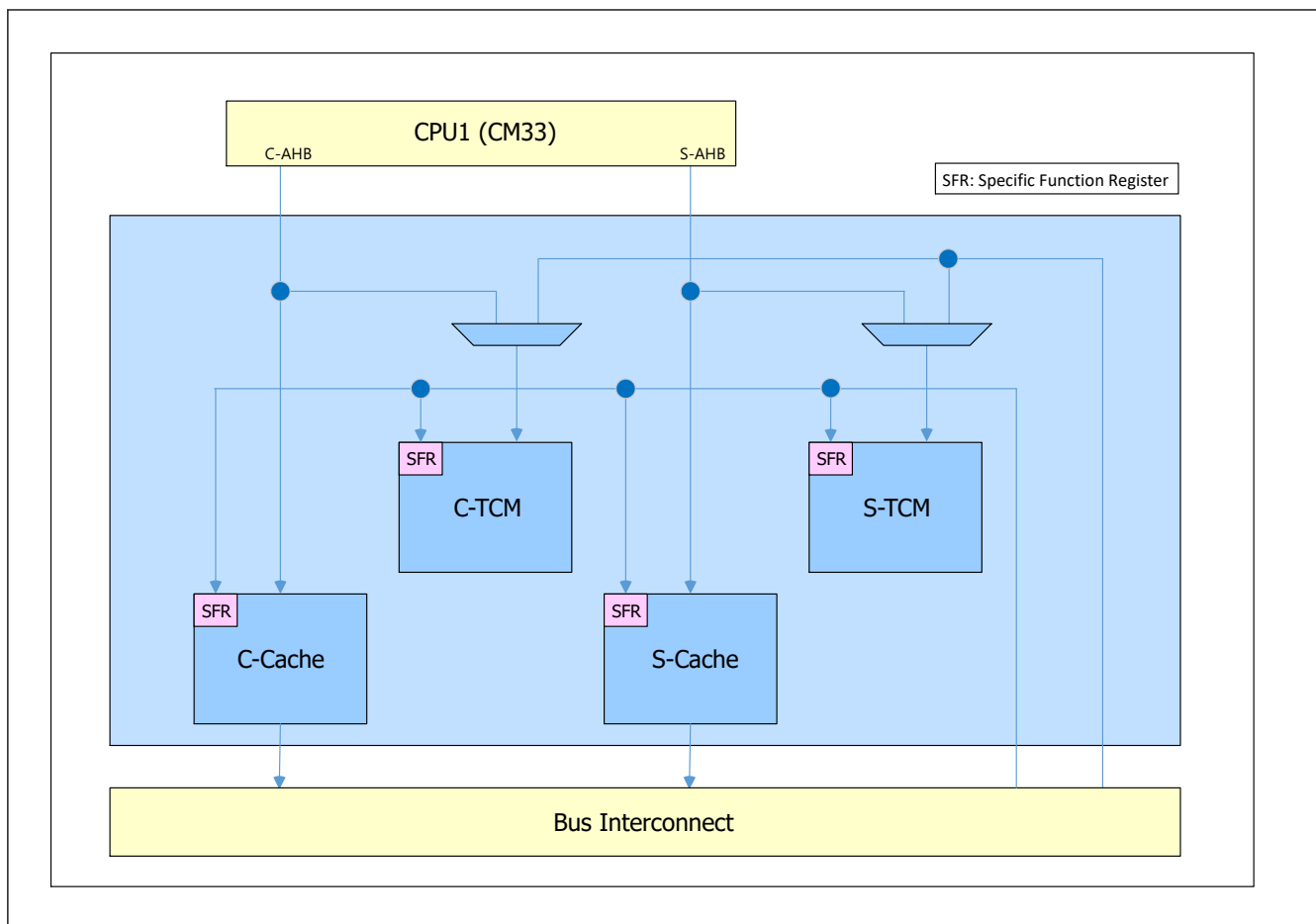


Figure 2.6 Cache block structure

Figure 2.7 shows the detailed cache structure.

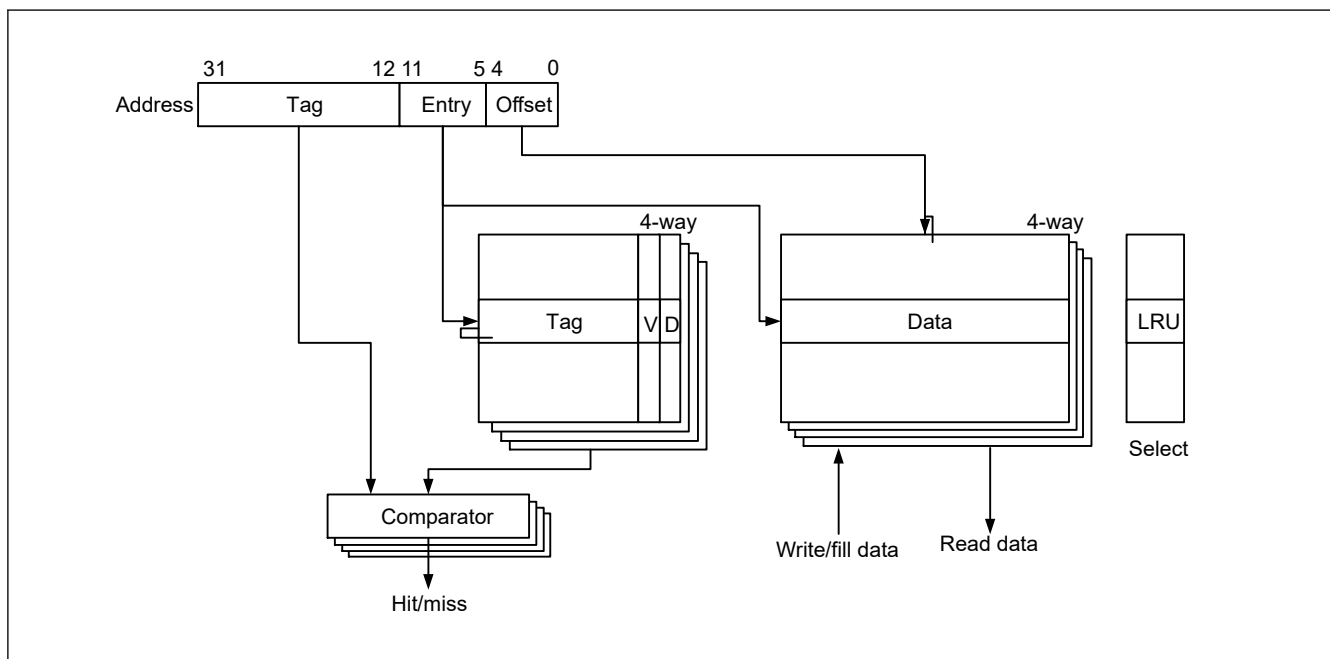


Figure 2.7 Detailed cache structure

Tag

Tag holds the cached address [31:12]. Tag also holds the ECC code for 32 bits with zero-extended address[31:12] and valid bit V.

Tag is cleared by the cache flush function.

V

V is a valid bit and exists in each cache line. It shows the validated data is stored in the cache line.

V bit is cleared by the cache flush function.

D

D is a dirty bit and exists in each cache line. It shows the validated data stored in the cache line is changed from the original value, and requires to be written back to bus.

Data

Data holds 256-bit data in each cache line.

LRU

LRU (Least Recently Used) is a way to replace each cache line.

Comparator

Comparator determines whether the data to be accessed is stored in the cache line.

If V is set and the address equals to the address stored in the tag, the cache determines the access is a hit. Otherwise, it is a miss. If there is an ECC error in the Hit tag memory line, it is treated as a miss.

2.16.2 Register List

Cache has registers which control the cache functions. [Table 2.35](#) shows all the cache related registers.

Table 2.35 Register list

Register name	Symbol	Address (for secure/non-secure access)
Cache Security Attribution Register	CACHESAR	CPSCU_BASE + 0x500
C-Cache Control Register	CCACTL	CACHE_BASE + 0x000
C-Cache Flush Control Register	CCAFCT	CACHE_BASE + 0x004
C-Cache Write Attribute	CCAWTA	CACHE_BASE + 0x00C
C-Cache Error Detection Status	CCAEDST	CACHE_BASE + 0x010
C-Cache Test Access Address	CCATAA	CACHE_BASE + 0x014
C-Cache Test Access Data	CCATAD	CACHE_BASE + 0x018
S-Cache Control Register	SCACTL	CACHE_BASE + 0x040
S-Cache Flush Control Register	SCAFCT	CACHE_BASE + 0x044
S-Cache Write Attribute	SCAWTA	CACHE_BASE + 0x04C
S-Cache Error Detection Status	SCAEDST	CACHE_BASE + 0x050
S-Cache Test Access Address	SCATAA	CACHE_BASE + 0x054
S-Cache Test Access Data	SCATAD	CACHE_BASE + 0x058
Cache ECC Error Operation After Detection Register	CAPOAD	CACHE_BASE + 0x200
Cache Protection Register	CAPRCR	CACHE_BASE + 0x204

CPSCU_BASE (Base address of CPSCU) is 0x4000_8000 (Secure) or 0x5000_8000 (Non-secure).

CACHE_BASE (Base address of Cache) is 0x4001_C000 (Secure) or 0x5001_C000 (Non-secure).

2.16.3 Register Descriptions

2.16.3.1 CACHESAR : Cache Security Attribution Register

Base address: CPSCU = 0x4000_8000
CPSCU_NS = 0x5000_8000

Offset address: 0x500

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	CACH EESA	—	CACH ESA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CACHESA	Security attributes of registers for cache control 0: Secure 1: Non-Secure	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
2	CACHEESA	Security attributes of registers for cache error 0: Secure 1: Non-Secure	R/W
31:3	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-1
Privilege: P-TYPE-1
This register is write-protected by PRPCR_S.PRC4.

Only Secure access can write to this register. Non-secure read access is allowed, non-secure write access is denied, and TrustZone access error is not generated.

CACHESA bit (Security attributes of registers for cache control)

Security attributes of registers for cache control. The target registers are as follows.

- CCACTL
- CCAWTA
- CCAEDST
- SCACTL
- SCAWTA
- SCAEDST

CACHEESA bit (Security attributes of registers for cache error)

Security attributes of registers for cache error. The target registers are as follows.

- CAPOAD
- CAPRCR

CACHESA and CACHEESA must be set to 0 when secure program/secure data is cacheable.

2.16.3.2 CCACTL : C-Cache Control Register

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	WB	FC	—	—	—	—	—	—	—	ENC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENC	C-cache enable bit Set C-Cache enable. 0: C-Cache disable 1: C-Cache enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	FC	C-Cache flush bit Set C-Cache line flush. 0: No action 1: C-Cache line flush (all lines invalidate)	R/W
9	WB	C-cache write-back Trigger to write back C-Cache. 0: No action 1: C-Cache write-back	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHESA)
Privilege: P-TYPE-2

ENC bit (C-cache enable bit)

The ENC bit controls the cache enable of C-Cache.

When C-Cache set to enable, whether it caches data or not depends on the access attribute setting for cacheable of CPU1.

FC bit (C-Cache flush bit)

The FC bit is an alias for CCAFCT.FC. See [section 2.16.3.3. CCAFCT : C-Cache Flush Control Register](#) for details.

WB bit (C-cache write-back)

The WB bit is an alias for CCAFCT.WB. See [section 2.16.3.3. CCAFCT : C-Cache Flush Control Register](#) for details.

2.16.3.3 CCAFCT : C-Cache Flush Control Register

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WB	FC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FC	C-Cache flush bit Set C-Cache line flush. 0: No action 1: C-Cache line flush (all lines invalidate)	R/W
1	WB	C-Cache write-back Trigger to write back C-Cache. 0: No action 1: C-Cache write-back	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-5
Privilege: P-TYPE-2

FC bit (C-Cache flush bit)

The FC bit controls the cache flush of C-Cache.

[Clearing condition]

- This bit is cleared automatically when a cache flush is completed.

[Setting condition]

- Writing 1 to this bit.

WB bit (C-Cache write-back)

The WB bit triggers a write-back of C-Cache.

[Clearing condition]

- This bit is cleared automatically when write back is completed.

[Setting condition]

- Writing 1 to this bit.

Note: Setting FC to 1 when CCAWTA.WT is 0 performs both write back and flush.

If flush or write back has already started (CCAFCT.FC = 1 or CCAFCT.WB = 1), CCAFCT and CCACTL cannot be written until CCAFCT.FC and CCAFCT.WB are cleared automatically, that is, the process completes.

Cache is disabled after reset.

Cache does not perform write back and flush cache memory automatically when cache is disabled or reset.

If there is a dirty line left in the cache while the cache is disabled, inconsistencies can occur between cache and memory.

To disable the cache, use the following procedure.

1. Write 0x0100 (if CCAWTA.WT = 1) or 0x0300 (if CCAWTA.WT = 0) to CCACTL.

2. Make sure that CCAFCT.FC and CCAFCT.WB are both 0.

To enable the cache after reset, use the following procedure.

1. Set CCAFCT.FC to 1 to flush cache memory.
2. Wait for the flush to complete until CCAFCT.FC is 0.
3. Set CCACTL.ENC to 1 to enable the cache.
4. Writing to CCACTL or CCAFCT after reset automatically flushes the cache.

To enable the cache after cache is disabled, use the following procedure.

1. Make sure CCAFCT.WB and CCAFCT.FC is 0, that is, the cache does not perform write back or flush.
2. Set CCACTL.ENC to 1 to enable the cache.

2.16.3.4 CCAWTA : C-Cache Write Attribute

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x00C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WA	WT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	WT	C-Cache write-through Setting the attribute of C-Cache for write-through or write-back. 0: Write-through or write-back depends on the cache attribute of the area 1: Always write-through mode	R/W
1	WA	C-Cache write-allocation Setting the attribute of C-Cache for write-allocation. 0: Always disable write-allocation 1: Write-allocation enable depends on the cache attribute of the area	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHESA)
Privilege: P-TYPE-2

WT bit (C-Cache write-through)

The WT bit specifies the write attribute of C-Cache for write-through or write-back.

C-Cache must be disabled (CCACTL.ENC = 0) when changing this bit value.

When WT is 0, the cache attribute is determined by the default memory map or MPU of the Cortex-M33 processor.

WA bit (C-Cache write-allocation)

The WA bit enables C-Cache write-allocation.

C-Cache must be disabled (CCACTL.ENC = 0) when changing this bit value.

2.16.3.5 CCAEDST : C-Cache Error Detection Status

Base address: CACHE = 0x4001_C000
 CACHE_NS = 0x5001_C000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	EST2	ESTD	ESTC	ESD1	ESD0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESD0	C-Cache data error status 0 Indicates status of whether ECC 1-bit error was detected when reading C-Cache data memory. Write value must be 0. Writing 0 clears this status. 0: ECC 1-bit error was not detected 1: ECC 1-bit error was detected and corrected	R/W
1	ESD1	C-Cache data error status 1 Indicates status of whether ECC 2-bit error was detected when reading C-Cache data memory. Write value must be 0. Writing 0 clears this status. 0: ECC 2-bit error was not detected 1: ECC 2-bit error was detected	R/W
2	ESTC	C-Cache tag clean line invalidate status Indicates status of whether clean line was invalidated by ECC 1-bit error of C-Cache tag memory. Write value must be 0. Writing 0 clears this status. 0: Clean line was not invalidated 1: Clean line was invalidated	R/W
3	ESTD	C-Cache tag dirty line invalidate status Indicates status of whether dirty line was invalidated by ECC 1-bit error of C-Cache tag memory. Write value must be 0. Writing 0 clears this status. 0: Dirty line was not invalidated 1: Dirty line was invalidated	R/W
4	EST2	C-Cache tag 2-bit error status Indicates status of whether ECC 2-bit error was detected when reading C-Cache tag memory. Write value must be 0. Writing 0 clears this status. 0: ECC 2-bit error was not detected 1: ECC 2-bit error was detected	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHESA)
 Privilege: P-TYPE-2

ESD0, ESD1 bit

These bits indicate the status of whether ECC errors were detected when reading C-Cache data memory.

Writing 0 clears this status. Resets other than Bus Error Reset or Memory Error Reset clear this status. If an error occurs and the status is cleared at the same time, the status is cleared, and the error is ignored.

ESTC, ESTD, EST2 bit

These bits indicate the status of whether uncorrectable ECC errors were detected when reading C-Cache tag memory.

ESTC : Invalidate clean line by ECC 1-bit error (correctable)

ESTD : invalidate dirty line by ECC 1-bit error (uncorrectable)

EST2 : Detect ECC 2-bit error (uncorrectable)

Writing 0 clears this status. If an error occurs and the status is cleared at the same time, the status is cleared, and the error is ignored.

If ESTD is 1, dirty line is invalidated without write back.

2.16.3.6 CCATAA : C-Cache Test Access Address

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	WAY[1:0]		—	—	—	—	—	—	RW	—	—	—	—	TARGET[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ENTRY[6:0]							OFFSET[2:0]			—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
4:2	OFFSET[2:0]	Address offset C-Cache test access address lower 3 bits [4:2] corresponds to upper 3 bits of 5-bit offset [4:0]. These bits are valid when cache data read/write and ECC code read/write.	R/W
11:5	ENTRY[6:0]	Address entry C-Cache test access address middle 7 bits [11:5] corresponds to entry.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
18:16	TARGET[2:0]	Access target These bits specify the C-Cache test access target. Access type is specified by bit RW[23]. 0 0 0: Cache data read or write 0 0 1: Data ECC code read or write 0 1 0: Tag, V, D read or write 0 1 1: LRU read or write 1 0 0: Tag ECC code read or write 1 0 1: Setting prohibited 1 1 1: —	R/W
22:19	—	These bits are read as 0. The write value should be 0.	R/W
23	RW	Read write This bit specifies whether C-Cache test access type is read or write. 0: Read 1: Write	R/W
29:24	—	These bits are read as 0. The write value should be 0.	R/W
31:30	WAY[1:0]	Address way C-Cache test access address higher 2 bits corresponds to way. These bits are valid when cache data/ecc code/tag read/write. These bits are invalid when LRU read/write.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

WAY[1:0], ENTRY[6:0], OFFSET[2:0] bit

These bits indicate the address for C-Cache test access.

WAY[1:0], ENTRY[6:0], OFFSET[2:0] are valid for ECC code/cache data read or write.

WAY[1:0], ENTRY[6:0] are valid, and OFFSET[2:0] is invalid for Tag and Tag ECC code read or write.

ENTRY[6:0] is valid and WAY[1:0], OFFSET[2:0] are invalid for LRU read or write.

TARGET[2:0] bits (Access target)

These bits specify the C-Cache test access target. The truth table is below.

RW bit (Read write)

The RW bit specifies whether C-Cache test access type is read or write. The truth table is below.

CCATAA.RW[23]	CCATAA.TARGET[18:16]	Command
0	000	Cache data read
1	000	Cache data write
0	001	Data ECC code read
1	001	Data ECC code write
0	010	Tag read
1	010	Tag write
0	011	LRU read
1	011	LRU write
0	100	Tag ECC code read
1	100	Tag ECC code write

Writing to CCATAA register triggers the test access. During test access writing, write to register CCATAD before writing to register CCATAA. During test access reading, read register CCATAD after writing to register CCATAA.

To perform read access, follow these steps:

1. Write to CCATAA with CCATAA.RW = 0.
2. Read CCATAA to check.
3. Read CCATAD.

To perform write access, follow these steps:

1. Write to CCATDA.
2. Write to CCATAA with CCATAA.RW = 1.

If CCATAD register is updated by the test access reading and the CCATAD register is written at the same time, CCATAD register update by the test access reading is ignored and CCATAD register is written.

2.16.3.7 CCATAD_DATA : C-Cache Test Access Data (DATA)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DATA[31:16]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	DATA[31:0]	C-Cache test access data These bits are valid when C-Cache test access cache data read/write (CCATAA[18:16] = 000).	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6
Privilege: P-TYPE-2

DATA[31:0] bits

These bits indicate each data when C-Cache test access.

Which bits are valid depends on the type of C-Cache test access.

C-Cache test access should be done with C-Cache turned off (CCACTL.ENC = 0).

If CCATAD register is updated by the test access reading and the CCATAD register is written at the same time, CCATAD register update by the test access reading is ignored and CCATAD register is written.

2.16.3.8 CCATAD_ECC : C-Cache Test Access Data (ECC)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	ECC[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	ECC[6:0]	C-Cache test access ECC code These bits are valid when C-Cache test access data ECC code of cache data read/write (CCATAA[18:16] = 001).	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6
Privilege: P-TYPE-2

ECC[6:0] bits

The ECC[6:0] bits indicate each data when C-Cache test access.

Which bits are valid depends on the type of C-Cache test access.

C-Cache test access should be done with C-Cache turned off (CCACTL.ENC = 0).

If CCATAD register is updated by the test access reading and the CCATAD register is written at the same time, CCATAD register update by the test access reading is ignored and CCATAD register is written.

2.16.3.9 CCATAD_TAG : C-Cache Test Access Data (TAG)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TAG[19:4]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TAG[3:0]				—	—	—	—	—	—	—	—	—	—	V	D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	D	C-Cache test access Dirty This bit is valid when C-Cache test access Tag, V, D read/write (CCATAA[18:16] = 010).	R/W
1	V	C-Cache test access Valid This bit is valid when C-Cache test access Tag, V, D read/write (CCATAA[18:16] = 010).	R/W
11:2	—	These bits are read as 0. The write value should be 0.	R/W
31:12	TAG[19:0]	C-Cache test access Tag data These bits are valid when C-Cache test access Tag, V, D read/write (CCATAA[18:16] = 010).	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

TAG[19:0], V, D bits

These bits indicate each data when C-Cache test access.

Which bits are valid depends on the type of C-Cache test access.

C-Cache test access should be done with C-Cache turned off (CCACTL.ENC = 0).

If CCATAD register is updated by the test access reading and the CCATAD register is written at the same time, CCATAD register update by the test access reading is ignored and CCATAD register is written.

2.16.3.10 CCATAD_LRU : C-Cache Test Access Data (LRU)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	LRU[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	LRU[4:0]	C-Cache test access LRU data These bits are valid when C-Cache test access LRU read/write (CCATAA[18:16] = 011).	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

LRU[4:0] bits

The LRU[4:0] bits indicate each data when C-Cache test access.

Which bits are valid depends on the type of C-Cache test access.

C-Cache test access should be done with C-Cache turned off (CCACTL.ENC = 0).

If CCATAD register is updated by the test access reading and the CCATAD register is written at the same time, CCATAD register update by the test access reading is ignored and CCATAD register is written.

2.16.3.11 CCATAD_TAGECC : C-Cache Test Access Data (TAGECC)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	TAG ECC[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	TAG ECC[6:0]	C-Cache test access Tag ECC code These bits are valid when C-Cache test access Tag ECC code read/write (CCATAA[18:16] = 100).	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

TAG ECC[6:0] bits (C-Cache test access Tag ECC code)

These bits indicate each data when C-Cache test access.

Which bits are valid depends on the type of C-Cache test access.

C-Cache test access should be done with C-Cache turned off (CCACTL.ENC = 0).

If CCATAD register is updated by the test access reading and the CCATAD register is written at the same time, CCATAD register update by the test access reading is ignored and CCATAD register is written.

2.16.3.12 SCACTL : S-Cache Control Register

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	WB	FS	—	—	—	—	—	—	—	ENS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ENS	S-Cache enable bit Set S-Cache enable. 0: S-Cache disable 1: S-Cache enable	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
8	FS	S-Cache flush bit Set S-Cache line flush. 0: No action 1: S-Cache line flush (all lines invalidate)	R/W
9	WB	S-Cache write-back Trigger to write back S-Cache. 0: No action 1: S-Cache write-back	R/W
31:10	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHESA)
Privilege: P-TYPE-2

ENS bit (S-Cache enable bit)

The ENS bit controls the cache enable of S-Cache.

When S-Cache set to enable, whether it caches the data or not depends on the access attribute setting of cacheable for CPU1.

FS bit (S-Cache flush bit)

The FS bit is an alias for SCAFCT.FS. See [section 2.16.3.13. SCAFCT : S-Cache Flush Control Register](#) for details.

WB bit (S-Cache write-back)

The WB bit is an alias for SCAFCT.WB. See [section 2.16.3.13. SCAFCT : S-Cache Flush Control Register](#) for details.

2.16.3.13 SCAFCT : S-Cache Flush Control Register

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WB	FS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FS	S-Cache flush bit Set S-Cache line flush. 0: No action 1: S-Cache line flush (all lines invalidate)	R/W
1	WB	S-Cache write-back Trigger to write back S-Cache. 0: No action 1: S-Cache write-back	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-5
Privilege: P-TYPE-2

FS bit (S-Cache flush bit)

The FS bit controls the cache flush of S-Cache.

[Clearing condition]

- This bit is cleared automatically when a cache flush is completed.

[Setting condition]

- Writing 1 to this bit.

WB bit (S-Cache write-back)

The WB bit triggers a write-back of S-Cache.

[Clearing condition]

- This bit is cleared automatically when write back is completed.

[Setting condition]

- Writing 1 to this bit.

Note: Setting FS to 1 when SCAWTA.WT is 0 performs both write back and flush.

If flush or write back has already started (SCAFCT.FS = 1 or SCAFCT.WB = 1), SCAFCT and SCACTL cannot be written until SCAFCT.FS and SCAFCT.WB are cleared automatically, that is, the process completes.

Cache is disabled after reset.

Cache does not write back and flush cache memory automatically when cache is disabled or reset.

If there is a dirty line left in the cache while cache is disabled, inconsistencies can occur between the cache and memory.

To disable the cache, use the following procedure.

1. Write 0x0100 (if SCAWTA.WT = 1) or 0x0300 (if SCAWTA.WT = 0) to SCACTL.

2. Make sure that SCAFCT.FC and SCAFCT.WB are both 0.

To enable the cache after reset, use the following procedure.

1. Set SCAFCT.FS to 1 to flush cache memory.
2. Wait for the flush to complete until SCAFCT.FS is 0.
3. Set SCACTL.ENS to 1 to enable the cache.
4. Writing to SCACTL or SCAFCT after reset automatically flushes the cache.

To enable the cache after cache is disabled, use the following procedure.

1. Make sure SCAFCT.WB and SCAFCT.FS is 0, that is, the cache does not perform write back or flush.
2. Set SCACTL.ENS to 1 to enable the cache.

2.16.3.14 SCAWTA : S-Cache Write Attribute

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x04C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WA	WT
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

Bit	Symbol	Function	R/W
0	WT	S-Cache write-through Setting the attribute of S-Cache for write-through or write-back. 0: Write-through or write-back depends on the cache attribute of the area 1: Always write-through mode	R/W
1	WA	S-Cache write-allocation Setting the attribute of S-Cache write-allocation. 0: Always disable write-allocation 1: Write-allocation enable depends on the cache attribute of the area	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R

Note: Security: S-TYPE-3 (CACHESAR.CACHESA)
Privilege: P-TYPE-2

WT bit (S-Cache write-through)

The WT bit specifies the write attribute of S-Cache for write-through or write-back.

S-Cache must be disabled (SCACTL.ENS = 0) when changing this bit value.

When WT is 0, the cache attribute is determined by the default memory map or MPU of the Cortex-M33 processor.

WA bit (S-Cache write-allocation)

The WA bit enables S-Cache write-allocation.

S-Cache must be disabled (SCACTL.ENS = 0) when changing this bit value.

2.16.3.15 SCAEDST : S-Cache Error Detection Status

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x050

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	EST2	ESTD	ESTC	ESD1	ESD0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESD0	S-Cache data error status 0 Indicates status of whether ECC 1-bit error was detected when reading S-Cache data memory. Write value must be 0. Writing 0 clears this status. 0: ECC 1-bit error was not detected 1: ECC 1-bit error was detected, and corrected	R/W
1	ESD1	S-Cache data error status 1 Indicates status of whether ECC 2-bit error was detected when reading S-Cache data memory. Write value must be 0. Writing 0 clears this status. 0: ECC 2-bit error was not detected 1: ECC 2-bit error was detected	R/W
2	ESTC	S-Cache Tag clean line invalidate status Indicates status of whether clean line was invalidated by ECC 1-bit error of S-Cache tag memory. Write value must be 0. Writing 0 clears this status. 0: Clean line was not invalidated 1: Clean line was invalidated	R/W
3	ESTD	S-Cache Tag dirty line invalidate status Indicates status of whether dirty line was invalidated by ECC 1-bit error of S-Cache tag memory. Write value must be 0. Writing 0 clears this status. 0: Dirty line was not invalidated 1: Dirty line was invalidated	R/W
4	EST2	S-Cache Tag 2-bit error status Indicates status of whether ECC 2-bit error was detected when reading S-Cache tag memory. Write value must be 0. Writing 0 clears this status. 0: ECC 2-bit error was not detected 1: ECC 2-bit error was detected	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHESA)
Privilege: P-TYPE-2

ESD0, ESD1

These bits indicate the status of whether ECC errors were detected when reading S-Cache data memory.

Writing 0 clears this status. Resets other than Bus Error Reset or Memory Error Reset clear this status. If an error occurs and the status is cleared at the same time, the status is cleared, and the error is ignored.

ESTC, ESTD, EST2

These bits indicate the status of whether uncorrectable ECC error were detected when reading S-Cache tag memory.

ESTC : Invalidate clean line by ECC 1-bit error (correctable)

ESTD : Invalidate dirty line by ECC 1-bit error (uncorrectable)

EST2 : Detect ECC 2-bit error (uncorrectable)

Writing 0 clears this status. If an error occurs and the status is cleared at the same time, the status is cleared, and the error is ignored.

If ESTD is 1, dirty line is invalidated without write back.

2.16.3.16 SCATAA : S-Cache Test Access Address

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x054

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	WAY[1:0]		—	—	—	—	—	—	RW	—	—	—	—	TARGET[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ENTRY[6:0]							OFFSET[2:0]			—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
4:2	OFFSET[2:0]	Address offset S-Cache test access address lower 3-bit [4:2] corresponds to upper 3-bit of 5-bit offset [4:0]. These bits are valid when cache data read/write and ECC code read/write.	R/W
11:5	ENTRY[6:0]	Address entry S-Cache test access address middle 7-bit [11:5] corresponds to entry.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
18:16	TARGET[2:0]	Access target These bits specify the S-Cache test access target. Access type is specified by bit RW[23]. 0 0 0: Cache data read or write 0 0 1: Data ECC code read or write 0 1 0: Tag, V, D read or write 0 1 1: LRU read or write 1 0 0: Tag ECC code read or write 1 0 1: Setting prohibited 1 1 1: —	R/W
22:19	—	These bits are read as 0. The write value should be 0.	R/W
23	RW	Read write This bit specifies whether S-Cache test access type is read or write. 0: read 1: write	R/W
29:24	—	These bits are read as 0. The write value should be 0.	R/W
31:30	WAY[1:0]	Address way S-Cache test access address higher 2-bit corresponds to way. These bits are valid when cache data/ecc code/tag read/write. These bits are invalid when LRU read/write.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6
Privilege: P-TYPE-2

WAY[1:0], ENTRY[6:0], OFFSET[2:0] bit

These bits indicate the address when S-Cache test access.

WAY[1:0], ENTRY[6:0], OFFSET[2:0] are valid for ECC code/cache data read or write.

WAY[1:0], ENTRY[6:0] are valid, and OFFSET[2:0] is invalid for TAG and TAG ECC code read or write.

ENTRY[6:0] is valid and WAY[1:0], OFFSET[2:0] are invalid for LRU read or write.

TARGET[2:0] bits

The TARGET[2:0] bits specify the S-Cache test access target. The truth table is below.

RW bit

The RW bit specifies whether S-Cache test access type is read or write. The truth table is below.

SCATAA.RW[23]	SCATAA.TARGET[18:16]	Command
0	000	Cache data read
1	000	Cache data write
0	001	Data ECC code read
1	001	Data ECC code write
0	010	Tag read
1	010	Tag write
0	011	LRU read
1	011	LRU write
0	100	Tag ECC code read
1	100	Tag ECC code write

Writing to SCATAA register triggers the test access. When the test access is writing, write to register SCATAD before writing to register SCATAA. When the test access is reading, read register SCATAD after writing to register SCATAA.

To perform read access, follow these steps:

1. Write to SCATAA with SCATAA.RW = 0.
2. Read SCATAA to check.
3. Read SCATAD.

To perform write access, follow these steps:

1. Write to SCATDA.
2. Write to SCATAA with SCATAA.RW = 1.

If SCATAD register is updated by the test access reading and the SCATAD register is written at the same time, SCATAD register update by the test access reading is ignored and SCATAD register is written.

2.16.3.17 SCATAD_DATA : S-Cache Test Access Data (DATA)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DATA[31:16]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DATA[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	DATA[31:0]	S-Cache test access data These bits are valid when S-Cache test access cache data read/write (SCATAA[18:16] = 000).	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6
Privilege: P-TYPE-2

DATA[31:0] bits

These bits indicate each data when S-Cache test access.

Which bits are valid depend on the type of S-Cache test access.

S-Cache test access should be done with S-Cache turned off (SCACTL.ENS = 0).

If SCATAD register is updated by the test access reading and the SCATAD register is written at the same time, SCATAD register update by the test access reading is ignored and SCATAD register is written.

2.16.3.18 SCATAD_ECC : S-Cache Test Access Data (ECC)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	ECC[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	ECC[6:0]	S-Cache test access ECC code These bits are valid when S-Cache test access data ECC code of cache data read/write (SCATAA[18:16] = 001).	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6
Privilege: P-TYPE-2

ECC[6:0] bits

These bits indicate each data when S-Cache test access.

Which bits are valid depend on the type of S-Cache test access.

S-Cache test access should be done with S-Cache turned off (SCACTL.ENS = 0).

If SCATAD register is updated by the test access reading and the SCATAD register is written at the same time, SCATAD register update by the test access reading is ignored and SCATAD register is written.

2.16.3.19 SCATAD_TAG : S-Cache Test Access Data (TAG)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	TAG[19:4]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TAG[3:0]				—	—	—	—	—	—	—	—	—	—	V	D
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	D	S-Cache test access Dirty This bit is valid when S-Cache test access Tag, V, D read/write (SCATAA[18:16] = 010).	R/W
1	V	S-Cache test access Valid This bit is valid when S-Cache test access Tag, V, D read/write (SCATAA[18:16] = 010).	R/W
11:2	—	These bits are read as 0. The write value should be 0.	R/W
31:12	TAG[19:0]	S-Cache test access Tag data These bits are valid when S-Cache test access Tag, V, D read/write (SCATAA[18:16] = 010).	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

TAG, V, D bits

These bits indicate each data when S-Cache test access.

Which bits are valid depend on the type of S-Cache test access.

S-Cache test access should be done with S-Cache turned off (SCACTL.ENS = 0).

If SCATAD register is updated by the test access reading and the SCATAD register is written at the same time, SCATAD register update by the test access reading is ignored and SCATAD register is written.

2.16.3.20 SCATAD_LRU : S-Cache Test Access Data (LRU)

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	LRU[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	LRU[4:0]	S-Cache test access LRU data These bits are valid when S-Cache test access LRU read/write (SCATAA[18:16] = 011).	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

LRU[4:0] bits

The LRU[4:0] bits indicate each data when S-Cache test access.

Which bits are valid depend on the type of S-Cache test access.

S-Cache test access should be done with S-Cache turned off (SCACTL.ENS = 0).

If SCATAD register is updated by the test access reading and the SCATAD register is written at the same time, SCATAD register update by the test access reading is ignored and SCATAD register is written.

2.16.3.21 SCATAD_TAGECC : S-Cache Test Access Data

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	TAGECC[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	TAGECC[6:0]	S-Cache test access Tag ECC code These bits are valid when S-Cache test access Tag ECC code read/write (CCATAA[18:16] = 100).	R/W
31:7	—	These bits are read as 0. The write value should be 0.	R/W

Note: It is necessary to write by word (32-bit) access. Half word/byte write access is prohibited.

Note: Security: S-TYPE-6

Privilege: P-TYPE-2

TAGECC[6:0] bits (S-Cache test access Tag ECC code)

The TAGECC[6:0] bits indicate each data when S-Cache test access.

Which bits are valid depend on the type of S-Cache test access.

S-Cache test access should be done with S-Cache turned off (SCACTL.ENC = 0).

If SCATAD register is updated by the test access reading and the SCATAD register is written at the same time, SCATAD register update by the test access reading is ignored and SCATAD register is written.

2.16.3.22 CAPOAD : Cache ECC Error Operation After Detection Register

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x200

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	E1STS EN	ECCM OD1	—	—	OAD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OAD	Operation after detection bit Selects the operation after error detection. 0: Interrupt 1: Reset	R/W
2:1	—	These bits are read as 0. The write value should be 0.	R/W
3	ECCMOD1	ECC enable Enable ECC (error correction code) on C-Cache and S-Cache. 0: Disable ECC 1: Enable ECC	R/W
4	E1STSEN	ECC correctable error information update enable Correctable ECC error is: 1-bit ECC error of Cache Data memory - Clean line of Cache Tag memory was invalidated by ECC 1-bit error 0: Disable updating of correctable ECC error information 1: Enable updating of correctable ECC error information	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHEESA)
Privilege: P-TYPE-2

The CAPOAD register is write-protected by CAPRCR. Set CAPRCR.PRCR before writing to this register. Do not write to this register when the cache is enabled.

OAD bit (Operation after detection bit)

The OAD bit specifies the operation after ECC error detection.

ECCMOD1 bit (ECC enable)

The ECCMOD1 bit enables detecting and correcting error with ECC on C-Cache and S-Cache.

E1STSEN bit (ECC correctable error information update enable)

The E1STSEN bit enables or disables updating of the correctable ECC error status CCAEDST.ESD0/SCAEDST.ESD0/CCAEDST.ESTC/SCAEDST.ESTC.

When CAPOAD.E1STSEN = 0, even if a correctable ECC error occurs, the error status is not updated and no actions such as interrupt or reset are taken after an error is detected, regardless of the value of CAPOAD.OAD.

2.16.3.23 CAPRCR : Cache Protection Register

Base address: CACHE = 0x4001_C000
CACHE_NS = 0x5001_C000

Offset address: 0x204

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	KW[6:0]							PRCR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PRCR	Register Write Control bit Set Register Write Control 0: Writing to the protected register is disabled 1: Writing to the protected register is enabled	R/W
7:1	KW[6:0]	Write Key Code bit These bits are used to enable or disable writing of the PRCR bit.	R/W
31:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: Security: S-TYPE-3 (CACHESAR.CACHEESA)
Privilege: P-TYPE-2

PRCR bit (Register Write Control bit)

The PRCR bit controls the writing of the CAPOAD register.

When the PRCR bit is set to 1, writing to the CAPOAD register is enabled.

When writing to this bit, write 0x78 in KW[6:0] at the same time.

KW[6:0]bits (Write Key Code bit)

The KW[6:0] bits are used to enable or disable writing of the PRCR bit.

When writing to PRCR bit, write 0x78 in KW[6:0] at the same time.

When the value of KW[6:0] bits except 0x78 is written in, PRCR bit is not updated.

The KW[6:0] bits are always read as 0x00.

2.16.4 Operation

2.16.4.1 C-Cache/S-Cache

Figure 2.8 shows the function flow to access C-Cache/S-Cache from CPU.

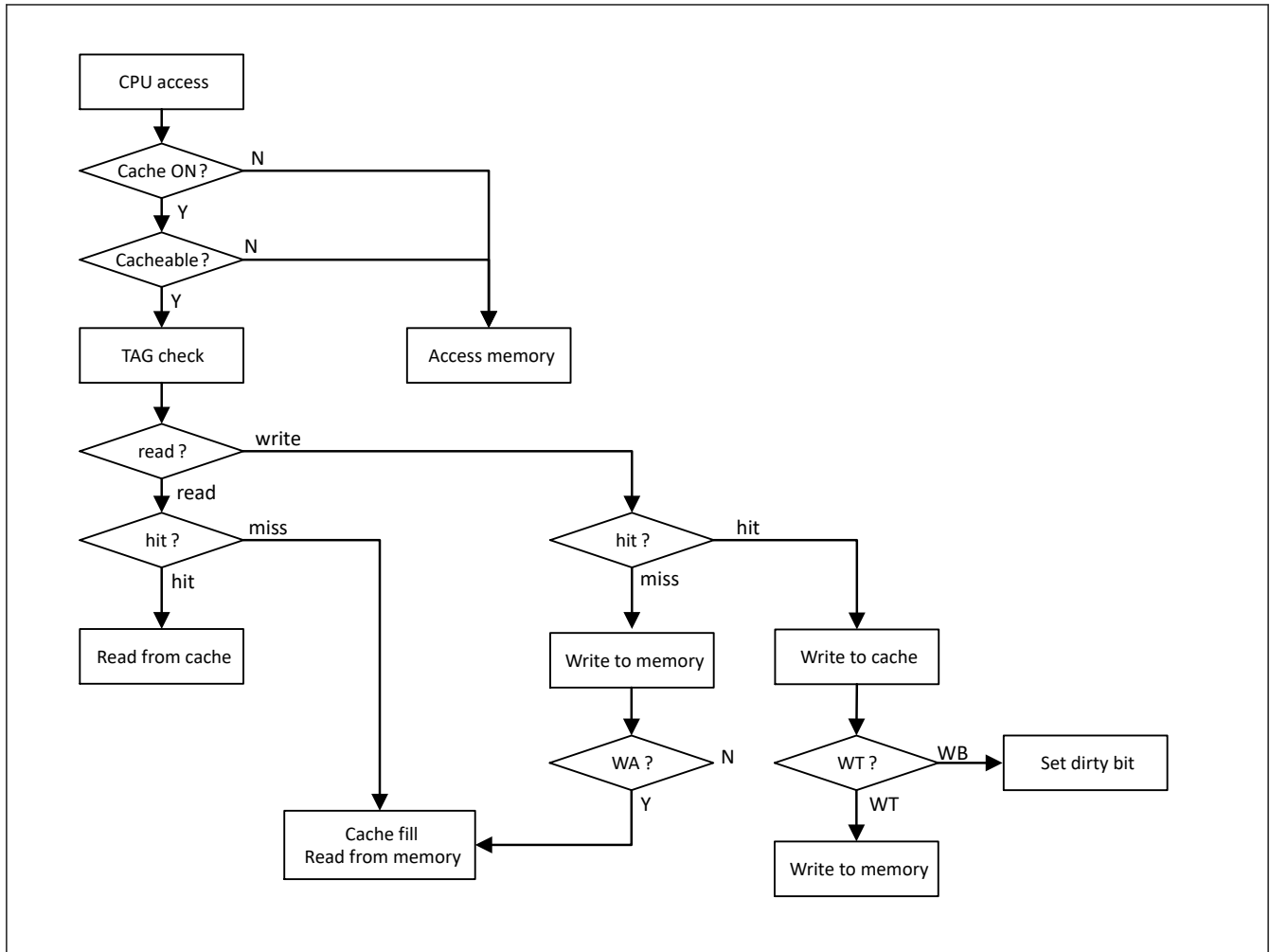


Figure 2.8 CPU access function flow of C-Cache/S-Cache

The cache function works when the cache is enabled (CCACTL.ENC = 1, SCACTL.ENS = 1) and cacheable access from the CPU. Cache checks the address from which the CPU access request and the one in the cache tag, and determines that the CPU access is a hit or miss.

- Read miss: Cache reads one cache line data from memory and stores it into the cache data. Then the cache returns the required data to the CPU.
- Read hit: Cache reads required data from the cache data and returns to the CPU. Then the access cycle is determined as a hit of zero-wait cycle.
- Write miss: Cache writes data to memory. When write-allocation is set (CCAWTA.WA = 1, SCAWTA.WA = 1) and the attribute of area is write-allocation, the cache fills a line from memory.
- Write hit: Cache writes the data to the cache. When write-through is set (CCAWTA.WT = 1, SCAWTA.WT = 1) or attribute of area is write-through, the cache writes the data to memory. However, when write-through is not set (CCAWTA.WT = 0, SCAWTA.WT = 0) and attribute of area is write-back, the cache sets the Dirty bit of the data in Tag.

2.16.4.2 Cache Flush

In order to flush the cache, set 1 to CCAFCT.FC and SCAFCT.FS bits. The Valid bit, Dirty bit and Tag data in Tag are then cleared. The Tag ECC code is initialized to the code generated by the value 0. Cache data is not cleared.

Note: If the same address is accessed with different cacheability (set by the Arm MPU), write back and flush the cache before any access after cacheability changes. If CCAWTA.WT and SCAWTA.WT are set to 1, no write back is required.

If the same address is accessed with different cacheability without write back and flush, the read/write data is undefined.

2.16.4.3 LRU and Replace

Cache uses LRU (Least Recently Used) mechanism as cache replacement algorithm. If a CPU access is determined as a hit or miss, the cache replaces the cache data which is not the last restored. The cache also marks as the latest data in LRU of the cache data. Therefore, even if cache line in cache ways are full, the cache can replace the cache data using LRU that shows the older cache.

2.16.4.4 ECC

2.16.4.4.1 Correction of ECC Errors of Cache Data Memory

Enabling and disabling of ECC error correction can be selected through the setting by the CAPOAD.ECCMOD1 bit. In the initial state, ECC error correction is disabled. The ECC check type is SEC-DED (Single-Error Correction and Double-Error Detection Code).

When ECC function is enabled, 7-bit ECC is generated from 32-bit data, 32-bit data and 7-bit ECC code are written to the cache data memory. For reading, 32-bit data and 7-bit ECC code are read out from the cache data memory.

When ECC function is enabled, if a 1-bit error occurs, error correction is performed and the CCAEDST.ESD0 or SCAEDST.ESD0 bit is set to 1, if the CAPOAD.E1STSEN bit is 1. If a 2-bit error occurs, error detection is performed and the CCAEDST.ESD1 or SCAEDST.ESD1 bit is set to 1, though error correction is not performed.

When ECC error is detected in cache data, it only performs error correction and CCAEDST, SCAEDST setting. It does not invalidate the cache like Tag ECC error.

When ECC function is disabled, error correction or error detection is not performed even though 1-bit or 2-bit error occurs. So the CCAEDST.ESD0 bit, the CCAEDST.ESD1 bit, the SCAEDST.ESD0 bit and the SCAEDST.ESD1 bit are not updated.

2.16.4.4.2 ECC Errors of Cache Tag Memory

Enabling and disabling of Tag ECC error detection can be selected through the setting by the CAPOAD.ECCMOD1 bit (the same as cache data). The ECC check type is SED-DED (Single-Error Detection and Double-Error Detection Code).

When ECC function is enabled, 7-bit ECC code is generated from 32-bit data (upper bit of tag/V extended 0), tag data and 7-bit ECC code are written to the cache tag memory. For read, tag data and 7-bit ECC code are read out from the cache tag memory.

When ECC function is disabled, no error detection is performed even though 1-bit or 2-bit error occurs. So, the CCAEDST.ESTC bit, the CCAEDST.ESTD bit, the CCAEDST.EST2 bit, the SCAEDST.ESTC and the SCAEDST.ESTD bits, and the SCAEDST.EST2 bit are not updated.

If Tag ECC error is detected, the line is invalidated and treated as a miss. For clean line error, the cache can be recovered by cache refilling. However, for dirty line error, the system cannot be recovered because dirty data cannot write back.

2.16.4.4.3 ECC Error Interrupt Function

When ECC function is enabled, error checking is possible in the cache tag/data memory.

For a 2-bit error of the cache data, the CCAEDST.ESD1 bit or the SCAEDST.ESD1 bit is set to 1. For a 1-bit error of the cache data, the CCAEDST.ESD0 bit or the SCAEDST.ESD0 bit is set to 1 when CAPOAD.E1STSEN bit is 1.

For a 2-bit error of the cache tag, the CCAEDST.EST2 bit or the SCAEDST.EST2 bit is set to 1. For invalidate of the cache tag dirty line by ECC 1-bit error, the CCAEDST.ESTD bit or the SCAEDST.ESTD bit is set to 1.

For invalidate of the cache tag clean line by ECC 1-bit error, the CCAEDST.ESTC bit or the SCAEDST.ESTC bit is set to 1 when CAPOAD.E1STSEN bit is 1.

When the correctable error is to be masked, it is necessary to set the CAPOAD.E1STSEN bit to 0 to disable updating the following bits to 1.

- CCAEDST.ESD0
- CCAEDST
- ESTC
- SCAEDST.ESD0

- SCAEDST.ESTC

An ECC error is not generated while ECC function is disabled

ECC error may result request of a reset or an interrupt. When an ECC error occurs and when the CAPOAD.OAD bit is 1, a reset request is output. When it is 0, an interrupt request is output to the ICU.

ECC Error interrupt occurs when there is at least one error status in the CCAEDST register and the SCAEDST register is set to 1. ECC Error interrupt continues to occur until all error status in the CCAEDST register and SCAEDST register are cleared.

2.16.4.4.4 ECC Decoder Testing

There are six ECC decoders in total, one for each way of the cache tag, and one each for the CPU read path and write-back path of cache data.

To test the ECC decoder in cache data, perform the flow of CPU read ([section 2.16.4.4.4.1. ECC Decoder of CPU Read](#)) and the flow of write-back ([section 2.16.4.4.4.2. ECC Decoder of Write-back](#)). To test the ECC decoder in cache tag, perform all of Way0 through Way3 of tag in the flow of CPU read ([section 2.16.4.4.4.1. ECC Decoder of CPU Read](#)).

Since the ECC decoder for cache data is shared between Way0 and Way3, it is possible to test the ECC decoder without testing both ways. For reference, both ways test are described.

2.16.4.4.4.1 ECC Decoder of CPU Read

[Figure 2.9](#) shows the testing flow of cache ECC decoder for CPU reading from the cache or decoder of cache tag.

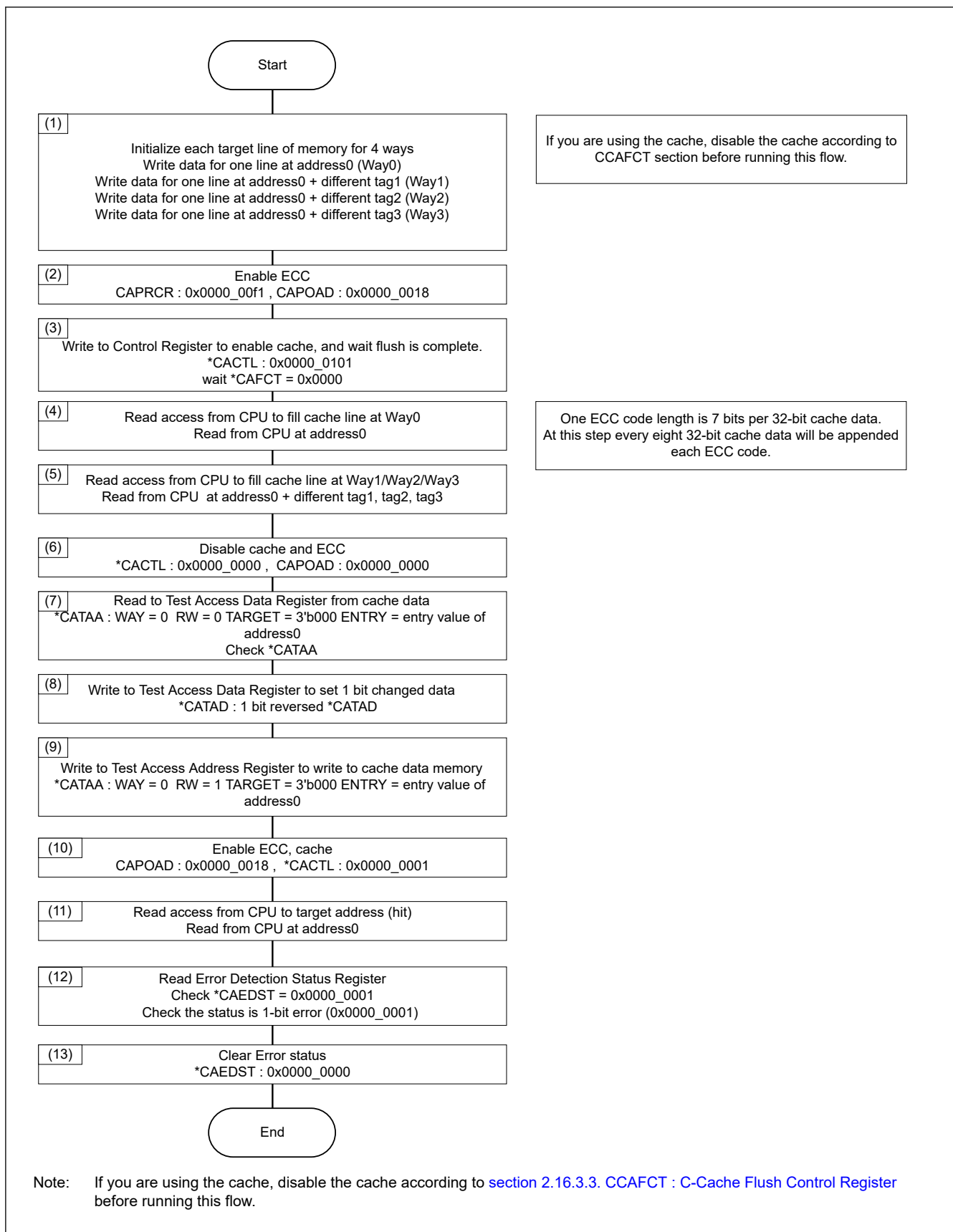


Figure 2.9 Test flow of cache ECC decoder for CPU reading from decoder of cache tag

Notes for each step in the operation flow are described as follows.

(1) To initialize the target line, write data at the address that offset value is from 0x00 to 0x1F.

To initialize data for Way1, write at address0 + different tag1 address.

Initialize data for Way2 and Way3 as well.

For example: address0 = 0x2100_0000

address0 + different tag1 = 0x2101_0000

address0 + different tag2 = 0x2102_0000

address0 + different tag3 = 0x2103_0000

(7)(9) The flow for these steps is about ECC decoder in Way0 of the cache data. For ECC decoders test except write-back ECC decoder (see [section 2.16.4.4.2. ECC Decoder of Write-back](#)), repeat steps (1) to (13) with the following item combination value written to *CATAA.

	*CATAA.TARGET
Data	000
Data ECC	001
Tag	010
Tag ECC	100

	*CATAA.WAY
Way0	00
Way1	01
Way2	10
Way3	11

Test all of Way0 through Way3 in TARGET = 3'b010 or 3'b100, because Tag ECC decoders are in each of all ways.

(8) To test 2-bit ECC error detection, write 2 bits reversed *CATAD.

(11) Read from CPU at address0 + different tag1 when testing Way1.

Read from CPU at address0 + different tag2 when testing Way2.

Read from CPU at address0 + different tag3 when testing Way3.

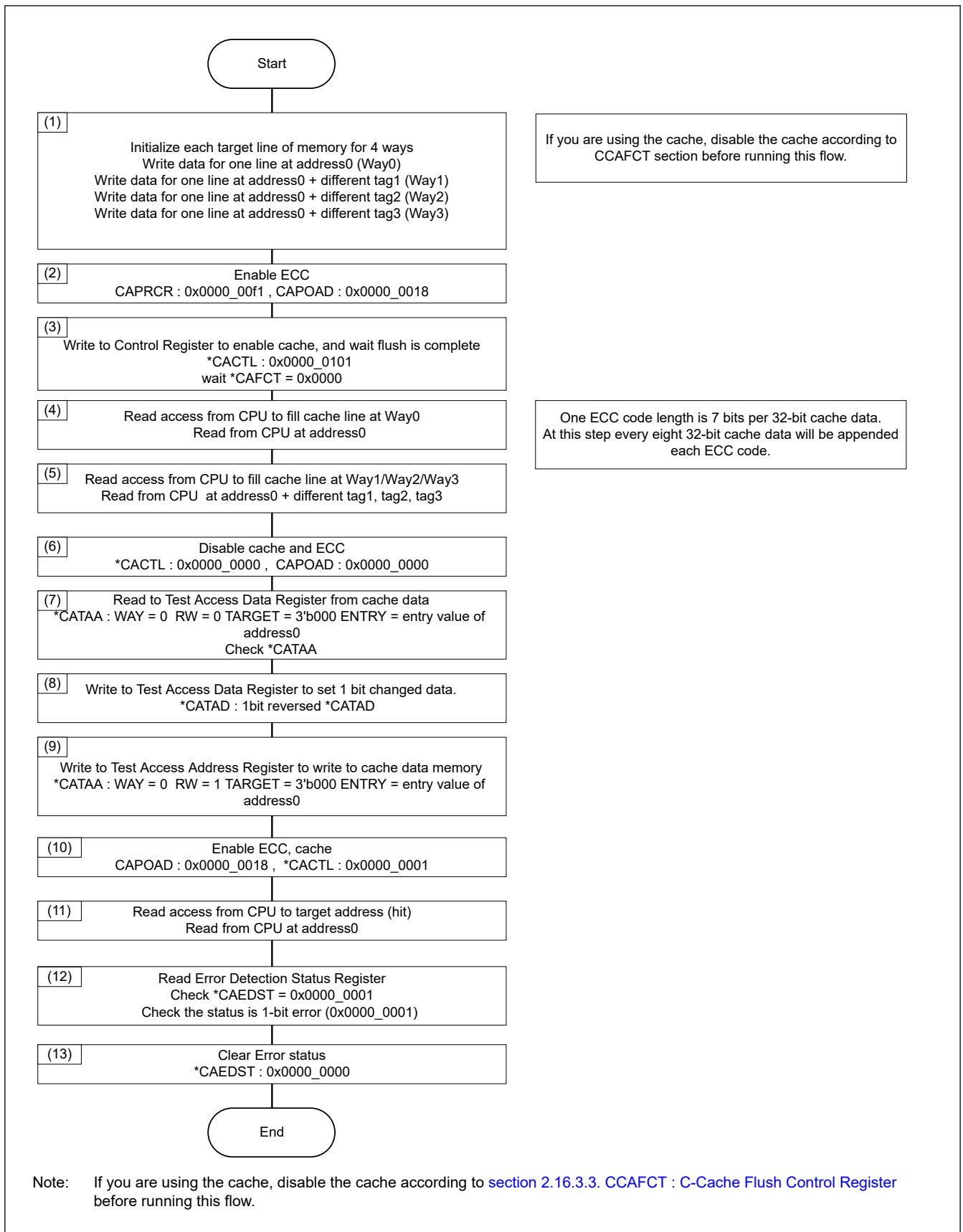
(12) To test 1-bit ECC error correction, check that the read value has been corrected.

Check *CAEDST is of the following value in each test.

Target	Error	Value of *CAEDST
Data	ECC 1-bit error	0x0000_0001
	ECC 2-bit error	0x0000_0002
Data ECC	ECC 1-bit error	0x0000_0001
	ECC 2-bit error	0x0000_0002
Tag	ECC 1-bit error	0x0000_0004 (in this flow, the cache line is clean)
	ECC 2-bit error	0x0000_0010
Tag ECC	ECC 1-bit error	0x0000_0004 (in this flow, the cache line is clean)
	ECC 2-bit error	0x0000_0010

2.16.4.4.4.2 ECC Decoder of Write-back

[Figure 2.10](#) shows the testing flow of cache ECC decoder for write-back.

**Figure 2.10 Test flow of cache ECC decoder for write-back**

Notes for each step in the operation flow are described as follows.

(1) To initialize the target line, write data at the address that offset value is from 0x00 to 0x1F.

To initialize data for Way1, write at address0 + different tag1 address.

Initialize data for Way2 and Way3 as well.

For example: address0 = 0x2100_0000

address0 + different tag1 = 0x2101_0000

address0 + different tag2 = 0x2102_0000

address0 + different tag3 = 0x2103_0000

(7)(9) The flow for these steps is about ECC decoder in Way0 of the cache data. For decoder in write-back, repeat steps (1) to (15) with the following item combination value written to *CATAA.

	*CATAA.TARGET
Data	000
Data ECC	001

	*CATAA.WAY
Way0	00
Way1	01
Way2	10
Way3	11

(8) To test 2-bit ECC error detection, write 2 bits reversed *CATAD.

(10)(11) The flow for these steps is about ECC decoder in Way0 of the cache data. For decoder in path of other ways, repeat steps (1) to (15) with the following value of Way0 through Way3.

	Way0	Way1	Way2	Way3
*CATAA.WAY	00	01	10	11
*CATAD of (11)	0x0000_0014	0x0000_0009	0x0000_001E	0x0000_0013

(13) Different tag (not in cache) has a different tag address from each Way address in (1).

For example: address0 + different tag (not in cache) = 0x2110_0000

(14) To test 1-bit ECC error, check that the read value at address0 (write-back value) has been corrected.

To test 2-bit ECC error, check that the read value at address0 (write-back value) has not been corrected.

Check *CAEDST is of the following value.

Target	Error	*CAEDST
Data	ECC 1-bit error	0x0000_0001
	ECC 2-bit error	0x0000_0002
Data ECC	ECC 1-bit error	0x0000_0001
	ECC 2-bit error	0x0000_0002

2.16.4.4.5 Note of ECC

When switching to enable ECC, follow the procedure ([section 2.16.3.3. CCAFCT : C-Cache Flush Control Register](#) and [section 2.16.3.13. SCAFCT : S-Cache Flush Control Register](#)) to disable C-Cache and S-Cache before switching. Switching to enable ECC while the cache is enabled can cause ECC errors due to incorrect ECC code.

2.16.4.5 Bus Error

The correspondence to a bus error from a bus slave is shown in the section that follows:

- For cache off: Cache returns the bus error to the CPU.
- For non-cacheable access: Cache returns the bus error to the CPU.

- For cache fill: Cache returns the bus error to the CPU. The cache does not write the line contains error to cache memory.
- For write hit status: Cache cannot return the bus error to the CPU. The cache does not clear validation.
- For write miss status: Cache cannot return the bus error to the CPU.

2.16.5 Usage Notes

2.16.5.1 Coherency

The coherency between the cache and the external memory and internal SRAM must be guaranteed by software. To share memory between multiple bus masters (such as between multiple CPUs or between CPU and DMAC) in the cache support area, perform write back and flush the cache data as needed.

2.16.5.2 Cache Flush/WB Operation

When CPU1 operates Cache Flush/WB by *CAFCT/*CACTL, the CPU1 must check that Flush/WB completes (*CAFCT = 0, * denotes C-Cache, or S-Cache, that is, CCAFCT/CCACTL or SCAFCT/SCACTL).

Do not operate Flush/WB from a bus master other than CPU1.

When Cache Flush/WB operation is executed, the priority of Cache Flush/WB operation is higher than that of access from CPU1.

2.16.5.3 Restrictions Relating to Security Attribution of C-Cache and S-Cache

When using C-Cache and S-Cache, one of the following conditions must be observed:

- CPU1 should be used only in secure state.
- CPU1 should be used only in non-secure state. It is only possible if SECEXT (Security Extension) of CPU1 is disabled.
- C-Cache and S-Cache should be used only in non-secure program. In addition, all of the following conditions must be observed:
 1. Set the MPU so that all secure memory space used by the secure program is non-cacheable.
 2. When secure program accesses non-secure memory space, execute WB and Flush of the cache before accessing it.
 3. When using on-chip WDT instead of an external WDT, the on-chip WDT should be set as a secure attribution and also set to apply interrupt to CPU0 or reset when a WDT error occurs.

2.17 CM33 TCM (Tightly Coupled Memory)

2.17.1 Overview

The MCU provides an on-chip TCM (tightly coupled memory) with Error Correction Code (ECC).

2.17.1.1 Specifications of TCM

Table 2.36 Specifications of TCM

Item	C-TCM	S-TCM
Capacity	64 KB	64 KB
Error detection	ECC (SECEDED)	ECC (SECEDED)
Security	Secure region/Non-secure region	Secure region/Non-secure region

Table 2.37 Address map of TCM (1 of 2)

Address	Usage	Master	SA
0x0000_0000 to 0x0000_FFFF	C-TCM	CPU1 (CM33)	Secure
0x1000_0000 to 0x1000_FFFF	C-TCM	CPU1 (CM33)	Non-secure

Table 2.37 Address map of TCM (2 of 2)

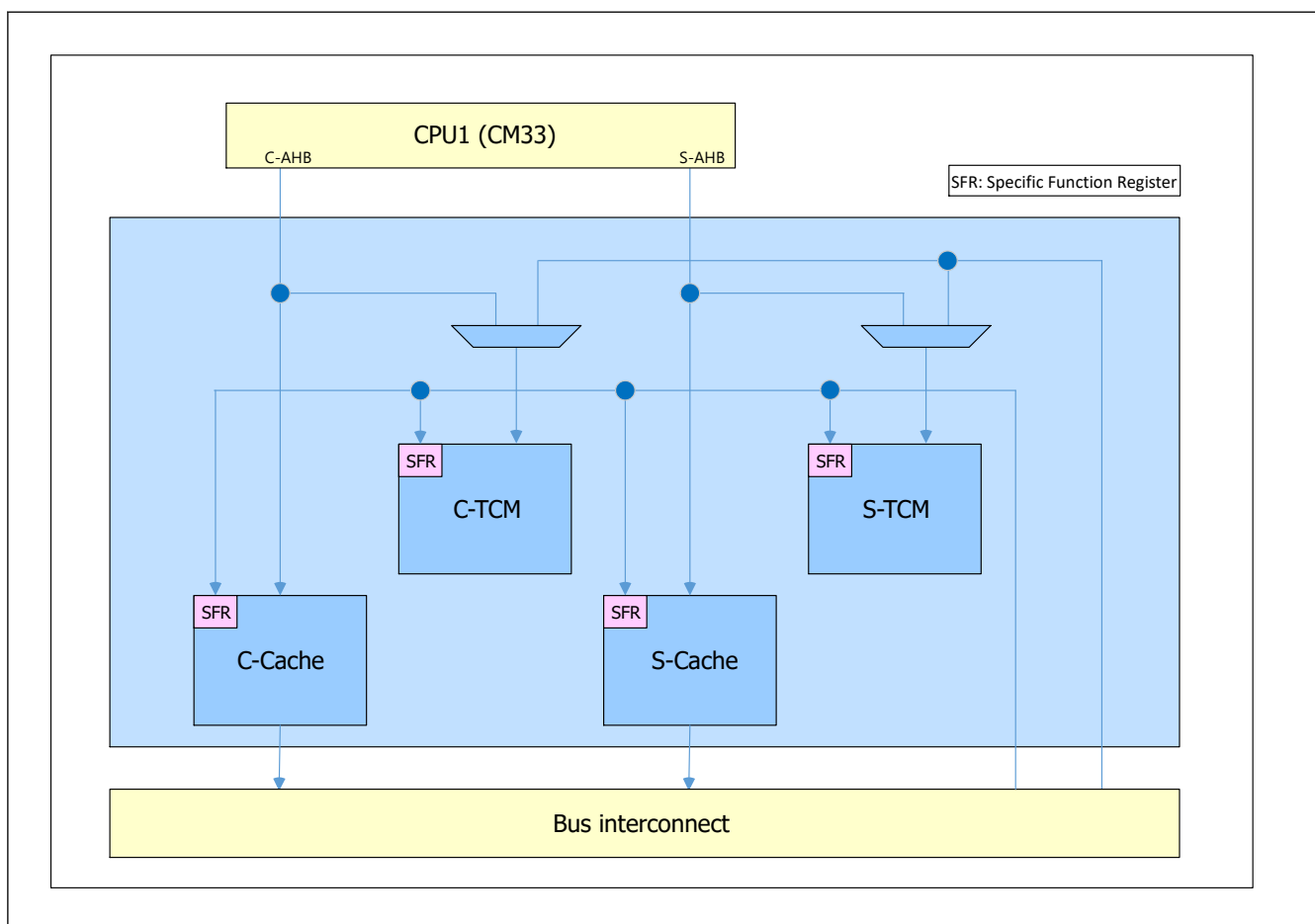
Address	Usage	Master	SA
0x2000_0000 to 0x2000_FFFF	S-TCM	CPU1 (CM33)	Secure
0x2A00_0000 to 0x2A00_FFFF	C-TCM	Bus master (example, CPU0)	Secure
0x2A01_0000 to 0x2A01_FFFF	S-TCM	Bus master (example, CPU0)	Secure
0x3000_0000 to 0x3000_FFFF	S-TCM	CPU1 (CM33)	Non-secure
0x3A00_0000 to 0x3A00_FFFF	C-TCM	Bus master (example, CPU0)	Non-secure
0x3A01_0000 to 0x3A01_FFFF	S-TCM	Bus master (example, CPU0)	Non-secure

Each C-TCM or S-TCM memory is one entity which capacity is 64 KB can be accessed from different master through different address.

If the access of bus master conflicts with the access of CPU1, the bus interface of TCM returns a wait cycle to the bus. See [section 2.17.4.4. Wait Cycle](#).

2.17.1.2 Block Diagram

[Figure 2.11](#) shows the block structure of Cache and TCM.

**Figure 2.11 Block structure of Cache and TCM**

2.17.2 Register List

TCM_BASE_S(Base address of TCM) is 0x4001_C800

TCM_BASE_NS(Base address of TCM) is 0x5001_C800

CPSCU_BASE_S (Base address of CPSCU) is 0x4000_8000.

CPSCU_BASE_NS (Base address of CPSCU) is 0x5000_8000.

Security Attribution Register (TCMSAR) is write protected by PRCR.PRCR4. TCMSAR controls the security attributes of all TCM registers. TCMPCR_S and TCMPCR_NS protects TCMCRn from secure and non-secure write accesses.

Table 2.38 Register list

Register name	Address (for secure access)		Address (for non-secure access)	
TCMSAR	CPSCU_BASE_S +	0x504	—	—
TCMSABARC TCMSABARS	CPSCU_BASE_S +	0x508 0x50C	—	—
TCMPCR_S	TCM_BASE_S +	0x00	—	—
TCMPCR_NS	—	—	TCM_BASE_NS +	0x04
TCMCRC TCMCRS	TCM_BASE_S +	0x10 0x14	TCM_BASE_NS +	0x10 0x14
TCMESR	TCM_BASE_S +	0x40	TCM_BASE_NS +	0x40
TCMESCLR	TCM_BASE_S +	0x48	TCM_BASE_NS +	0x48
TCMEARC0 TCMEARC1 TCMEARS0 TCMEARS1	TCM_BASE_S +	0x50 0x54 0x58 0x5C	TCM_BASE_NS +	0x50 0x54 0x58 0x5C

2.17.3 Register Descriptions

2.17.3.1 TCMSAR : TCM Security Attribution Register

Base address: CPSCU = 0x4000_8000

Offset address: 0x504

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TCMSA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	TCMSA	Security attributes of registers for TCM control 0: Secure 1: Non-Secure	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: This register is write-protected by PRCR_S.PRC4.

Security: S-TYPE-1

Privilege: P-TYPE-1

Only secure access can write to this register. Non-secure read access is allowed, non-secure write access is denied, TrustZone access error is not generated.

TCMSA bit (Security attributes of registers for TCM control)

Security attributes of registers for TCM control. The target registers are as follow.

- TCMCRx
- TCMECLR

The TCMSA bit must be set to 0 when secure program/secure data is located at TCM.

2.17.3.2 TCMSABARx : TCM Security Attribute Boundary Address Register (x = C or S)

Base address: CPSCU = 0x4000_8000

Offset address: 0x508 or 0x50C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	TCMSABA		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TCMSABA			—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	—	These bits are read as 0. The write value should be 0.	R/W
18:13	TCMSABA	Boundary address between secure and non-secure (start address of non-secure region).	R/W
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note: This register is write-protected by PRCR_S.PRC4.

Security: S-TYPE-1

Privilege: P-TYPE-1

TCMSABARx specifies the boundary address between the secure and non-secure regions of each TCM. TCMSABARx indicates the start address of non-secure regions. TCMSABARx 32-bit value corresponds to the absolute address. When writing to TCMSABARx, the write value of bits [31:19] and bits [12:0] should be 0.

The region lower than the boundary address is marked as Secure, and the region higher than or equal to the boundary address is marked as Non-secure.

TCMSABARC

When the boundary address is 0x0000_0000, the whole C-TCM is marked as Non-secure.

When the boundary address is 0x0001_0000 or higher, the whole C-TCM is marked as Secure.

Since the initial boundary address is 0x0007_E000, the whole of C-TCM (64 KB) is marked as Secure after reset.

The boundary address setting affects not only the CPU1 address but also the bus master address (alias address).

Table 2.39 Address map of C-TCM

Address	Master	SA
0x0000_0000 to 0x0000_0000 + TCMSABARC - 0x1	CPU1 (CM33)	Secure
0x0000_0000 + TCMSABARC to 0x0000_FFFF	CPU1 (CM33)	Non-secure
0x1000_0000 to 0x1000_0000 + TCMSABARC - 0x1	CPU1 (CM33)	Secure
0x1000_0000 + TCMSABARC to 0x1000_FFFF	CPU1 (CM33)	Non-secure
0x2A00_0000 to 0x2A00_0000 + TCMSABARC - 0x1	Bus master	Secure
0x2A00_0000 + TCMSABARC to 0x2A00_FFFF	Bus master	Non-secure
0x3A00_0000 to 0x3A00_0000 + TCMSABARC - 0x1	Bus master	Secure
0x3A00_0000 + TCMSABARC to 0x3A00_FFFF	Bus master	Non-secure

TCMSABARS

When the boundary address is 0x0000_0000, the whole S-TCM is marked as Non-secure.

When the boundary address is 0x0001_0000 or higher, the whole S-TCM is marked as Secure.

Since the initial boundary address is 0x0007_E000, the whole of S-TCM (64 KB) is marked as Secure after reset. The boundary address setting affects not only the CPU1 address but also the bus master address (alias address).

Table 2.40 Address map of S-TCM

Address	Master	SA
0x2000_0000 to 0x2000_0000 + TCMSABARS - 0x1	CPU1 (CM33)	Secure
0x2000_0000 + TCMSABARS to 0x2000_FFFF	CPU1 (CM33)	Non-secure
0x2A01_0000 to 0x2A01_0000 + TCMSABARS - 0x1	Bus master	Secure
0x2A01_0000 + TCMSABARS to 0x2A01_FFFF	Bus master	Non-secure
0x3000_0000 to 0x3000_0000 + TCMSABARS - 0x1	CPU1 (CM33)	Secure
0x3000_0000 + TCMSABARS to 0x3000_FFFF	CPU1 (CM33)	Non-secure
0x3A01_0000 to 0x3A01_0000 + TCMSABARS - 0x1	Bus master	Secure
0x3A01_0000 + TCMSABARS to 0x3A01_FFFF	Bus master	Non-secure

2.17.3.3 TCMPCR_S : TCM Protection Control Register for Secure

Base address: TCM = 0x4001_C800

Offset address: 0x00

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KW[7:0]								—	—	—	—	—	—	—	PR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PR	Register Write Control 0: Write registers are disabled 1: Write registers are enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KW[7:0]	Write Key Code Key code protection to the PR bit.	R/W

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

Security: S-TYPE-1

Privilege: P-TYPE-1

PR bit (Register Write Control)

The PR bit controls the write mode of the TCMCRx registers that are marked as Secure by TCMSAR register. When this bit is set to 1, writing to the TCMCRx marked as Secure is enabled.

KW[7:0] bits (Write Key Code)

The KW[7:0] bits enable or disable writes to the PR bit. When writing to the PR bit, write 0xA5 to the KW[7:0] bits simultaneously. When a value other than 0xA5 is written to KW[7:0], the PR bit is not updated. The KW[7:0] bits are always read as 0x00.

2.17.3.4 TCMPCR_NS : TCM Protection Control Register for Non-Secure

Base address: TCM_NS = 0x5001_C800

Offset address: 0x04

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	KW[7:0]								—	—	—	—	—	—	—	PR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PR	Register Write Control 0: Write registers are disabled 1: Write registers are enabled	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W
15:8	KW[7:0]	Write Key Code Key code protection to the PR bit.	R/W

Note: It is necessary to write by half word access.

Byte write access is prohibited. When byte write access is executed, operation is not guaranteed.

Security: S-TYPE-1

Privilege: P-TYPE-1

PR bit (Register Write Control)

The PR bit controls the write mode of the TCMCRx registers that are marked as Non-secure by TCMSAR register. When this bit is set to 1, writing to the TCMCRx marked as Non-secure is enabled.

KW[7:0] bits (Write Key Code)

The KW[7:0] bits enable or disable writes to the PR bit. When writing to the PR bit, write 0xA5 to the KW[7:0] bits simultaneously. When a value other than 0xA5 is written to KW[7:0], the PR bit is not updated. The KW[7:0] bits are always read as 0x00.

2.17.3.5 TCMCRx : TCM Control Register (x = C or S)

Base address: TCM = 0x4001_C800
TCM_NS = 0x5001_C800

Offset address: 0x10 or 0x14

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TSTB YP	—	—	E1STS EN	ECCMOD[1:0]	—	—	OAD
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	OAD	Operation after ECC error detection 0: Interrupt 1: Reset	R/W
1	—	This bit is read as 0. The write value should be 0.	R/W
3:2	ECCMOD[1:0]	ECC Operating Mode Select 0 0: Disable ECC function 0 1: Setting prohibited 1 0: Enable ECC function without error checking 1 1: Enable ECC function with error checking	R/W
4	E1STSEN	ECC 1-Bit Error Information Update Enable 0: Disable updating of 1-bit ECC error information 1: Enable updating of 1-bit ECC error information	R/W

Bit	Symbol	Function	R/W
6:5	—	These bits are read as 0. The write value should be 0.	R/W
7	TSTBYP	ECC Test Enable/ECC Bypass Select 0: Disable ECC bypass 1: Enable ECC bypass	R/W

Security: S-TYPE-3 (TCMSAR.TCMSA)

Privilege: P-TYPE-2

The protection register (TCMPRCR_S or TCMPRCR_NS) protects this register against writing. This register can be written only when the PR bit in the TCMPRCR_S or TCMPRCR_NS register is 1.

Do not write to this register while access to TCM is in progress.

If CPU0 and CPU1 issue speculative access to an uninitialized TCM with ECC enabled, it may trigger an interrupt or reset. After enabling ECC function with error checking, initialize the TCM before setting the interrupt in the IELSRn register or setting the reset in the OAD bit.

OAD bit (Operation after ECC error detection)

The OAD bit selects whether to generate a reset or an interrupt when ECC error is detected.

ECCMOD[1:0] bits (ECC Operating Mode Select)

The ECCMOD[1:0] bits set the access mode to the ECC area.

E1STSEN bit (ECC 1-Bit Error Information Update Enable)

The E1STSEN bit enables or disables updating of the ECC 1-bit error status TCMSR.ERRC0, TCMSR.ERRS0.

When TCMCRx.E1STSEN = 0, the operation after error (such as NMI or reset) is not performed regardless of the value TCMCRx.OAD, even when ECC 1-bit error correction is done.

TSTBYP bit (ECC Test Enable/ECC Bypass Select)

The TSTBYP bit enables direct access to ECC code by bypassing the ECC function.

The ECC bypass function is used along with, setting the ECCMOD[1:0] bits in the same register to 00b.

The ECC code length is 7 bits for 32-bit data. When TSTBYP bit is 1, bits [6:0] of read data is ECC code, bits [31:7] of read data is undefined, bits [6:0] of write data are written to the ECC region of the address, bits [31:7] of write data are ignored.

For details about ECC test, see [section 2.17.4.1. ECC Function](#).

2.17.3.6 TCMSR : TCM Error Status Register

Base address: TCM = 0x4001_C800
TCM_NS = 0x5001_C800

Offset address: 0x40

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	ERRS 1	ERRS 0	ERRC 1	ERRC 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ERRC0	C-TCM 1-bit ECC Error Status 0: 1-bit ECC error has not occurred 1: 1-bit ECC error has occurred	R
1	ERRC1	C-TCM 2-bit ECC Error Status 0: 2-bit ECC error has not occurred 1: 2-bit ECC error has occurred	R

Bit	Symbol	Function	R/W
2	ERRS0	S-TCM 1-bit ECC Error Status 0: 1-bit ECC error has not occurred 1: 1-bit ECC error has occurred	R
3	ERRS1	S-TCM 2-bit ECC Error Status 0: 2-bit ECC error has not occurred 1: 2-bit ECC error has occurred	R
15:4	—	These bits are read as 0.	R

Note: This register is initialized after returning from Software Standby mode while the debugger is connected.

Security: S-TYPE-5

Privilege: P-TYPE-2

The TCMESR register is cleared by the corresponding bit in TCMESCLR register or resets other than Bus Error Reset and Memory Error Reset. Also during access from the debugger, the TCMESR register is not updated.

ERRx0 bit (x-TCM 1-bit ECC Error Status) (x = C or S)

The ERRx0 bit shows whether there is a 1-bit ECC error in x-TCM.

When ECC function with error checking is enabled (TCMCRx.ECCMOD = 2'b11) and, updating of the 1-bit error information is enabled (TCMCRx.E1STSEN = 1), the ERRx0 bit is set to 1 if a 1-bit error is detected. When ERRx0 bit is set to 1, a reset or an interrupt request is generated according to the TCMCRx.OAD.

ERRx1 bit (x-TCM 2-bit ECC Error Status) (x = C or S)

The ERRx1 bit shows whether there is a 2-bit ECC error in x-TCM.

When ECC function with error checking is enabled (TCMCRx.ECCMOD = 2'b11), the ERRx1 bit is set to 1 if a 2-bit error is detected.

When ERRx1 bit is set to 1, a reset or an interrupt request is generated according to the TCMCRx.OAD.

If the status is cleared and an error occurs at the same time, the error is ignored and the status is cleared.

2.17.3.7 TCMESCLR : TCM Error Status Clear Register

Base address: TCM = 0x4001_C800
TCM_NS = 0x5001_C800

Offset address: 0x48

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	CLRS 1	CLRS 0	CLRC 1	CLRC 0

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	CLRC0	C-TCM 1-bit ECC Error Status Clear Writing to the CLRC0 clears TCMESR.ERRC0 and TCMEARC0.EAR. 1: Clear error	R/W ^{*1}
1	CLRC1	C-TCM 2-bit ECC Error Status Clear Writing to the CLRC1 clears TCMESR.ERRC1 and TCMEARC1.EAR. 1: Clear error	R/W ^{*1}
2	CLRS0	S-TCM 1-bit ECC Error Status Clear Writing to the CLRS0 clears TCMESR.ERS0 and TCMEARS0.EAR. 1: Clear error	R/W ^{*1}
3	CLRS1	S-TCM 2-bit ECC Error Status Clear Writing to the CLRS1 clears TCMESR.ERS1 and TCMEARS1.EAR. 1: Clear error	R/W ^{*1}
15:4	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only writing 1 is valid. Writing value is not retained, read value is 0.

Security: S-TYPE-3 (TCMSAR.TCMSA)

Privilege: P-TYPE-2

CLR_x0 bit (x-TCM 1-bit ECC Error Status Clear) (x = C or S)

The CLR_x0 bit can clear 1-bit ECC error status bit in the TCMESR.

TCMESR.ERR_x0 can be cleared when writing 1 to CLR_x0.

CLR_x1 bit (x-TCM 2-bit ECC Error Status Clear) (x = C or S)

The CLR_x1 bit can clear 2-bit ECC error status bit in the TCMESR.

TCMESR.ERR_x1 can be cleared when writing 1 to CLR_x1.

If the status is cleared and an error occurs at the same time, the error is ignored and the status is cleared.

2.17.3.8 TCMEAR_{xm} : TCM Error Address Register (x = C or S) (m = 0 to 1)

Base address: TCM = 0x4001_C800
TCM_NS = 0x5001_C800

Offset address: 0x50, 0x54, 0x58 or 0x5C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EAR[17:2]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EAR[17:2]														—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	—	These bits are read as 0. The write value should be 0.	R/W
17:2	EAR[17:2]	When an SRAM error occurs, an error address is stored.	R/W
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note: This register is initialized after returning from Software Standby mode while the debugger is connected.

Security: S-TYPE-5

Privilege: P-TYPE-2

This register is cleared by the corresponding bit in TCMESCLR register, or resets other than Bus Error Reset and Memory Error Reset. Also during access from the debugger, this register is not updated.

TCMEAR_x0 bits

This register stores the error address where 1-bit ECC error is detected. These bits hold the error address that occurred first. These bits are cleared by clearing 1-bit ECC error from TCMESCLR.CLR_x0.

TCMEAR_x1 bits

This register stores the error address where 2-bit ECC error is detected. These bits hold the error address that occurred first. These bits are cleared by clearing 2-bit ECC error from TCMESCLR.CLR_x1.

If the status is cleared and an error occurs at the same time, the error is ignored and the status is cleared.

2.17.4 Operation

2.17.4.1 ECC Function

2.17.4.1.1 Correction of ECC Errors

Enabling and disabling of ECC error correction can be selected through the setting by the TCMCRx.ECCMOD[1:0] bits. In the initial state, ECC error correction is disabled. The ECC check type is SEC-DED (Single-Error Correction and Double-Error Detection Code).

When the ECC function is enabled, 7-bit ECC code is generated from 32-bit data. Writes to the TCM are 32-bit data and 7-bit ECC code. For reads, 32-bit data and 7-bit ECC code are read out from the TCM.

When ECC function with error checking is enabled, if a 1-bit error occurs, error correction is performed and the TCMEsr.ERRx0 bit is set to 1, if the TCMCRx.E1STSEN bit is 1. If a 2-bit error occurs, error detection is performed and the TCMEsr.ERRx1 bit is set to 1, although error correction is not performed.

When ECC function without error checking is enabled, if a 1-bit error occurs, error correction is performed but the TCMEsr.ERRx0 bit is not updated even though the TCMCRx.E1STSEN bit is 1. If a 2-bit error occurs, error detection is performed but the TCMEsr.ERRx1 bit is not updated, and error correction is not performed.

When the ECC function is disabled, neither error correction nor error detection is performed even though 1-bit or 2-bit error occurs. Therefore, the TCMEsr.ERRx0 bit and the TCMEsr.ERRx1 bit are not updated.

When updating the data after the occurrence of an error, the only support of 32-bit data writing. Since the TCM data is undefined after power on and release from Deep Software Standby mode, accessing the TCM when the ECC function is enabled and error checking is selected causes an ECC error to occur. Therefore, before using ECC function, initial writing with 32-bit data size, or initial writing with ECC function enabled and error checking disabled, to the area to be used in the TCM should be performed.

2.17.4.1.2 ECC Error Interrupt Function

When the ECC function with error checking is enabled, error checking is possible in the TCM. For a 2-bit error, the TCMEsr.ERRx1 bit set to 1. For a 1-bit error, the TCMEsr.ERRx0 bit set to 1 when TCMCRx.E1STSEN bit is 1.

When the ECC 1-bit error is to be masked, it is necessary to set the TCMCRx.E1STSEN bit to 0 to disable the setting of TCMEsr.ERRx0 bit to 1. An ECC error is not generated while the ECC function is disabled or the ECC function without error checking is enabled.

An ECC error may result in a request for a reset or an interrupt. When an ECC error occurs and when the TCMCRx.OAD bit is 1, a reset request is output. When the TCMCRx.OAD bit is 0, an interrupt request is output to the ICU.

An ECC error interrupt occurs when an error status in the TCMEsr register is set to 1. ECC error interrupt continues to occur until the error status in the TCMEsr register is cleared.

2.17.4.1.3 ECC Decoder Testing

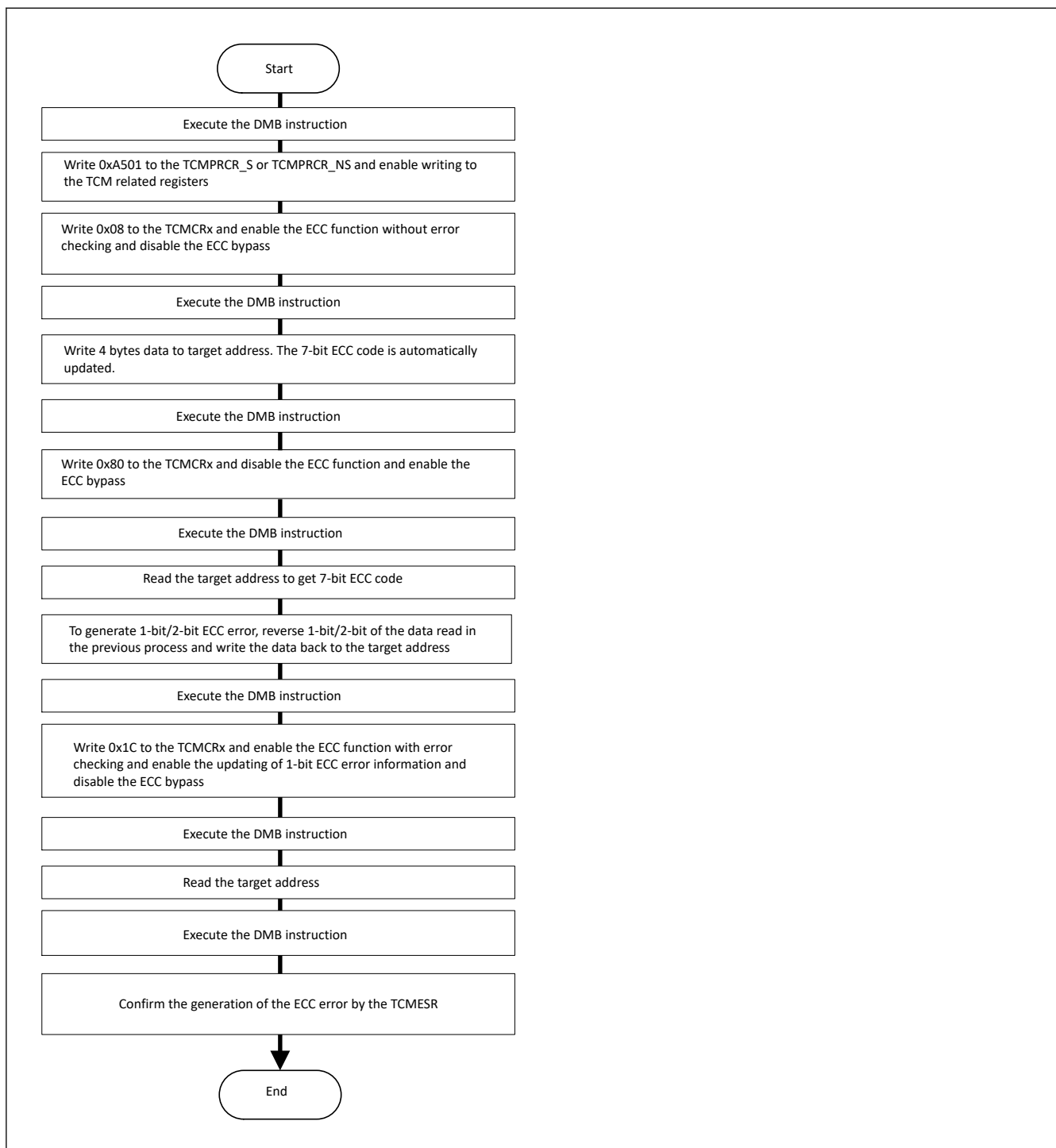


Figure 2.12 ECC decoder testing flowchart

2.17.4.2 TrustZone Filter Function

2.17.4.2.1 TrustZone Filter for TCM Registers

TCM registers can be protected with a Security Attribution (SA) from non-secure or secure transaction. When SA indicates that TCM registers are secure status, non-secure transaction cannot access them because TrustZone Filter detects an error and protects the access. When SA indicates that TCM registers are non-secure status, secure transaction cannot access them

because TrustZone Filter detects an error and protects the access. SA for TCM registers is one to be used commonly among TCM registers.

For a failed access due to a TrustZone error, an error response is generated.

SA	Transaction	Write access	Read access
Secure	Secure	Permitted	Permitted
	Non-secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
Non-secure	Secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
	Non-secure	Permitted	Permitted

2.17.4.2.2 TrustZone Filter for TCM Memory Regions

C-TCM and S-TCM regions can divide into secure, non-secure independently. The access permissions for these regions are as follows.

SA	Transaction	Write access	Read access
Secure	Secure	Permitted	Permitted
	Non-secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
Non-secure	Secure	Protected (TrustZone Filter error)	Protected (TrustZone Filter error)
	Non-secure	Permitted	Permitted

A TrustZone Filter error for TCM memory, generates an error notification.

For details, see [section 52, Security Features](#).

2.17.4.3 Interrupt Source

The TCM interrupt source includes an ECC error and a TrustZone Filter error.

ECC error can choose a reset request or an interrupt request by the TCMCRx.OAD bit.

The TCM interrupt occurs when an error status in the TCMEISR register is set to 1. The TCM interrupt continues to occur until the error status in the TCMEISR register is cleared.

When local memory error occurs (NMISR.LMST = 1 or RSTSR1.LM1RF = 1), read TCMEISR and check the TCM interrupt source. When access is from the debugger, the error is detected and corrected, but no error status is set, and no reset request or interrupt request is output.

For more details on Debug mode, see [section 2, CPU](#).

Table 2.41 TCM interrupt source

Name	Interrupt source	DTC activation	DMAC activation
ECCERR	ECC error (x-TCM with ECC)	Not possible	Not possible
TZFLT	TrustZone Filter error	Not possible	Not possible

2.17.4.4 Wait Cycle

For write access with ECC function enabled (TCMCRx.ECCMOD[1] = 1) and access size is other than 32-bit (for example, 8-bit or 16-bit), a wait cycle occurs.

If access of the bus master conflicts with access of CPU1, the bus interface of TCM returns a wait cycle to the bus master until CPU1 finishes the access, because CPU1 has priority.

If there is a security breach in CPU1 or access of the bus master to TCM, TCM responds by inserting a wait cycle with an AHB error. In other case, no wait cycle is required.

2.17.5 Usage Notes

2.17.5.1 Note on TCM Memory Cell Test

Because the TCM has a write buffer, if reading from the same address after writing, data in the write buffer may be read instead of the TCM memory cell. When reading data written for testing of the TCM memory cell, insert the write according to the following procedure to avoid reading from a write buffer.

1. Write data to the test target address.
2. Inside the same TCM, write to an address 4 or more away from the address in step 1.
3. Read the data from the address in step 1.